AI Memory Tool Comparison

This notebook benchmarks four memory layers for LLM applications — Zep, mem0, Supermemory, and Cognee — across setup, ingestion, and retrieval. The comparison is scoped to free-tier, beginner-friendly configurations, so results reflect what a developer gets without hitting a paywall, not each provider's full feature set.

Input data and retrieval tasks are held constant across all four providers to isolate differences in memory architecture rather than dataset artifacts. Each provider requires its own API key, set in a local `.env` file before running the notebook.

Before you run this notebook

- [ ] **API keys**: create a `.env` file in this notebook's directory with `ZEP_API_KEY`, `MEM0_API_KEY`, `SUPERMEMORY_API_KEY`, `COGNEE_API_KEY`, `COGNEE_URL`, and `OPENAI_API_KEY`.
- [ ] **Data path**: in section 2a ("Loading shared dataset"), update `DATA_ROOT` to point at your local copy of the `deterministic/` and `non_deterministic/` folders.

Contents

1. [Download & Setup](#1-download--setup)
- [Zep](#zep-installation)
- [mem0](#mem0-installation)
- [Supermemory](#supermemory-installation)
- [Cognee](#cognee-installation)

2. [Data Ingestion](#2-data-ingestion)
- [a. Loading shared dataset](#2a-loading-shared-dataset)
- [b. Ingestion per provider](#2b-ingestion-for-each-provider)

3. [Retrieval & Evaluation](#3-retrieval--eval)
- [a. Ground-truth QA](#a-ground-truth-qa) — absolute truth, temporal, conflict resolution, multi-hop
- [b. Ambiguity & contradiction handling](#b-ambiguity--contradiction-handling)
- [c. Needle in haystack](#c-needle-in-haystack)
- [d. Run evaluations](#d-run-evaluations)
- [e. Metrics & comparison](#e-metrics--comparison)
- [f. Manual spot-check](#f-manual-spot-check)

---

1. Download & Setup

Each provider's setup ships with a basic ingestion/retrieval smoke test; those are omitted here to keep this section focused on installation and configuration.

Zep Installation

In [1]:

```
%pip install zep-cloud
%pip install python-dotenv
import os
import uuid
from dotenv import load_dotenv
from zep_cloud.client import Zep
from zep_cloud.types import Message

load_dotenv()
zep_client = Zep(api_key=os.getenv("ZEP_API_KEY"))
```

```
Requirement already satisfied: zep-cloud in
./.venv/lib/python3.14/site-packages (3.25.0)
Requirement already satisfied: httpx>=0.21.2 in
./.venv/lib/python3.14/site-packages (from zep-cloud) (0.28.1)
Requirement already satisfied: pydantic>=1.9.2 in
./.venv/lib/python3.14/site-packages (from zep-cloud) (2.13.4)
Requirement already satisfied: pydantic-core>=2.18.2 in
./.venv/lib/python3.14/site-packages (from zep-cloud) (2.46.4)
Requirement already satisfied: python-dateutil<3.0.0,>=2.9.0 in
./.venv/lib/python3.14/site-packages (from zep-cloud)
(2.9.0.post0)
Requirement already satisfied: typing_extensions>=4.0.0 in
./.venv/lib/python3.14/site-packages (from zep-cloud) (4.16.0)
Requirement already satisfied: six>=1.5 in
./.venv/lib/python3.14/site-packages (from python-
dateutil<3.0.0,>=2.9.0->zep-cloud) (1.17.0)
Requirement already satisfied: anyio in
./.venv/lib/python3.14/site-packages (from httpx>=0.21.2->zep-
cloud) (4.14.2)
Requirement already satisfied: certifi in
./.venv/lib/python3.14/site-packages (from httpx>=0.21.2->zep-
cloud) (2026.6.17)
Requirement already satisfied: httpcore==1.* in
./.venv/lib/python3.14/site-packages (from httpx>=0.21.2->zep-
cloud) (1.0.9)
Requirement already satisfied: idna in
./.venv/lib/python3.14/site-packages (from httpx>=0.21.2->zep-
cloud) (3.18)
Requirement already satisfied: h11>=0.16 in
./.venv/lib/python3.14/site-packages (from httpcore==1.*-
>httpx>=0.21.2->zep-cloud) (0.16.0)
Requirement already satisfied: annotated-types>=0.6.0 in
./.venv/lib/python3.14/site-packages (from pydantic>=1.9.2->zep-
cloud) (0.7.0)
Requirement already satisfied: typing-inspection>=0.4.2 in
./.venv/lib/python3.14/site-packages (from pydantic>=1.9.2->zep-
cloud) (0.4.2)
Note: you may need to restart the kernel to use updated packages.
Requirement already satisfied: python-dotenv in
./.venv/lib/python3.14/site-packages (1.2.2)
Note: you may need to restart the kernel to use updated packages.
```

mem0 Installation

In [2]:

```
%pip install mem0ai
%pip install python-dotenv
%pip install mem0ai[nlp]
%pip install "mem0ai[extras]"
import os
from dotenv import load_dotenv
from mem0 import MemoryClient

load_dotenv()
mem0_client = MemoryClient(api_key=os.getenv("MEM0_API_KEY"))
```

```
Requirement already satisfied: mem0ai in
./.venv/lib/python3.14/site-packages (2.0.12)
Requirement already satisfied: httpx>=0.28.0 in
./.venv/lib/python3.14/site-packages (from mem0ai) (0.28.1)
Requirement already satisfied: openai>=1.90.0 in
./.venv/lib/python3.14/site-packages (from mem0ai) (2.46.0)
Requirement already satisfied: posthog>=7.14.0 in
./.venv/lib/python3.14/site-packages (from mem0ai) (7.27.0)
Requirement already satisfied: protobuf<7.0.0,>=5.29.6 in
./.venv/lib/python3.14/site-packages (from mem0ai) (6.33.6)
Requirement already satisfied: pydantic>=2.7.3 in
./.venv/lib/python3.14/site-packages (from mem0ai) (2.13.4)
Requirement already satisfied: pytz>=2024.1 in
./.venv/lib/python3.14/site-packages (from mem0ai) (2026.2)
Requirement already satisfied: qdrant-client>=1.12.0 in
./.venv/lib/python3.14/site-packages (from mem0ai) (1.18.0)
Requirement already satisfied: sqlalchemy>=2.0.31 in
./.venv/lib/python3.14/site-packages (from mem0ai) (2.0.51)
Requirement already satisfied: anyio in
./.venv/lib/python3.14/site-packages (from httpx>=0.28.0->mem0ai)
(4.14.2)
Requirement already satisfied: certifi in
./.venv/lib/python3.14/site-packages (from httpx>=0.28.0->mem0ai)
(2026.6.17)
Requirement already satisfied: httpcore==1.* in
./.venv/lib/python3.14/site-packages (from httpx>=0.28.0->mem0ai)
(1.0.9)
Requirement already satisfied: idna in
./.venv/lib/python3.14/site-packages (from httpx>=0.28.0->mem0ai)
(3.18)
Requirement already satisfied: h11>=0.16 in
./.venv/lib/python3.14/site-packages (from httpcore==1.*-
>httpx>=0.28.0->mem0ai) (0.16.0)
Requirement already satisfied: distro<2,>=1.7.0 in
./.venv/lib/python3.14/site-packages (from openai>=1.90.0-
>mem0ai) (1.9.0)
Requirement already satisfied: jiter<1,>=0.10.0 in
./.venv/lib/python3.14/site-packages (from openai>=1.90.0-
>mem0ai) (0.13.0)
Requirement already satisfied: sniffio in
./.venv/lib/python3.14/site-packages (from openai>=1.90.0-
>mem0ai) (1.3.1)
Requirement already satisfied: tqdm>4 in
./.venv/lib/python3.14/site-packages (from openai>=1.90.0-
>mem0ai) (4.69.0)
Requirement already satisfied: typing-extensions<5,>=4.14 in
./.venv/lib/python3.14/site-packages (from openai>=1.90.0-
>mem0ai) (4.16.0)
Requirement already satisfied: annotated-types>=0.6.0 in
./.venv/lib/python3.14/site-packages (from pydantic>=2.7.3-
>mem0ai) (0.7.0)
Requirement already satisfied: pydantic-core==2.46.4 in
./.venv/lib/python3.14/site-packages (from pydantic>=2.7.3-
>mem0ai) (2.46.4)
Requirement already satisfied: typing-inspection>=0.4.2 in
./.venv/lib/python3.14/site-packages (from pydantic>=2.7.3-
>mem0ai) (0.4.2)
Requirement already satisfied: requests<3.0,>=2.7 in
./.venv/lib/python3.14/site-packages (from posthog>=7.14.0-
>mem0ai) (2.34.2)
Requirement already satisfied: backoff>=1.10.0 in
./.venv/lib/python3.14/site-packages (from posthog>=7.14.0-
>mem0ai) (2.2.1)
Requirement already satisfied: charset_normalizer<4,>=2 in
./.venv/lib/python3.14/site-packages (from requests<3.0,>=2.7-
>posthog>=7.14.0->mem0ai) (3.4.9)
Requirement already satisfied: urllib3<3,>=1.26 in
./.venv/lib/python3.14/site-packages (from requests<3.0,>=2.7-
>posthog>=7.14.0->mem0ai) (2.7.0)
Requirement already satisfied: grpcio>=1.41.0 in
./.venv/lib/python3.14/site-packages (from qdrant-client>=1.12.0-
>mem0ai) (1.82.1)
Requirement already satisfied: numpy>=2.3.0 in
./.venv/lib/python3.14/site-packages (from qdrant-client>=1.12.0-
>mem0ai) (2.5.1)
Requirement already satisfied: portalocker<4.0,>=2.7.0 in
./.venv/lib/python3.14/site-packages (from qdrant-client>=1.12.0-
>mem0ai) (3.2.0)
Requirement already satisfied: h2<5,>=3 in
./.venv/lib/python3.14/site-packages (from httpx[http2]>=0.20.0-
>qdrant-client>=1.12.0->mem0ai) (4.3.0)
Requirement already satisfied: hyperframe<7,>=6.1 in
./.venv/lib/python3.14/site-packages (from h2<5,>=3-
>httpx[http2]>=0.20.0->qdrant-client>=1.12.0->mem0ai) (6.1.0)
Requirement already satisfied: hpack<5,>=4.1 in
./.venv/lib/python3.14/site-packages (from h2<5,>=3-
>httpx[http2]>=0.20.0->qdrant-client>=1.12.0->mem0ai) (4.2.0)
Note: you may need to restart the kernel to use updated packages.
Requirement already satisfied: python-dotenv in
./.venv/lib/python3.14/site-packages (1.2.2)
Note: you may need to restart the kernel to use updated packages.
zsh:1: no matches found: mem0ai[nlp]
Note: you may need to restart the kernel to use updated packages.
Requirement already satisfied: mem0ai[extras] in
./.venv/lib/python3.14/site-packages (2.0.12)
Requirement already satisfied: httpx>=0.28.0 in
./.venv/lib/python3.14/site-packages (from mem0ai[extras])
(0.28.1)
Requirement already satisfied: openai>=1.90.0 in
./.venv/lib/python3.14/site-packages (from mem0ai[extras])
(2.46.0)
Requirement already satisfied: posthog>=7.14.0 in
./.venv/lib/python3.14/site-packages (from mem0ai[extras])
(7.27.0)
Requirement already satisfied: protobuf<7.0.0,>=5.29.6 in
./.venv/lib/python3.14/site-packages (from mem0ai[extras])
(6.33.6)
Requirement already satisfied: pydantic>=2.7.3 in
./.venv/lib/python3.14/site-packages (from mem0ai[extras])
(2.13.4)
Requirement already satisfied: pytz>=2024.1 in
./.venv/lib/python3.14/site-packages (from mem0ai[extras])
(2026.2)
Requirement already satisfied: qdrant-client>=1.12.0 in
./.venv/lib/python3.14/site-packages (from mem0ai[extras])
(1.18.0)
Requirement already satisfied: sqlalchemy>=2.0.31 in
./.venv/lib/python3.14/site-packages (from mem0ai[extras])
(2.0.51)
Requirement already satisfied: boto3>=1.34.0 in
./.venv/lib/python3.14/site-packages (from mem0ai[extras])
(1.43.52)
Requirement already satisfied: elasticsearch<9.0.0,>=8.0.0 in
./.venv/lib/python3.14/site-packages (from mem0ai[extras])
(8.19.3)
Requirement already satisfied: fastembed>=0.3.1 in
./.venv/lib/python3.14/site-packages (from mem0ai[extras])
(0.8.0)
Requirement already satisfied: langchain-community<1.0.0,>=0.3.27
in ./.venv/lib/python3.14/site-packages (from mem0ai[extras])
(0.3.31)
Requirement already satisfied: langchain-core<1.0.0,>=0.3.85 in
./.venv/lib/python3.14/site-packages (from mem0ai[extras])
(0.3.86)
Requirement already satisfied: langchain<1.0.0,>=0.3.30 in
./.venv/lib/python3.14/site-packages (from mem0ai[extras])
(0.3.30)
Requirement already satisfied: opensearch-py>=2.0.0 in
./.venv/lib/python3.14/site-packages (from mem0ai[extras])
(3.2.0)
Requirement already satisfied: sentence-transformers>=5.2.0 in
./.venv/lib/python3.14/site-packages (from mem0ai[extras])
(5.6.0)
Requirement already satisfied: transformers>=5.3.0 in
./.venv/lib/python3.14/site-packages (from mem0ai[extras])
(5.14.1)
Requirement already satisfied: elastic-transport<9,>=8.15.1 in
./.venv/lib/python3.14/site-packages (from
elasticsearch<9.0.0,>=8.0.0->mem0ai[extras]) (8.19.0)
Requirement already satisfied: python-dateutil in
./.venv/lib/python3.14/site-packages (from
elasticsearch<9.0.0,>=8.0.0->mem0ai[extras]) (2.9.0.post0)
Requirement already satisfied: typing-extensions in
./.venv/lib/python3.14/site-packages (from
elasticsearch<9.0.0,>=8.0.0->mem0ai[extras]) (4.16.0)
Requirement already satisfied: urllib3<3,>=1.26.2 in
./.venv/lib/python3.14/site-packages (from elastic-
transport<9,>=8.15.1->elasticsearch<9.0.0,>=8.0.0-
>mem0ai[extras]) (2.7.0)
Requirement already satisfied: certifi in
./.venv/lib/python3.14/site-packages (from elastic-
transport<9,>=8.15.1->elasticsearch<9.0.0,>=8.0.0-
>mem0ai[extras]) (2026.6.17)
Requirement already satisfied: langchain-text-
splitters<1.0.0,>=0.3.9 in ./.venv/lib/python3.14/site-packages
(from langchain<1.0.0,>=0.3.30->mem0ai[extras]) (0.3.11)
Requirement already satisfied: langsmith<1.0.0,>=0.1.17 in
./.venv/lib/python3.14/site-packages (from
langchain<1.0.0,>=0.3.30->mem0ai[extras]) (0.10.9)
Requirement already satisfied: requests<3.0.0,>=2.0.0 in
./.venv/lib/python3.14/site-packages (from
langchain<1.0.0,>=0.3.30->mem0ai[extras]) (2.34.2)
Requirement already satisfied: PyYAML<7.0.0,>=5.3.0 in
./.venv/lib/python3.14/site-packages (from
langchain<1.0.0,>=0.3.30->mem0ai[extras]) (6.0.3)
Requirement already satisfied: aiohttp<4.0.0,>=3.8.3 in
./.venv/lib/python3.14/site-packages (from langchain-
community<1.0.0,>=0.3.27->mem0ai[extras]) (3.14.1)
Requirement already satisfied: tenacity!=8.4.0,<10.0.0,>=8.1.0 in
./.venv/lib/python3.14/site-packages (from langchain-
community<1.0.0,>=0.3.27->mem0ai[extras]) (9.1.4)
Requirement already satisfied: dataclasses-json<0.7.0,>=0.6.7 in
./.venv/lib/python3.14/site-packages (from langchain-
community<1.0.0,>=0.3.27->mem0ai[extras]) (0.6.7)
Requirement already satisfied: pydantic-settings<3.0.0,>=2.10.1
in ./.venv/lib/python3.14/site-packages (from langchain-
community<1.0.0,>=0.3.27->mem0ai[extras]) (2.14.2)
Requirement already satisfied: httpx-sse<1.0.0,>=0.4.0 in
./.venv/lib/python3.14/site-packages (from langchain-
community<1.0.0,>=0.3.27->mem0ai[extras]) (0.4.3)
Requirement already satisfied: numpy>=2.1.0 in
./.venv/lib/python3.14/site-packages (from langchain-
community<1.0.0,>=0.3.27->mem0ai[extras]) (2.5.1)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in
./.venv/lib/python3.14/site-packages (from aiohttp<4.0.0,>=3.8.3-
>langchain-community<1.0.0,>=0.3.27->mem0ai[extras]) (2.7.1)
Requirement already satisfied: aiosignal>=1.4.0 in
./.venv/lib/python3.14/site-packages (from aiohttp<4.0.0,>=3.8.3-
>langchain-community<1.0.0,>=0.3.27->mem0ai[extras]) (1.4.0)
Requirement already satisfied: attrs>=17.3.0 in
./.venv/lib/python3.14/site-packages (from aiohttp<4.0.0,>=3.8.3-
>langchain-community<1.0.0,>=0.3.27->mem0ai[extras]) (26.1.0)
Requirement already satisfied: frozenlist>=1.1.1 in
./.venv/lib/python3.14/site-packages (from aiohttp<4.0.0,>=3.8.3-
>langchain-community<1.0.0,>=0.3.27->mem0ai[extras]) (1.8.0)
Requirement already satisfied: multidict<7.0,>=4.5 in
./.venv/lib/python3.14/site-packages (from aiohttp<4.0.0,>=3.8.3-
>langchain-community<1.0.0,>=0.3.27->mem0ai[extras]) (6.7.1)
Requirement already satisfied: propcache>=0.2.0 in
./.venv/lib/python3.14/site-packages (from aiohttp<4.0.0,>=3.8.3-
>langchain-community<1.0.0,>=0.3.27->mem0ai[extras]) (0.5.2)
Requirement already satisfied: yarl<2.0,>=1.17.0 in
./.venv/lib/python3.14/site-packages (from aiohttp<4.0.0,>=3.8.3-
>langchain-community<1.0.0,>=0.3.27->mem0ai[extras]) (1.24.5)
Requirement already satisfied: marshmallow<4.0.0,>=3.18.0 in
./.venv/lib/python3.14/site-packages (from dataclasses-
json<0.7.0,>=0.6.7->langchain-community<1.0.0,>=0.3.27-
>mem0ai[extras]) (3.26.2)
Requirement already satisfied: typing-inspect<1,>=0.4.0 in
./.venv/lib/python3.14/site-packages (from dataclasses-
json<0.7.0,>=0.6.7->langchain-community<1.0.0,>=0.3.27-
>mem0ai[extras]) (0.9.0)
Requirement already satisfied: jsonpatch<2.0.0,>=1.33.0 in
./.venv/lib/python3.14/site-packages (from langchain-
core<1.0.0,>=0.3.85->mem0ai[extras]) (1.33)
Requirement already satisfied: packaging<26.0.0,>=23.2.0 in
./.venv/lib/python3.14/site-packages (from langchain-
core<1.0.0,>=0.3.85->mem0ai[extras]) (24.2)
Requirement already satisfied: uuid-utils<1.0,>=0.12.0 in
./.venv/lib/python3.14/site-packages (from langchain-
core<1.0.0,>=0.3.85->mem0ai[extras]) (0.17.0)
Requirement already satisfied: jsonpointer>=1.9 in
./.venv/lib/python3.14/site-packages (from
jsonpatch<2.0.0,>=1.33.0->langchain-core<1.0.0,>=0.3.85-
>mem0ai[extras]) (3.1.1)
Requirement already satisfied: anyio>=3.5.0 in
./.venv/lib/python3.14/site-packages (from
langsmith<1.0.0,>=0.1.17->langchain<1.0.0,>=0.3.30-
>mem0ai[extras]) (4.14.2)
Requirement already satisfied: distro>=1.7.0 in
./.venv/lib/python3.14/site-packages (from
langsmith<1.0.0,>=0.1.17->langchain<1.0.0,>=0.3.30-
>mem0ai[extras]) (1.9.0)
Requirement already satisfied: orjson>=3.9.14 in
./.venv/lib/python3.14/site-packages (from
langsmith<1.0.0,>=0.1.17->langchain<1.0.0,>=0.3.30-
>mem0ai[extras]) (3.11.9)
Requirement already satisfied: requests-toolbelt>=1.0.0 in
./.venv/lib/python3.14/site-packages (from
langsmith<1.0.0,>=0.1.17->langchain<1.0.0,>=0.3.30-
>mem0ai[extras]) (1.0.0)
Requirement already satisfied: sniffio>=1.1 in
./.venv/lib/python3.14/site-packages (from
langsmith<1.0.0,>=0.1.17->langchain<1.0.0,>=0.3.30-
>mem0ai[extras]) (1.3.1)
Requirement already satisfied: websockets>=15.0 in
./.venv/lib/python3.14/site-packages (from
langsmith<1.0.0,>=0.1.17->langchain<1.0.0,>=0.3.30-
>mem0ai[extras]) (15.0.1)
Requirement already satisfied: xxhash>=3.0.0 in
./.venv/lib/python3.14/site-packages (from
langsmith<1.0.0,>=0.1.17->langchain<1.0.0,>=0.3.30-
>mem0ai[extras]) (3.8.1)
Requirement already satisfied: zstandard>=0.23.0 in
./.venv/lib/python3.14/site-packages (from
langsmith<1.0.0,>=0.1.17->langchain<1.0.0,>=0.3.30-
>mem0ai[extras]) (0.25.0)
Requirement already satisfied: httpcore==1.* in
./.venv/lib/python3.14/site-packages (from httpx>=0.28.0-
>mem0ai[extras]) (1.0.9)
Requirement already satisfied: idna in
./.venv/lib/python3.14/site-packages (from httpx>=0.28.0-
>mem0ai[extras]) (3.18)
Requirement already satisfied: h11>=0.16 in
./.venv/lib/python3.14/site-packages (from httpcore==1.*-
>httpx>=0.28.0->mem0ai[extras]) (0.16.0)
Requirement already satisfied: annotated-types>=0.6.0 in
./.venv/lib/python3.14/site-packages (from pydantic>=2.7.3-
>mem0ai[extras]) (0.7.0)
Requirement already satisfied: pydantic-core==2.46.4 in
./.venv/lib/python3.14/site-packages (from pydantic>=2.7.3-
>mem0ai[extras]) (2.46.4)
Requirement already satisfied: typing-inspection>=0.4.2 in
./.venv/lib/python3.14/site-packages (from pydantic>=2.7.3-
>mem0ai[extras]) (0.4.2)
Requirement already satisfied: python-dotenv>=0.21.0 in
./.venv/lib/python3.14/site-packages (from pydantic-
settings<3.0.0,>=2.10.1->langchain-community<1.0.0,>=0.3.27-
>mem0ai[extras]) (1.2.2)
Requirement already satisfied: charset_normalizer<4,>=2 in
./.venv/lib/python3.14/site-packages (from
requests<3.0.0,>=2.0.0->langchain<1.0.0,>=0.3.30->mem0ai[extras])
(3.4.9)
Requirement already satisfied: mypy-extensions>=0.3.0 in
./.venv/lib/python3.14/site-packages (from typing-
inspect<1,>=0.4.0->dataclasses-json<0.7.0,>=0.6.7->langchain-
community<1.0.0,>=0.3.27->mem0ai[extras]) (1.1.0)
Requirement already satisfied: botocore<1.44.0,>=1.43.52 in
./.venv/lib/python3.14/site-packages (from boto3>=1.34.0-
>mem0ai[extras]) (1.43.52)
Requirement already satisfied: jmespath<2.0.0,>=0.7.1 in
./.venv/lib/python3.14/site-packages (from boto3>=1.34.0-
>mem0ai[extras]) (1.1.0)
Requirement already satisfied: s3transfer<0.20.0,>=0.19.0 in
./.venv/lib/python3.14/site-packages (from boto3>=1.34.0-
>mem0ai[extras]) (0.19.1)
Requirement already satisfied: six>=1.5 in
./.venv/lib/python3.14/site-packages (from python-dateutil-
>elasticsearch<9.0.0,>=8.0.0->mem0ai[extras]) (1.17.0)
Requirement already satisfied: huggingface-hub<2.0,>=0.20 in
./.venv/lib/python3.14/site-packages (from fastembed>=0.3.1-
>mem0ai[extras]) (1.24.0)
Requirement already satisfied: loguru<0.8.0,>=0.7.2 in
./.venv/lib/python3.14/site-packages (from fastembed>=0.3.1-
>mem0ai[extras]) (0.7.3)
Requirement already satisfied: mmh3<6.0.0,>=4.1.0 in
./.venv/lib/python3.14/site-packages (from fastembed>=0.3.1-
>mem0ai[extras]) (5.2.1)
Requirement already satisfied: onnxruntime>=1.24.2 in
./.venv/lib/python3.14/site-packages (from fastembed>=0.3.1-
>mem0ai[extras]) (1.27.0)
Requirement already satisfied: pillow<13.0,>=12.0.0 in
./.venv/lib/python3.14/site-packages (from fastembed>=0.3.1-
>mem0ai[extras]) (12.3.0)
Requirement already satisfied: py-rust-stemmers<0.2.0,>=0.1.0 in
./.venv/lib/python3.14/site-packages (from fastembed>=0.3.1-
>mem0ai[extras]) (0.1.8)
Requirement already satisfied: tokenizers<1.0,>=0.15 in
./.venv/lib/python3.14/site-packages (from fastembed>=0.3.1-
>mem0ai[extras]) (0.22.2)
Requirement already satisfied: tqdm<5.0,>=4.66 in
./.venv/lib/python3.14/site-packages (from fastembed>=0.3.1-
>mem0ai[extras]) (4.69.0)
Requirement already satisfied: click<9.0.0,>=8.4.2 in
./.venv/lib/python3.14/site-packages (from huggingface-
hub<2.0,>=0.20->fastembed>=0.3.1->mem0ai[extras]) (8.4.2)
Requirement already satisfied: filelock>=3.10.0 in
./.venv/lib/python3.14/site-packages (from huggingface-
hub<2.0,>=0.20->fastembed>=0.3.1->mem0ai[extras]) (3.31.1)
Requirement already satisfied: fsspec>=2023.5.0 in
./.venv/lib/python3.14/site-packages (from huggingface-
hub<2.0,>=0.20->fastembed>=0.3.1->mem0ai[extras]) (2026.6.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.5.1 in
./.venv/lib/python3.14/site-packages (from huggingface-
hub<2.0,>=0.20->fastembed>=0.3.1->mem0ai[extras]) (1.5.2)
Requirement already satisfied: flatbuffers in
./.venv/lib/python3.14/site-packages (from onnxruntime>=1.24.2-
>fastembed>=0.3.1->mem0ai[extras]) (25.12.19)
Requirement already satisfied: jiter<1,>=0.10.0 in
./.venv/lib/python3.14/site-packages (from openai>=1.90.0-
>mem0ai[extras]) (0.13.0)
Requirement already satisfied: Events in
./.venv/lib/python3.14/site-packages (from opensearch-py>=2.0.0-
>mem0ai[extras]) (0.5)
Requirement already satisfied: opensearch-protobufs==1.2.0 in
./.venv/lib/python3.14/site-packages (from opensearch-py>=2.0.0-
>mem0ai[extras]) (1.2.0)
Requirement already satisfied: grpcio>=1.70.0 in
./.venv/lib/python3.14/site-packages (from opensearch-
protobufs==1.2.0->opensearch-py>=2.0.0->mem0ai[extras]) (1.82.1)
Requirement already satisfied: backoff>=1.10.0 in
./.venv/lib/python3.14/site-packages (from posthog>=7.14.0-
>mem0ai[extras]) (2.2.1)
Requirement already satisfied: portalocker<4.0,>=2.7.0 in
./.venv/lib/python3.14/site-packages (from qdrant-client>=1.12.0-
>mem0ai[extras]) (3.2.0)
Requirement already satisfied: h2<5,>=3 in
./.venv/lib/python3.14/site-packages (from httpx[http2]>=0.20.0-
>qdrant-client>=1.12.0->mem0ai[extras]) (4.3.0)
Requirement already satisfied: hyperframe<7,>=6.1 in
./.venv/lib/python3.14/site-packages (from h2<5,>=3-
>httpx[http2]>=0.20.0->qdrant-client>=1.12.0->mem0ai[extras])
(6.1.0)
Requirement already satisfied: hpack<5,>=4.1 in
./.venv/lib/python3.14/site-packages (from h2<5,>=3-
>httpx[http2]>=0.20.0->qdrant-client>=1.12.0->mem0ai[extras])
(4.2.0)
Requirement already satisfied: torch>=1.11.0 in
./.venv/lib/python3.14/site-packages (from sentence-
transformers>=5.2.0->mem0ai[extras]) (2.13.0)
Requirement already satisfied: scikit-learn>=0.22.0 in
./.venv/lib/python3.14/site-packages (from sentence-
transformers>=5.2.0->mem0ai[extras]) (1.9.0)
Requirement already satisfied: scipy>=1.0.0 in
./.venv/lib/python3.14/site-packages (from sentence-
transformers>=5.2.0->mem0ai[extras]) (1.18.0)
Requirement already satisfied: regex>=2025.10.22 in
./.venv/lib/python3.14/site-packages (from transformers>=5.3.0-
>mem0ai[extras]) (2026.7.19)
Requirement already satisfied: typer in
./.venv/lib/python3.14/site-packages (from transformers>=5.3.0-
>mem0ai[extras]) (0.27.0)
Requirement already satisfied: safetensors>=0.8.0 in
./.venv/lib/python3.14/site-packages (from transformers>=5.3.0-
>mem0ai[extras]) (0.8.0)
Requirement already satisfied: joblib>=1.4.0 in
./.venv/lib/python3.14/site-packages (from scikit-learn>=0.22.0-
>sentence-transformers>=5.2.0->mem0ai[extras]) (1.5.3)
Requirement already satisfied: narwhals>=2.0.1 in
./.venv/lib/python3.14/site-packages (from scikit-learn>=0.22.0-
>sentence-transformers>=5.2.0->mem0ai[extras]) (2.24.0)
Requirement already satisfied: threadpoolctl>=3.5.0 in
./.venv/lib/python3.14/site-packages (from scikit-learn>=0.22.0-
>sentence-transformers>=5.2.0->mem0ai[extras]) (3.6.0)
Requirement already satisfied: setuptools>=77.0.3 in
./.venv/lib/python3.14/site-packages (from torch>=1.11.0-
>sentence-transformers>=5.2.0->mem0ai[extras]) (83.0.0)
Requirement already satisfied: sympy>=1.13.3 in
./.venv/lib/python3.14/site-packages (from torch>=1.11.0-
>sentence-transformers>=5.2.0->mem0ai[extras]) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in
./.venv/lib/python3.14/site-packages (from torch>=1.11.0-
>sentence-transformers>=5.2.0->mem0ai[extras]) (3.6.1)
Requirement already satisfied: jinja2 in
./.venv/lib/python3.14/site-packages (from torch>=1.11.0-
>sentence-transformers>=5.2.0->mem0ai[extras]) (3.1.6)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in
./.venv/lib/python3.14/site-packages (from sympy>=1.13.3-
>torch>=1.11.0->sentence-transformers>=5.2.0->mem0ai[extras])
(1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in
./.venv/lib/python3.14/site-packages (from jinja2->torch>=1.11.0-
>sentence-transformers>=5.2.0->mem0ai[extras]) (3.0.3)
Requirement already satisfied: shellingham>=1.3.0 in
./.venv/lib/python3.14/site-packages (from typer-
>transformers>=5.3.0->mem0ai[extras]) (1.5.4)
Requirement already satisfied: rich>=13.8.0 in
./.venv/lib/python3.14/site-packages (from typer-
>transformers>=5.3.0->mem0ai[extras]) (14.3.4)
Requirement already satisfied: annotated-doc>=0.0.2 in
./.venv/lib/python3.14/site-packages (from typer-
>transformers>=5.3.0->mem0ai[extras]) (0.0.4)
Requirement already satisfied: markdown-it-py>=2.2.0 in
./.venv/lib/python3.14/site-packages (from rich>=13.8.0->typer-
>transformers>=5.3.0->mem0ai[extras]) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
./.venv/lib/python3.14/site-packages (from rich>=13.8.0->typer-
>transformers>=5.3.0->mem0ai[extras]) (2.20.0)
Requirement already satisfied: mdurl~=0.1 in
./.venv/lib/python3.14/site-packages (from markdown-it-py>=2.2.0-
>rich>=13.8.0->typer->transformers>=5.3.0->mem0ai[extras])
(0.1.2)
Note: you may need to restart the kernel to use updated packages.
```

```
/Users/amandademoura/VSCode/AI-Memory
Comparison/.venv/lib/python3.14/site-packages/tqdm/auto.py:21:
TqdmWarning: IProgress not found. Please update jupyter and
ipywidgets. See
https://ipywidgets.readthedocs.io/en/stable/user_install.html
  from .autonotebook import tqdm as notebook_tqdm
```

Supermemory Installation

In [3]:

```
%pip install supermemory
import os
from supermemory import Supermemory
from dotenv import load_dotenv


load_dotenv()
sm_client = Supermemory(api_key=os.getenv("SUPERMEMORY_API_KEY"))
```

```
/opt/homebrew/Cellar/python@3.14/3.14.6/Frameworks/Python.fram
ework/Versions/3.14/lib/python3.14/pty.py:66:
DeprecationWarning: This process (pid=31795) is multi-
threaded, use of forkpty() may lead to deadlocks in the child.
  pid, fd = os.forkpty()
```

```
Requirement already satisfied: supermemory in
./.venv/lib/python3.14/site-packages (3.51.0)
Requirement already satisfied: anyio<5,>=3.5.0 in
./.venv/lib/python3.14/site-packages (from supermemory) (4.14.2)
Requirement already satisfied: distro<2,>=1.7.0 in
./.venv/lib/python3.14/site-packages (from supermemory) (1.9.0)
Requirement already satisfied: httpx<1,>=0.23.0 in
./.venv/lib/python3.14/site-packages (from supermemory) (0.28.1)
Requirement already satisfied: pydantic<3,>=1.9.0 in
./.venv/lib/python3.14/site-packages (from supermemory) (2.13.4)
Requirement already satisfied: sniffio in
./.venv/lib/python3.14/site-packages (from supermemory) (1.3.1)
Requirement already satisfied: typing-extensions<5,>=4.14 in
./.venv/lib/python3.14/site-packages (from supermemory) (4.16.0)
Requirement already satisfied: idna>=2.8 in
./.venv/lib/python3.14/site-packages (from anyio<5,>=3.5.0-
>supermemory) (3.18)
Requirement already satisfied: certifi in
./.venv/lib/python3.14/site-packages (from httpx<1,>=0.23.0-
>supermemory) (2026.6.17)
Requirement already satisfied: httpcore==1.* in
./.venv/lib/python3.14/site-packages (from httpx<1,>=0.23.0-
>supermemory) (1.0.9)
Requirement already satisfied: h11>=0.16 in
./.venv/lib/python3.14/site-packages (from httpcore==1.*-
>httpx<1,>=0.23.0->supermemory) (0.16.0)
Requirement already satisfied: annotated-types>=0.6.0 in
./.venv/lib/python3.14/site-packages (from pydantic<3,>=1.9.0-
>supermemory) (0.7.0)
Requirement already satisfied: pydantic-core==2.46.4 in
./.venv/lib/python3.14/site-packages (from pydantic<3,>=1.9.0-
>supermemory) (2.46.4)
Requirement already satisfied: typing-inspection>=0.4.2 in
./.venv/lib/python3.14/site-packages (from pydantic<3,>=1.9.0-
>supermemory) (0.4.2)
Note: you may need to restart the kernel to use updated packages.
```

Cognee Installation

In [4]:

```
%pip install cognee
import os
from dotenv import load_dotenv
import cognee

load_dotenv()

async def main():
    await cognee.serve(
        url=os.getenv("COGNEE_URL"),
        api_key=os.getenv("COGNEE_API_KEY")
    )

await main()
```

```
/opt/homebrew/Cellar/python@3.14/3.14.6/Frameworks/Python.fram
ework/Versions/3.14/lib/python3.14/pty.py:66:
DeprecationWarning: This process (pid=31795) is multi-
threaded, use of forkpty() may lead to deadlocks in the child.
  pid, fd = os.forkpty()
```

```
Requirement already satisfied: cognee in
./.venv/lib/python3.14/site-packages (1.4.0)
Requirement already satisfied: aiofiles>=23.2.1 in
./.venv/lib/python3.14/site-packages (from cognee) (25.1.0)
Requirement already satisfied: aiohttp<4.0.0,>=3.13.5 in
./.venv/lib/python3.14/site-packages (from cognee) (3.14.1)
Requirement already satisfied: aiolimiter>=1.2.1 in
./.venv/lib/python3.14/site-packages (from cognee) (1.2.1)
Requirement already satisfied: aiosqlite<1.0.0,>=0.20.0 in
./.venv/lib/python3.14/site-packages (from cognee) (0.22.1)
Requirement already satisfied: alembic<2,>=1.13.3 in
./.venv/lib/python3.14/site-packages (from cognee) (1.18.5)
Requirement already satisfied: cbor2>=5.8.0 in
./.venv/lib/python3.14/site-packages (from cognee) (6.1.3)
Requirement already satisfied: datamodel-code-generator>=0.54.0
in ./.venv/lib/python3.14/site-packages (from cognee) (0.69.0)
Requirement already satisfied: diskcache>=5.6.3 in
./.venv/lib/python3.14/site-packages (from cognee) (5.6.3)
Requirement already satisfied: fakeredis>=2.32.0 in
./.venv/lib/python3.14/site-packages (from
fakeredis[lua]>=2.32.0->cognee) (2.36.2)
Requirement already satisfied: fastapi-users>=15.0.2 in
./.venv/lib/python3.14/site-packages (from fastapi-
users[sqlalchemy]>=15.0.2->cognee) (15.0.5)
Requirement already satisfied: fastapi<1.0.0,>=0.116.2 in
./.venv/lib/python3.14/site-packages (from cognee) (0.139.2)
Requirement already satisfied: filelock<4.0.0,>=3.12.0 in
./.venv/lib/python3.14/site-packages (from cognee) (3.31.1)
Requirement already satisfied: filetype<2.0.0,>=1.2.0 in
./.venv/lib/python3.14/site-packages (from cognee) (1.2.0)
Requirement already satisfied: gunicorn<24,>=20.1.0 in
./.venv/lib/python3.14/site-packages (from cognee) (23.0.0)
Requirement already satisfied: instructor<1.15.3,>=1.9.1 in
./.venv/lib/python3.14/site-packages (from cognee) (1.15.1)
Requirement already satisfied: jinja2<4,>=3.1.3 in
./.venv/lib/python3.14/site-packages (from cognee) (3.1.6)
Requirement already satisfied: ladybug<0.18,>=0.16.0 in
./.venv/lib/python3.14/site-packages (from cognee) (0.17.1)
Requirement already satisfied: lancedb<1.0.0,>=0.24.3 in
./.venv/lib/python3.14/site-packages (from cognee) (0.34.0)
Requirement already satisfied: langdetect>=1.0.9 in
./.venv/lib/python3.14/site-packages (from cognee) (1.0.9)
Requirement already satisfied: limits<5,>=4.4.1 in
./.venv/lib/python3.14/site-packages (from cognee) (4.8.0)
Requirement already satisfied: litellm>=1.83.7 in
./.venv/lib/python3.14/site-packages (from cognee) (1.93.0)
Requirement already satisfied: nbformat<6.0.0,>=5.7.0 in
./.venv/lib/python3.14/site-packages (from cognee) (5.10.4)
Requirement already satisfied: networkx<4,>=3.4.2 in
./.venv/lib/python3.14/site-packages (from cognee) (3.6.1)
Requirement already satisfied: numpy<=4.0.0,>=1.26.4 in
./.venv/lib/python3.14/site-packages (from cognee) (2.5.1)
Requirement already satisfied: openai>=1.80.1 in
./.venv/lib/python3.14/site-packages (from cognee) (2.46.0)
Requirement already satisfied: pydantic-
settings!=2.12.*,!=2.13.*,!=2.14.0,!=2.14.1,<3,>=2.2.1 in
./.venv/lib/python3.14/site-packages (from cognee) (2.14.2)
Requirement already satisfied: pydantic>=2.10.5 in
./.venv/lib/python3.14/site-packages (from cognee) (2.13.4)
Requirement already satisfied: pylance<=0.36.0,>=0.22.0 in
./.venv/lib/python3.14/site-packages (from cognee) (0.36.0)
Requirement already satisfied: pympler<2.0.0,>=1.1 in
./.venv/lib/python3.14/site-packages (from cognee) (1.1)
Requirement already satisfied: pypdf<7.0.0,>=6.6.2 in
./.venv/lib/python3.14/site-packages (from cognee) (6.14.2)
Requirement already satisfied: python-dotenv<2.0.0,>=1.0.1 in
./.venv/lib/python3.14/site-packages (from cognee) (1.2.2)
Requirement already satisfied: python-multipart<1.0.0,>=0.0.22 in
./.venv/lib/python3.14/site-packages (from cognee) (0.0.32)
Requirement already satisfied: rdflib<7.2.0,>=7.1.4 in
./.venv/lib/python3.14/site-packages (from cognee) (7.1.4)
Requirement already satisfied: sqlalchemy<3.0.0,>=2.0.39 in
./.venv/lib/python3.14/site-packages (from cognee) (2.0.51)
Requirement already satisfied: starlette>=0.48 in
./.venv/lib/python3.14/site-packages (from cognee) (1.3.1)
Requirement already satisfied: structlog<26,>=25.2.0 in
./.venv/lib/python3.14/site-packages (from cognee) (25.5.0)
Requirement already satisfied: tenacity>=9.0.0 in
./.venv/lib/python3.14/site-packages (from cognee) (9.1.4)
Requirement already satisfied: tiktoken<1.0.0,>=0.8.0 in
./.venv/lib/python3.14/site-packages (from cognee) (0.13.0)
Requirement already satisfied: typing-extensions<5.0.0,>=4.12.2
in ./.venv/lib/python3.14/site-packages (from cognee) (4.16.0)
Requirement already satisfied: urllib3>=2.6.0 in
./.venv/lib/python3.14/site-packages (from cognee) (2.7.0)
Requirement already satisfied: uvicorn<1.0.0,>=0.34.0 in
./.venv/lib/python3.14/site-packages (from cognee) (0.51.0)
Requirement already satisfied: websockets<16.0.0,>=15.0.1 in
./.venv/lib/python3.14/site-packages (from cognee) (15.0.1)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in
./.venv/lib/python3.14/site-packages (from
aiohttp<4.0.0,>=3.13.5->cognee) (2.7.1)
Requirement already satisfied: aiosignal>=1.4.0 in
./.venv/lib/python3.14/site-packages (from
aiohttp<4.0.0,>=3.13.5->cognee) (1.4.0)
Requirement already satisfied: attrs>=17.3.0 in
./.venv/lib/python3.14/site-packages (from
aiohttp<4.0.0,>=3.13.5->cognee) (26.1.0)
Requirement already satisfied: frozenlist>=1.1.1 in
./.venv/lib/python3.14/site-packages (from
aiohttp<4.0.0,>=3.13.5->cognee) (1.8.0)
Requirement already satisfied: multidict<7.0,>=4.5 in
```

```
./.venv/lib/python3.14/site-packages (from
aiohttp<4.0.0,>=3.13.5->cognee) (6.7.1)
Requirement already satisfied: propcache>=0.2.0 in
./.venv/lib/python3.14/site-packages (from
aiohttp<4.0.0,>=3.13.5->cognee) (0.5.2)
Requirement already satisfied: yarl<2.0,>=1.17.0 in
./.venv/lib/python3.14/site-packages (from
aiohttp<4.0.0,>=3.13.5->cognee) (1.24.5)
Requirement already satisfied: Mako in
./.venv/lib/python3.14/site-packages (from alembic<2,>=1.13.3-
>cognee) (1.3.12)
Requirement already satisfied: typing-inspection>=0.4.2 in
./.venv/lib/python3.14/site-packages (from
fastapi<1.0.0,>=0.116.2->cognee) (0.4.2)
Requirement already satisfied: annotated-doc>=0.0.2 in
./.venv/lib/python3.14/site-packages (from
fastapi<1.0.0,>=0.116.2->cognee) (0.0.4)
Requirement already satisfied: packaging in
./.venv/lib/python3.14/site-packages (from gunicorn<24,>=20.1.0-
>cognee) (24.2)
Requirement already satisfied: docstring-parser<1.0,>=0.16 in
./.venv/lib/python3.14/site-packages (from
instructor<1.15.3,>=1.9.1->cognee) (0.18.0)
Requirement already satisfied: jiter<0.14,>=0.6.1 in
./.venv/lib/python3.14/site-packages (from
instructor<1.15.3,>=1.9.1->cognee) (0.13.0)
Requirement already satisfied: pydantic-core<3.0.0,>=2.18.0 in
./.venv/lib/python3.14/site-packages (from
instructor<1.15.3,>=1.9.1->cognee) (2.46.4)
Requirement already satisfied: requests<3.0.0,>=2.32.3 in
./.venv/lib/python3.14/site-packages (from
instructor<1.15.3,>=1.9.1->cognee) (2.34.2)
Requirement already satisfied: rich<15.0.0,>=13.7.0 in
./.venv/lib/python3.14/site-packages (from
instructor<1.15.3,>=1.9.1->cognee) (14.3.4)
Requirement already satisfied: typer<1.0.0,>=0.9.0 in
./.venv/lib/python3.14/site-packages (from
instructor<1.15.3,>=1.9.1->cognee) (0.27.0)
Requirement already satisfied: MarkupSafe>=2.0 in
./.venv/lib/python3.14/site-packages (from jinja2<4,>=3.1.3-
>cognee) (3.0.3)
Requirement already satisfied: deprecation>=2.1.0 in
./.venv/lib/python3.14/site-packages (from
lancedb<1.0.0,>=0.24.3->cognee) (2.1.0)
Requirement already satisfied: pyarrow>=16 in
./.venv/lib/python3.14/site-packages (from
lancedb<1.0.0,>=0.24.3->cognee) (25.0.0)
Requirement already satisfied: tqdm>=4.27.0 in
./.venv/lib/python3.14/site-packages (from
lancedb<1.0.0,>=0.24.3->cognee) (4.69.0)
Requirement already satisfied: lance-namespace>=0.3.2 in
./.venv/lib/python3.14/site-packages (from
lancedb<1.0.0,>=0.24.3->cognee) (0.9.0)
Requirement already satisfied: deprecated>=1.2 in
./.venv/lib/python3.14/site-packages (from limits<5,>=4.4.1-
>cognee) (1.3.1)
Requirement already satisfied: fastjsonschema>=2.15 in
./.venv/lib/python3.14/site-packages (from
nbformat<6.0.0,>=5.7.0->cognee) (2.21.2)
Requirement already satisfied: jsonschema>=2.6 in
./.venv/lib/python3.14/site-packages (from
nbformat<6.0.0,>=5.7.0->cognee) (4.26.0)
Requirement already satisfied: jupyter-core!=5.0.*,>=4.12 in
./.venv/lib/python3.14/site-packages (from
nbformat<6.0.0,>=5.7.0->cognee) (5.9.1)
Requirement already satisfied: traitlets>=5.1 in
./.venv/lib/python3.14/site-packages (from
nbformat<6.0.0,>=5.7.0->cognee) (5.15.1)
Requirement already satisfied: anyio<5,>=3.5.0 in
./.venv/lib/python3.14/site-packages (from openai>=1.80.1-
>cognee) (4.14.2)
Requirement already satisfied: distro<2,>=1.7.0 in
./.venv/lib/python3.14/site-packages (from openai>=1.80.1-
>cognee) (1.9.0)
Requirement already satisfied: httpx<1,>=0.23.0 in
./.venv/lib/python3.14/site-packages (from openai>=1.80.1-
>cognee) (0.28.1)
Requirement already satisfied: sniffio in
./.venv/lib/python3.14/site-packages (from openai>=1.80.1-
>cognee) (1.3.1)
Requirement already satisfied: idna>=2.8 in
./.venv/lib/python3.14/site-packages (from anyio<5,>=3.5.0-
>openai>=1.80.1->cognee) (3.18)
Requirement already satisfied: certifi in
./.venv/lib/python3.14/site-packages (from httpx<1,>=0.23.0-
>openai>=1.80.1->cognee) (2026.6.17)
Requirement already satisfied: httpcore==1.* in
./.venv/lib/python3.14/site-packages (from httpx<1,>=0.23.0-
>openai>=1.80.1->cognee) (1.0.9)
Requirement already satisfied: h11>=0.16 in
./.venv/lib/python3.14/site-packages (from httpcore==1.*-
>httpx<1,>=0.23.0->openai>=1.80.1->cognee) (0.16.0)
Requirement already satisfied: annotated-types>=0.6.0 in
./.venv/lib/python3.14/site-packages (from pydantic>=2.10.5-
>cognee) (0.7.0)
Requirement already satisfied: pyparsing<4,>=2.1.0 in
./.venv/lib/python3.14/site-packages (from rdflib<7.2.0,>=7.1.4-
>cognee) (3.3.2)
Requirement already satisfied: charset_normalizer<4,>=2 in
./.venv/lib/python3.14/site-packages (from
requests<3.0.0,>=2.32.3->instructor<1.15.3,>=1.9.1->cognee)
(3.4.9)
Requirement already satisfied: markdown-it-py>=2.2.0 in
./.venv/lib/python3.14/site-packages (from rich<15.0.0,>=13.7.0-
>instructor<1.15.3,>=1.9.1->cognee) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
./.venv/lib/python3.14/site-packages (from rich<15.0.0,>=13.7.0-
>instructor<1.15.3,>=1.9.1->cognee) (2.20.0)
Requirement already satisfied: regex in
./.venv/lib/python3.14/site-packages (from
tiktoken<1.0.0,>=0.8.0->cognee) (2026.7.19)
Requirement already satisfied: shellingham>=1.3.0 in
./.venv/lib/python3.14/site-packages (from typer<1.0.0,>=0.9.0-
>instructor<1.15.3,>=1.9.1->cognee) (1.5.4)
Requirement already satisfied: click>=7.0 in
./.venv/lib/python3.14/site-packages (from
uvicorn<1.0.0,>=0.34.0->cognee) (8.4.2)
Requirement already satisfied: argcomplete<4,>=2.10.1 in
./.venv/lib/python3.14/site-packages (from datamodel-code-
generator>=0.54.0->cognee) (3.7.0)
Requirement already satisfied: black>=19.10b0 in
./.venv/lib/python3.14/site-packages (from datamodel-code-
generator>=0.54.0->cognee) (26.5.1)
Requirement already satisfied: genson<2,>=1.2.1 in
./.venv/lib/python3.14/site-packages (from datamodel-code-
generator>=0.54.0->cognee) (1.4.0)
Requirement already satisfied: inflect<8,>=4.1 in
./.venv/lib/python3.14/site-packages (from datamodel-code-
generator>=0.54.0->cognee) (7.5.0)
Requirement already satisfied: isort<9,>=4.3.21 in
./.venv/lib/python3.14/site-packages (from datamodel-code-
generator>=0.54.0->cognee) (8.0.1)
Requirement already satisfied: pyyaml>=6.0.1 in
./.venv/lib/python3.14/site-packages (from datamodel-code-
generator>=0.54.0->cognee) (6.0.3)
Requirement already satisfied: more_itertools>=8.5.0 in
./.venv/lib/python3.14/site-packages (from inflect<8,>=4.1-
>datamodel-code-generator>=0.54.0->cognee) (11.1.0)
Requirement already satisfied: typeguard>=4.0.1 in
./.venv/lib/python3.14/site-packages (from inflect<8,>=4.1-
>datamodel-code-generator>=0.54.0->cognee) (4.5.2)
Requirement already satisfied: mypy-extensions>=0.4.3 in
./.venv/lib/python3.14/site-packages (from black>=19.10b0-
>datamodel-code-generator>=0.54.0->cognee) (1.1.0)
Requirement already satisfied: pathspec>=1.0.0 in
./.venv/lib/python3.14/site-packages (from black>=19.10b0-
>datamodel-code-generator>=0.54.0->cognee) (1.1.1)
Requirement already satisfied: platformdirs>=2 in
./.venv/lib/python3.14/site-packages (from black>=19.10b0-
>datamodel-code-generator>=0.54.0->cognee) (4.10.1)
Requirement already satisfied: pytokens~=0.4.0 in
./.venv/lib/python3.14/site-packages (from black>=19.10b0-
>datamodel-code-generator>=0.54.0->cognee) (0.4.1)
Requirement already satisfied: wrapt<3,>=1.10 in
./.venv/lib/python3.14/site-packages (from deprecated>=1.2-
>limits<5,>=4.4.1->cognee) (2.2.2)
Requirement already satisfied: redis>=4.3 in
./.venv/lib/python3.14/site-packages (from fakeredis>=2.32.0-
>fakeredis[lua]>=2.32.0->cognee) (8.0.1)
Requirement already satisfied: sortedcontainers>=2 in
./.venv/lib/python3.14/site-packages (from fakeredis>=2.32.0-
>fakeredis[lua]>=2.32.0->cognee) (2.4.0)
Requirement already satisfied: lupa>=2.1 in
./.venv/lib/python3.14/site-packages (from
fakeredis[lua]>=2.32.0->cognee) (2.8)
Requirement already satisfied: email-validator<2.4,>=1.1.0 in
./.venv/lib/python3.14/site-packages (from fastapi-users>=15.0.2-
>fastapi-users[sqlalchemy]>=15.0.2->cognee) (2.3.0)
Requirement already satisfied: makefun<2.0.0,>=1.11.2 in
./.venv/lib/python3.14/site-packages (from fastapi-users>=15.0.2-
>fastapi-users[sqlalchemy]>=15.0.2->cognee) (1.16.0)
Requirement already satisfied: pwdlib==0.3.0 in
./.venv/lib/python3.14/site-packages (from
pwdlib[argon2,bcrypt]==0.3.0->fastapi-users>=15.0.2->fastapi-
users[sqlalchemy]>=15.0.2->cognee) (0.3.0)
Requirement already satisfied: pyjwt<3.0.0,>=2.12.0 in
./.venv/lib/python3.14/site-packages (from pyjwt[crypto]
<3.0.0,>=2.12.0->fastapi-users>=15.0.2->fastapi-
users[sqlalchemy]>=15.0.2->cognee) (2.13.0)
Requirement already satisfied: argon2-cffi<26,>=23.1.0 in
./.venv/lib/python3.14/site-packages (from
pwdlib[argon2,bcrypt]==0.3.0->fastapi-users>=15.0.2->fastapi-
users[sqlalchemy]>=15.0.2->cognee) (25.1.0)
Requirement already satisfied: bcrypt<6,>=4.1.2 in
./.venv/lib/python3.14/site-packages (from
pwdlib[argon2,bcrypt]==0.3.0->fastapi-users>=15.0.2->fastapi-
users[sqlalchemy]>=15.0.2->cognee) (5.0.0)
Requirement already satisfied: argon2-cffi-bindings in
./.venv/lib/python3.14/site-packages (from argon2-
cffi<26,>=23.1.0->pwdlib[argon2,bcrypt]==0.3.0->fastapi-
users>=15.0.2->fastapi-users[sqlalchemy]>=15.0.2->cognee)
(25.1.0)
Requirement already satisfied: dnspython>=2.0.0 in
./.venv/lib/python3.14/site-packages (from email-
validator<2.4,>=1.1.0->fastapi-users>=15.0.2->fastapi-
users[sqlalchemy]>=15.0.2->cognee) (2.8.0)
Requirement already satisfied: cryptography>=3.4.0 in
./.venv/lib/python3.14/site-packages (from pyjwt[crypto]
<3.0.0,>=2.12.0->fastapi-users>=15.0.2->fastapi-
users[sqlalchemy]>=15.0.2->cognee) (49.0.0)
Requirement already satisfied: cffi>=2.0.0 in
./.venv/lib/python3.14/site-packages (from cryptography>=3.4.0-
>pyjwt[crypto]<3.0.0,>=2.12.0->fastapi-users>=15.0.2->fastapi-
users[sqlalchemy]>=15.0.2->cognee) (2.1.0)
Requirement already satisfied: pycparser in
./.venv/lib/python3.14/site-packages (from cffi>=2.0.0-
>cryptography>=3.4.0->pyjwt[crypto]<3.0.0,>=2.12.0->fastapi-
users>=15.0.2->fastapi-users[sqlalchemy]>=15.0.2->cognee) (3.0)
Requirement already satisfied: fastapi-users-db-sqlalchemy>=7.0.0
in ./.venv/lib/python3.14/site-packages (from fastapi-
users[sqlalchemy]>=15.0.2->cognee) (7.0.0)
Requirement already satisfied: greenlet>=1 in
./.venv/lib/python3.14/site-packages (from sqlalchemy[asyncio]
<2.1.0,>=2.0.0->fastapi-users-db-sqlalchemy>=7.0.0->fastapi-
users[sqlalchemy]>=15.0.2->cognee) (3.5.3)
Requirement already satisfied: jsonschema-
specifications>=2023.03.6 in ./.venv/lib/python3.14/site-packages
(from jsonschema>=2.6->nbformat<6.0.0,>=5.7.0->cognee) (2025.9.1)
Requirement already satisfied: referencing>=0.28.4 in
./.venv/lib/python3.14/site-packages (from jsonschema>=2.6-
>nbformat<6.0.0,>=5.7.0->cognee) (0.37.0)
Requirement already satisfied: rpds-py>=0.25.0 in
./.venv/lib/python3.14/site-packages (from jsonschema>=2.6-
>nbformat<6.0.0,>=5.7.0->cognee) (2026.6.3)
Requirement already satisfied: lance-namespace-urllib3-
client==0.9.0 in ./.venv/lib/python3.14/site-packages (from
lance-namespace>=0.3.2->lancedb<1.0.0,>=0.24.3->cognee) (0.9.0)
Requirement already satisfied: python-dateutil>=2.8.2 in
./.venv/lib/python3.14/site-packages (from lance-namespace-
urllib3-client==0.9.0->lance-namespace>=0.3.2-
>lancedb<1.0.0,>=0.24.3->cognee) (2.9.0.post0)
Requirement already satisfied: six in
./.venv/lib/python3.14/site-packages (from langdetect>=1.0.9-
>cognee) (1.17.0)
Requirement already satisfied: fastuuid<1.0,>=0.14.0 in
./.venv/lib/python3.14/site-packages (from litellm>=1.83.7-
>cognee) (0.14.0)
Requirement already satisfied: importlib-metadata<9.0,>=8.0.0 in
./.venv/lib/python3.14/site-packages (from litellm>=1.83.7-
>cognee) (8.9.0)
Requirement already satisfied: tokenizers<1.0,>=0.21.0 in
./.venv/lib/python3.14/site-packages (from litellm>=1.83.7-
>cognee) (0.22.2)
Requirement already satisfied: zipp>=3.20 in
./.venv/lib/python3.14/site-packages (from importlib-
metadata<9.0,>=8.0.0->litellm>=1.83.7->cognee) (4.1.0)
Requirement already satisfied: huggingface-hub<2.0,>=0.16.4 in
./.venv/lib/python3.14/site-packages (from
tokenizers<1.0,>=0.21.0->litellm>=1.83.7->cognee) (1.24.0)
Requirement already satisfied: fsspec>=2023.5.0 in
./.venv/lib/python3.14/site-packages (from huggingface-
hub<2.0,>=0.16.4->tokenizers<1.0,>=0.21.0->litellm>=1.83.7-
>cognee) (2026.6.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.5.1 in
./.venv/lib/python3.14/site-packages (from huggingface-
hub<2.0,>=0.16.4->tokenizers<1.0,>=0.21.0->litellm>=1.83.7-
>cognee) (1.5.2)
Requirement already satisfied: mdurl~=0.1 in
./.venv/lib/python3.14/site-packages (from markdown-it-py>=2.2.0-
>rich<15.0.0,>=13.7.0->instructor<1.15.3,>=1.9.1->cognee) (0.1.2)
Note: you may need to restart the kernel to use updated packages.
```

```
[2m2026-07-24T15:55:05.764431[0m [[32m[1minfo     [0m]
[1mDeleted old log file:
/Users/amandademoura/.cognee/logs/2026-07-22_09-59-41.log[0m
[[0m[1m[34mcognee.shared.logging_utils[0m][0m

[2m2026-07-24T15:55:05.764789[0m [[32m[1minfo     [0m]
[1mLog file created at:
/Users/amandademoura/.cognee/logs/2026-07-24_17-55-05.log[0m
[[0m[1m[34mcognee.shared.logging_utils[0m][0m
[36mlog_file[0m=[35m/Users/amandademoura/.cognee/logs/2026-
07-24_17-55-05.log[0m

[2m2026-07-24T15:55:05.765084[0m [[33m[1mwarning  [0m]
[1mCognee 1.0 changes: New API —
remember/recall/forget/improve (V1 add/cognify/search still
work). Session memory enabled by default (CACHING=false to
disable). Multi-user access control on by default
(ENABLE_BACKEND_ACCESS_CONTROL=false to disable). Agents
(@cognee.agent) auto-verified on registration. See
https://docs.cognee.ai/[0m
[[0m[1m[34mcognee.shared.logging_utils[0m][0m

[2m2026-07-24T15:55:05.765341[0m [[32m[1minfo     [0m]
[1mLogging initialized           [0m
[[0m[1m[34mcognee.shared.logging_utils[0m][0m
[36mcognee_version[0m=[35m1.4.0[0m
[36mdatabase_path[0m=[35m'/Users/amandademoura/VSCode/AI-
Memory Comparison/.venv/lib/python3.14/site-
packages/cognee/.cognee_system/databases'[0m
[36mos_info[0m=[35m'Darwin 25.5.0 (Darwin Kernel Version
25.5.0: Tue Jun  9 22:28:17 PDT 2026; root:xnu-
12377.121.10~1/RELEASE_ARM64_T8142)'[0m
[36mpython_version[0m=[35m3.14.6[0m
[36mstructlog_version[0m=[35m25.5.0[0m

[2m2026-07-24T15:55:05.765642[0m [[32m[1minfo     [0m]
[1mDatabase storage: /Users/amandademoura/VSCode/AI-Memory
Comparison/.venv/lib/python3.14/site-
packages/cognee/.cognee_system/databases[0m
[[0m[1m[34mcognee.shared.logging_utils[0m][0m

[2m2026-07-24T15:55:05.998481[0m [[32m[1minfo     [0m]
[1mauth posture: authentication=disabled,
multi_tenant=disabled (explicit REQUIRE_AUTHENTICATION)[0m
[[0m[1m[34mget_authenticated_user[0m][0m
/Users/amandademoura/VSCode/AI-Memory
Comparison/.venv/lib/python3.14/site-
packages/cognee/modules/observability/get_observe.py:67:
DeprecationWarning: 'asyncio.iscoroutinefunction' is
deprecated and slated for removal in Python 3.16; use
inspect.iscoroutinefunction() instead
  if asyncio.iscoroutinefunction(func):
/Users/amandademoura/VSCode/AI-Memory
Comparison/.venv/lib/python3.14/site-
packages/cognee/infrastructure/databases/graph/config.py:45:
PydanticDeprecatedSince20: Using extra keyword arguments on
`Field` is deprecated and will be removed. Use
`json_schema_extra` instead. (Extra keys: 'env'). Deprecated
in Pydantic V2.0 to be removed in V3.0. See Pydantic V2
Migration Guide at https://errors.pydantic.dev/2.13/migration/
  graph_database_provider: str = Field("ladybug",
env="GRAPH_DATABASE_PROVIDER")
/Users/amandademoura/VSCode/AI-Memory
Comparison/.venv/lib/python3.14/site-
packages/cognee/infrastructure/databases/graph/config.py:63:
PydanticDeprecatedSince20: Using extra keyword arguments on
`Field` is deprecated and will be removed. Use
`json_schema_extra` instead. (Extra keys: 'env'). Deprecated
in Pydantic V2.0 to be removed in V3.0. See Pydantic V2
Migration Guide at https://errors.pydantic.dev/2.13/migration/
  kuzu_num_threads: int = Field(0, env="KUZU_NUM_THREADS")
/Users/amandademoura/VSCode/AI-Memory
Comparison/.venv/lib/python3.14/site-
packages/cognee/infrastructure/databases/graph/config.py:64:
PydanticDeprecatedSince20: Using extra keyword arguments on
`Field` is deprecated and will be removed. Use
`json_schema_extra` instead. (Extra keys: 'env'). Deprecated
in Pydantic V2.0 to be removed in V3.0. See Pydantic V2
Migration Guide at https://errors.pydantic.dev/2.13/migration/
  kuzu_buffer_pool_size: int =
Field(DEFAULT_KUZU_BUFFER_POOL_SIZE,
env="KUZU_BUFFER_POOL_SIZE")
/Users/amandademoura/VSCode/AI-Memory
Comparison/.venv/lib/python3.14/site-
packages/cognee/infrastructure/databases/graph/config.py:65:
PydanticDeprecatedSince20: Using extra keyword arguments on
`Field` is deprecated and will be removed. Use
`json_schema_extra` instead. (Extra keys: 'env'). Deprecated
in Pydantic V2.0 to be removed in V3.0. See Pydantic V2
Migration Guide at https://errors.pydantic.dev/2.13/migration/
  kuzu_max_db_size: int = Field(DEFAULT_KUZU_MAX_DB_SIZE,
env="KUZU_MAX_DB_SIZE")
/Users/amandademoura/VSCode/AI-Memory
Comparison/.venv/lib/python3.14/site-
packages/pydantic/_internal/_generate_schema.py:325:
PydanticDeprecatedSince20: `json_encoders` is deprecated. See
https://docs.pydantic.dev/2.13/concepts/serialization/#custom-
serializers for alternatives. Deprecated in Pydantic V2.0 to
be removed in V3.0. See Pydantic V2 Migration Guide at
https://errors.pydantic.dev/2.13/migration/
  warnings.warn(
```

```
 Connected to Cognee (remote) at https://tenant-8668bdea-88d3-
46fb-909e-faab503d43e2.aws.cognee.ai
```

2. Data Ingestion

The shared dataset is loaded once, then ingested into each provider individually. After each ingestion, we run a quick check to confirm the data landed correctly before moving on to Section 3: Retrieval & Evaluation.

Shared dataset structure:

`
memory_provider_eval/
├─── README.md
├─── memory_provider_comparison.ipynb
├─── deterministic/
│ ├─── spec.json
│ ├─── documents/ (15 .md files)
│ └─── ground_truth_qa.json
└─── non_deterministic/
├─── index.json
├─── persona-01_priya.json ... persona-11_ravi.json
└─── README.md
`

2.a Loading shared dataset

In [5]:

```python
import glob
import json
import os

#replace with own path to data root
DATA_ROOT = "/Users/amandademoura/VSCode/AI-Memory Comparison/"

DET_DIR = os.path.join(DATA_ROOT, "deterministic")
DET_DOC_PATHS = sorted(glob.glob(os.path.join(DET_DIR, "documents",
"*.md")))
DET_DOCS = []
for p in DET_DOC_PATHS:
    with open(p) as f:
        DET_DOCS.append({"filename": os.path.basename(p), "text":
f.read()})

with open(os.path.join(DET_DIR, "ground_truth_qa.json")) as f:
    DET_QA = json.load(f)

ND_DIR = os.path.join(DATA_ROOT, "non_deterministic")
ND_PATHS = sorted(glob.glob(os.path.join(ND_DIR, "persona-*.json")))
ND_DOCS = []
for p in ND_PATHS:
    with open(p) as f:
        ND_DOCS.append(json.load(f))

print(f"Loaded {len(DET_DOCS)} deterministic documents, {len(DET_QA)}
deterministic QA pairs")
print(f"Loaded {len(ND_DOCS)} custom contradiction personas")
```

```
Loaded 15 deterministic documents, 29 deterministic QA pairs
Loaded 11 custom contradiction personas
```

2.b Ingestion for each provider

In this order:
1. Zep
2. mem0
3. Supermemory
5. Cognee

In [6]:

```python
from zep_cloud.client import Zep
from zep_cloud.types import Message
from zep_cloud.core.api_error import ApiError

zep_client = Zep(api_key=os.environ["ZEP_API_KEY"])

def _ensure_user(user_id, first_name, last_name):
    try:
        zep_client.user.add(user_id=user_id, first_name=first_name,
last_name=last_name)
    except ApiError as e:
        if "already exists" not in str(e.body):
            raise
def _role_for_speaker(speaker, persona_name):
    return "user" if speaker == persona_name else "assistant"

def _ensure_thread(thread_id, user_id):
    try:
        zep_client.thread.create(thread_id=thread_id, user_id=user_id)
    except ApiError as e:
        if "already exists" not in str(e.body):
            raise

def zep_ingest_documents():
    _ensure_user("solvane_docs", "Solvane", "Docs")
    for doc in DET_DOCS:
        zep_client.graph.add(user_id="solvane_docs", type="text",
data=doc["text"])

def zep_ingest_conversations():
    for persona in ND_DOCS:
        user_id = persona["persona_id"]
        persona_name = persona["persona_name"]
        _ensure_user(user_id, persona_name.split()[0], "Persona")
        for session in persona["sessions"]:
            thread_id = f"
{persona['persona_id']}_{session['session_id']}"
            _ensure_thread(thread_id, user_id)
            messages = [
                Message(
                    name=t["speaker"],
                    content=t["text"],
                    role=_role_for_speaker(t["speaker"],
persona_name),
                )
                for t in session["turns"]
                if t["text"] != "PLACEHOLDER_RESPONSE_TO_EVALUATE"
            ]
            if messages:
                zep_client.thread.add_messages(thread_id=thread_id,
messages=messages)

zep_ingest_documents()
zep_ingest_conversations()

print(f"Zep ingestion complete: {len(DET_DOCS)} documents and
{len(ND_DOCS)} personas loaded")
```

```
Zep ingestion complete: 15 documents and 11 personas loaded
```

In [7]:

```python
from mem0 import MemoryClient
import os
from dotenv import load_dotenv

load_dotenv()
mem0_client = MemoryClient(api_key=os.getenv("MEM0_API_KEY"))

def mem0_ingest_documents():
    for doc in DET_DOCS:
        mem0_client.add(
            user_id="solvane_docs",
            run_id="solvane_docs_thread",
            messages=[{"role": "system", "content": doc["text"]}],
        )

def mem0_ingest_conversations():
    for persona in ND_DOCS:
        user_id = persona["persona_id"]
        for session in persona["sessions"]:
            run_id = f"
{persona['persona_id']}_{session['session_id']}"
            messages = [
                {"role": t["speaker"], "content": t["text"]}
                for t in session["turns"] if t["text"] !=
"PLACEHOLDER_RESPONSE_TO_EVALUATE"
            ]
            if messages:
                mem0_client.add(user_id=user_id, run_id=run_id,
messages=messages)

mem0_ingest_documents()
mem0_ingest_conversations()

print(f"mem0 ingestion complete: {len(DET_DOCS)} documents and
{len(ND_DOCS)} personas loaded")
```

```
mem0 ingestion complete: 15 documents and 11 personas loaded
```

In [8]:

```python
from supermemory import Supermemory
import os
from dotenv import load_dotenv

load_dotenv()
sm_client = Supermemory(api_key=os.getenv("SUPERMEMORY_API_KEY"))

def supermemory_ingest_documents():
    for doc in DET_DOCS:
        sm_client.add(content=doc["text"],
container_tag="solvane_docs")

def supermemory_ingest_conversations():
    for persona in ND_DOCS:
        tag = persona["persona_id"]
        for session in persona["sessions"]:
            for t in session["turns"]:
                if t["text"] == "PLACEHOLDER_RESPONSE_TO_EVALUATE":
                    continue
                sm_client.add(content=f"{t['speaker']}: {t['text']}",
container_tag=tag)

supermemory_ingest_documents()
supermemory_ingest_conversations()

print(f"Supermemory ingestion complete: {len(DET_DOCS)} documents and
{len(ND_DOCS)} personas loaded")
```

```
Supermemory ingestion complete: 15 documents and 11 personas
loaded
```

In [9]:

```python
import os
import cognee

async def cognee_ingest_documents():
    for doc in DET_DOCS:
        await cognee.remember(
            f"[{doc['filename']}]\n{doc['text']}",
            dataset_name="solvane_docs",
        )

async def cognee_ingest_conversations():
    for persona in ND_DOCS:
        for session in persona["sessions"]:
            for turn in session["turns"]:
                if turn["text"] == "PLACEHOLDER_RESPONSE_TO_EVALUATE":
                    continue
                await cognee.remember(
                    f"[{persona['persona_id']} /
{session['session_id']} / {session['date']}] "
                    f"{turn['speaker']}: {turn['text']}",
                    dataset_name=persona["persona_id"],
                )

await cognee.serve(
    url=os.getenv("COGNEE_URL"),
    api_key=os.getenv("COGNEE_API_KEY"),
)

await cognee_ingest_documents()
await cognee_ingest_conversations()

print(f"Cognee ingestion complete: {len(DET_DOCS)} documents and
{len(ND_DOCS)} personas loaded")
```

```
[1mUnclosed client session
client_session: <aiohttp.client.ClientSession object at
0x122505d30>[0m
```

```
 Connected to Cognee (remote) at https://tenant-8668bdea-88d3-
46fb-909e-faab503d43e2.aws.cognee.ai
Cognee ingestion complete: 15 documents and 11 personas loaded
```

Confirm data ingestion actually landed and is queryable:

Several of these providers build their memory representation (graph edges, embeddings, indices) asynchronously after an add/remember call returns, so the call finishing without an error doesn't guarantee the data is queryable yet. Querying too soon after ingestion is a plausible source of run-to-run variance in the results below (a provider that hasn't finished indexing looks worse than one queried a minute later, even on identical data, independent of any LLM temperature setting). The cell below polls each provider a few times with a short wait between attempts, rather than checking once and moving on.

In [10]:

```python
import time
import asyncio

_CHECK_PERSONA = ND_DOCS[0]["persona_id"]  # representative sample,
not exhaustive over all 11

def _poll_until_nonzero(label, check_fn, attempts=4, delay=5):
    for attempt in range(attempts):
        try:
            count = check_fn()
        except Exception as exc:
            count = None
            print(f"  {label}: ERROR on attempt {attempt +
1}/{attempts} -- {exc}")
        else:
            print(f"  {label}: {count} result(s) on attempt {attempt +
1}/{attempts}")
            if count and count > 0:
                return True
        if attempt < attempts - 1:
            time.sleep(delay)
    print(f"  {label}: WARNING -- still 0/erroring after {attempts}
attempts x {delay}s. "
          f"Ingestion may not have finished indexing, or didn't
actually land.")
    return False

async def _poll_until_nonzero_async(label, check_fn, attempts=4,
delay=5):
    for attempt in range(attempts):
        try:
            count = await check_fn()
        except Exception as exc:
            count = None
            print(f"  {label}: ERROR on attempt {attempt +
1}/{attempts} -- {exc}")
        else:
            print(f"  {label}: {count} result(s) on attempt {attempt +
1}/{attempts}")
            if count and count > 0:
                return True
        if attempt < attempts - 1:
            await asyncio.sleep(delay)
    print(f"  {label}: WARNING -- still 0/erroring after {attempts}
attempts x {delay}s. "
          f"Ingestion may not have finished indexing, or didn't
actually land.")
    return False

print("--- Zep ---")
for uid in ["solvane_docs", _CHECK_PERSONA]:
    _poll_until_nonzero(uid, lambda uid=uid: len(
        zep_client.graph.search(user_id=uid, query="summary",
scope="edges", limit=3).edges
    ))

print("--- mem0 ---")
for uid in ["solvane_docs", _CHECK_PERSONA]:
    _poll_until_nonzero(uid, lambda uid=uid: len(
        mem0_client.search("What do you know so far?", filters=
{"user_id": uid})["results"]
    ))

print("--- Supermemory ---")
for tag in ["solvane_docs", _CHECK_PERSONA]:
    _poll_until_nonzero(tag, lambda tag=tag: len(
        sm_client.search.memories(q=tag, container_tag=tag,
search_mode="hybrid", threshold=0, limit=5).results
    ))

print("--- Cognee ---")
async def _cognee_count(dataset_name):
    results = await cognee.recall(query_text="summary", datasets=
[dataset_name], only_context=True)
    if isinstance(results, str):
        return 1 if results.strip() else 0
    return len(results) if results else 0

for dataset_name in ["solvane_docs", _CHECK_PERSONA]:
    await _poll_until_nonzero_async(dataset_name, lambda
dataset_name=dataset_name: _cognee_count(dataset_name))
```

```
--- Zep ---
  solvane_docs: 3 result(s) on attempt 1/4
  persona-01: 3 result(s) on attempt 1/4
--- mem0 ---
  solvane_docs: 10 result(s) on attempt 1/4
  persona-01: 7 result(s) on attempt 1/4
--- Supermemory ---
  solvane_docs: 5 result(s) on attempt 1/4
  persona-01: 5 result(s) on attempt 1/4
--- Cognee ---
  solvane_docs: 1 result(s) on attempt 1/4
  persona-01: 1 result(s) on attempt 1/4
```

3. Retrieval & Eval

With data ingested across all four providers, we query each system with three independent evaluation sets: deterministic ground-truth QA, ambiguous/contradictory persona probes, and needle-in-haystack recall. The per-provider query functions defined below (`zep_ask` , `mem0_ask` , `supermemory_ask` , `cognee_ask`) are shared across all three: they route every answer through one shared `_synthesize` call, switched between a "resolve" and "hedge" system prompt depending on which section is calling it (see note below).

#### a. Ground-Truth QA

Questions from `ground_truth_qa.json` , each with one defined correct answer. Automatically split by type into: plain factual recall, temporal reasoning (facts that change over time), conflict resolution (superseded facts as of a given date), and multi-hop reasoning.

Retrieval + synthesis, shared across sections (a)-(c)

Each `*_ask` function wraps that provider's native retrieval call, then routes the result through `_synthesize` . This synthesis step is what's actually graded, not raw memory output, so all four providers are judged on the same basis.

`_synthesize` runs in one of two modes, toggled by the global `_SYNTHESIS_MODE` :
- **resolve** (default, used in sections a and c): instructs the model to commit to one answer, picking the fact valid at the target date if several conflict. Every question here has one correct answer, so hedging would itself be a wrong answer.
- **hedge** (set only inside `evaluate_ambiguity` / `evaluate_ambiguity_async` , section b): instructs the model to surface unresolved contradictions instead of silently picking a side.

In [11]:

```python
import os
from openai import OpenAI
import re
import time


_llm = OpenAI(api_key=os.environ["OPENAI_API_KEY"])
_DATE_CLAUSE_RE = re.compile(r"\s*as of \d{4}-\d{2}-\d{2}\??",
re.IGNORECASE)

# _SYNTHESIS_MODE controls which system prompt _synthesize uses
_SYNTHESIS_MODE = "resolve"

_RESOLVE_SYSTEM_PROMPT = (
    "Answer using ONLY the facts listed below. Facts may include a
validity range "
    "like '(valid: DATE to DATE)' where 'present/current' means still
true. Ignore "
    "today's real-world date. If facts conflict, pick the one valid at
the target date. "
    "Be concise: state just the answer.\n\n"
    "Special case: if the question asks what a specific past document,
snapshot, or "
    "report stated (e.g. 'according to the April 2024 org
snapshot...') AND whether that "
    "is 'still accurate' or 'still true', answer BOTH parts: (1) what
the fact was at "
    "that past point in time, and (2) whether it has since been
superseded -- citing the "
    "newer fact if so. Do not just give the current state; the
question is explicitly "
    "asking you to compare the historical claim against the current
one."
)

_HEDGE_SYSTEM_PROMPT = (
    "Answer using ONLY the facts listed below. Some of these facts may
be genuinely "
    "ambiguous or contradictory with no resolution given -- do not
invent one. If the facts "
    "conflict and nothing in them tells you which one is current or
correct, say so "
    "explicitly and present both sides rather than confidently picking
one. If the facts "
    "are not actually in conflict, or an explicit resolution IS given
(e.g. a later fact "
    "that plainly supersedes an earlier one), answer directly and do
not hedge unnecessarily."
)

# --- Retry-with-backoff for every external network call -------------
------
# This notebook makes a very large number of sequential live API calls
(OpenAI +
# whichever provider is under test). A single transient
timeout/connection error used to
# crash the entire run_and_log cell partway through, losing every
provider that hadn't run
# yet. _with_retries / _with_retries_async retry on
timeout/connection-shaped errors with
# exponential backoff (2s, 4s, 8s) before giving up. Tune
max_attempts/base_delay if your
# network or a specific provider is consistently slower than this.
def _is_retryable(exc):
    msg = str(exc).lower()
    return any(s in msg for s in ("timeout", "timed out",
"connection", "temporarily unavailable"))

def _with_retries(fn, *args, max_attempts=4, base_delay=2, **kwargs):
    last_exc = None
    for attempt in range(max_attempts):
        try:
            return fn(*args, **kwargs)
        except Exception as exc:
            last_exc = exc
            if not _is_retryable(exc) or attempt == max_attempts - 1:
                raise
            wait = base_delay * (2 ** attempt)
            print(f"  [retry] {type(exc).__name__}: {exc} -- attempt
{attempt + 1}/{max_attempts}, retrying in {wait}s")
            time.sleep(wait)
    raise last_exc

async def _with_retries_async(fn, *args, max_attempts=4, base_delay=2,
**kwargs):
    import asyncio
    last_exc = None
    for attempt in range(max_attempts):
        try:
            return await fn(*args, **kwargs)
        except Exception as exc:
            last_exc = exc
            if not _is_retryable(exc) or attempt == max_attempts - 1:
                raise
            wait = base_delay * (2 ** attempt)
            print(f"  [retry] {type(exc).__name__}: {exc} -- attempt
{attempt + 1}/{max_attempts}, retrying in {wait}s")
            await asyncio.sleep(wait)
    raise last_exc

def _strip_date_clause(question):
    return _DATE_CLAUSE_RE.sub("", question).strip()

def _synthesize(question, snippets, query_date=None):
    if isinstance(snippets, str):
        context = snippets or "(no facts retrieved)"
    else:
        context = "\n".join(f"- {s}" for s in snippets) if snippets
else "(no facts retrieved)"

    date_note = (f"\n\nAnswer as of: {query_date}." if query_date
                 else "\n\nNo date given -- answer with the
current/most recent state.")

    system_prompt = _HEDGE_SYSTEM_PROMPT if _SYNTHESIS_MODE == "hedge"
else _RESOLVE_SYSTEM_PROMPT

    resp = _with_retries(
        _llm.chat.completions.create,
        model="gpt-4o-mini",
        temperature=0,
        messages=[
            {"role": "system", "content": system_prompt},
            {"role": "user", "content":
f"Facts:\n{context}\n\nQuestion: {question}{date_note}"},
        ],
    )
    return resp.choices[0].message.content.strip()


def mem0_ask(question, query_date=None, user_id="solvane_docs"):
    search_query = _strip_date_clause(question)
    results = _with_retries(mem0_client.search, search_query, filters=
{"user_id": user_id})
    snippets = [r["memory"] for r in results["results"]]
    return _synthesize(question, snippets, query_date)

def zep_ask(question, query_date=None, user_id="solvane_docs"):
    search_query = _strip_date_clause(question)
    thread_id = f"eval-{uuid.uuid4().hex}"
    _with_retries(zep_client.thread.create, thread_id=thread_id,
user_id=user_id)
    _with_retries(
        zep_client.thread.add_messages,
        thread_id,
        messages=[Message(name="Evaluator", role="user",
content=search_query)],
    )

    ctx = _with_retries(zep_client.thread.get_user_context,
thread_id=thread_id)
    return _synthesize(question, ctx.context, query_date)


def supermemory_ask(question, query_date=None, tag="solvane_docs"):
    search_query = _strip_date_clause(question)
    r = _with_retries(
        sm_client.search.memories,
        q=search_query, container_tag=tag, search_mode="hybrid",
threshold=0, limit=10,
    )
    snippets = [res.memory or res.chunk for res in r.results]
    return _synthesize(question, snippets, query_date)
```

#### Grading utilities

`ask_safe` : catches provider errors so one failure doesn't kill the run

`llm_judge` : grades free-text answers when they don't match the reference verbatim

`_contains_whole` : word-boundary substring check used before falling back to `llm_judge` . A plain `in` check would let a wrong answer like "75 kg" pass when the reference is "5 kg", since the shorter string sits inside the longer one

`CATEGORY_BY_TYPE` : buckets each ground-truth question into Temporal / Conflict Resolution / Multi-hop

These four are reused in section (c); `evaluate_provider` below is the only part specific to ground-truth QA.

Limitation: the same model family (gpt-4o-mini) both answers questions (via `_synthesize`) and grades them (via `llm_judge` / `annotation_judge`). That's a self-grading setup, see section (f) for a manual spot-check sample to validate the judge isn't rubber-stamping itself.

In [12]:

```
from collections import defaultdict
import re

CATEGORY_BY_TYPE = {
    "as_of": "Temporal Reasoning",
    "current_undated": "Conflict Resolution",
    "multi_hop": "Multi-hop Reasoning",
    "distractor_check": "Multi-hop Reasoning",
}

def normalize_answer(text):
    return " ".join(str(text or "").strip().lower().split())

def _contains_whole(needle, haystack):
    """Word-boundary substring check. Plain `in` would match '5 kg'
inside '75 kg' or '15 kg';
    this requires the needle to sit on a word boundary on both
sides."""
    if not needle:
        return False
    pattern = r"(?<!\w)" + re.escape(needle) + r"(?!\w)"
    return re.search(pattern, haystack) is not None

def ask_safe(question, query_date, ask_fn, *extra_args):
    try:
        return str(ask_fn(question, query_date, *extra_args))
    except Exception as exc:
        return f"<ERROR: {exc}>"

async def ask_safe_async(question, query_date, ask_fn, *extra_args):
    try:
        return str(await ask_fn(question, query_date, *extra_args))
    except Exception as exc:
        return f"<ERROR: {exc}>"

def llm_judge(question, expected, got):
    try:
        resp = _with_retries(
            _llm.chat.completions.create,
            model="gpt-4o-mini",
            temperature=0,
            messages=[
                {"role": "system", "content": (
                    "Grade a candidate answer against a reference
answer. Mark CORRECT if the "
                    "candidate's core factual claim matches the
reference - even if phrased very "
                    "differently, more verbosely, with extra true
context, or using equivalent "
                    "wording/units (e.g. '$150 per seat per month' =
'$150/seat/month'; 'Tuesday to "
                    "Thursday' = 'Tue/Wed/Thu').\n"
                    "Do not penalize: paraphrase, extra correct
detail, or additional caveats/exceptions "
                    "that coexist with the core claim without
contradicting it (e.g. if the reference says "
                    "'3 days/week in office (Tue/Wed/Thu)' and the
candidate adds 'Monday and Friday are "
                    "remote-flexible, with full-remote by manager
exception,' that is CORRECT - it doesn't "
                    "change the number of in-office days or which days
they are).\n"
                    "Mark INCORRECT only if the candidate: states a
different fact/value than the "
                    "reference, changes the specific
number/days/amount the reference commits to, omits "
                    "part of a multi-part reference answer, actually
contradicts the reference, or claims "
                    "no information when the reference has a specific
answer.\n"
                    "Respond in exactly this format:\n"
                    "VERDICT: CORRECT or INCORRECT\n"
                    "REASON: one short sentence"
                )},
                {"role": "user", "content": f"Question:
{question}\nReference answer: {expected}\nCandidate answer: {got}"},
            ],
        )
    except Exception as exc:
        # Retries exhausted -- don't crash the whole evaluate_provider
loop over one
        # stubborn timeout. Marked as a distinguishable FAIL so it's
easy to filter out
        # of the accuracy numbers rather than silently counting as a
real wrong answer.
        return False, f"<JUDGE_ERROR: {exc}>"

    text = resp.choices[0].message.content.strip()
    verdict_line = next((l for l in text.splitlines() if
l.upper().startswith("VERDICT")), "")
    reason_line = next((l for l in text.splitlines() if
l.upper().startswith("REASON")), "")
    is_correct = "CORRECT" in verdict_line.upper() and "INCORRECT" not
in verdict_line.upper()
    return is_correct, reason_line.split(":", 1)[-1].strip()

def qa_category(qa):
    return CATEGORY_BY_TYPE.get(qa["type"], "Other")


def evaluate_provider(name, ask_fn):
    correct = 0
    cat_correct, cat_total = defaultdict(int), defaultdict(int)

    for qa in DET_QA:
        question, query_date = qa["question"], qa.get("query_date")
        display_q = f"{question} (Answer as of {query_date})" if
query_date else question

        answer = ask_safe(question, query_date, ask_fn)
        norm_answer, norm_expected = normalize_answer(answer),
normalize_answer(qa["expected_answer"])

        if norm_answer == norm_expected or
_contains_whole(norm_expected, norm_answer):
            is_correct, reason = True, "exact/substring match"
        else:
            is_correct, reason = llm_judge(display_q,
qa["expected_answer"], answer)

        status = "PASS" if is_correct else "FAIL"
        correct += is_correct
        category = qa_category(qa)
        cat_total[category] += 1
        cat_correct[category] += is_correct

        print(f"{name} {qa['id']} [{category}] {status} - {reason}")
        print(" Q:", display_q)
        print(" expected:", qa["expected_answer"])
        print(" got     :", answer)
        print()

    print(f"{name} overall: {correct}/{len(DET_QA)}
({correct/len(DET_QA):.1%})")
    for category in ["Temporal Reasoning", "Conflict Resolution",
"Multi-hop Reasoning"]:
        c, t = cat_correct[category], cat_total[category]
        if t:
            print(f"  {category}: {c}/{t} ({c/t:.1%})")

    return {
        "overall": (correct, len(DET_QA)),
        "by_category": {cat: (cat_correct[cat], cat_total[cat]) for
cat in cat_total},
    }
```

Cognee's retrieval and eval loop are defined separately since its SDK is async-only. `cognee_ask` and `evaluate_provider_async` mirror the synchronous versions above one-for-one.

In [13]:

```
import cognee

async def cognee_ask(question, query_date=None,
dataset_name="solvane_docs"):
    search_query = _strip_date_clause(question)

    results = await _with_retries_async(
        cognee.recall,
        query_text=search_query,
        datasets=[dataset_name],
        only_context=True,
    )

    if isinstance(results, str):
        context = results
    elif results and isinstance(results[0], dict):
        context = "\n\n".join(r.get("text", "") for r in results if
r.get("text"))
    else:
        context = "\n".join(str(r) for r in results) if results else
""

    return _synthesize(question, context, query_date)


async def evaluate_provider_async(name, ask_fn):
    correct = 0
    cat_correct, cat_total = defaultdict(int), defaultdict(int)

    for qa in DET_QA:
        question, query_date = qa["question"], qa.get("query_date")
        display_q = f"{question} (Answer as of {query_date})" if
query_date else question

        answer = await ask_safe_async(question, query_date, ask_fn)
        norm_answer, norm_expected = normalize_answer(answer),
normalize_answer(qa["expected_answer"])

        if norm_answer == norm_expected or
_contains_whole(norm_expected, norm_answer):
            is_correct, reason = True, "exact/substring match"
        else:
            is_correct, reason = llm_judge(display_q,
qa["expected_answer"], answer)

        status = "PASS" if is_correct else "FAIL"
        correct += is_correct
        category = qa_category(qa)
        cat_total[category] += 1
        cat_correct[category] += is_correct

        print(f"{name} {qa['id']} [{category}] {status} - {reason}")
        print(" Q:", display_q)
        print(" expected:", qa["expected_answer"])
        print(" got     :", answer)
        print()

    print(f"{name} overall: {correct}/{len(DET_QA)}
({correct/len(DET_QA):.1%})")
    for category in ["Temporal Reasoning", "Conflict Resolution",
"Multi-hop Reasoning"]:
        c, t = cat_correct[category], cat_total[category]
        if t:
            print(f"  {category}: {c}/{t} ({c/t:.1%})")

    return {
        "overall": (correct, len(DET_QA)),
        "by_category": {cat: (cat_correct[cat], cat_total[cat]) for
cat in cat_total},
    }
```

#### b. Ambiguity & Contradiction Handling

Persona-based probes over genuinely contradictory user history, no single correct answer. Graded on whether the response hedges appropriately, silently picks a side, invents a resolution, or over-hedges when the conflict was actually resolved.

`evaluate_ambiguity` / `evaluate_ambiguity_async` switch `_synthesize` into "hedge" mode for the duration of their loop (see the note in section a). Without this, every provider will be graded on ambiguity while still being told by the shared prompt to pick a side, which would make the section unusable as a comparison signal.

What the four verdicts mean:

Every probe response gets classified into exactly one of these by `annotation_judge` , based on the specific tension documented for that persona (see each persona's `annotation` block):

- **GOOD** — the response surfaces the tension instead of silently picking a side (e.g. "you mentioned being vegetarian, but also had a ribeye recently — still keeping meals meat-free, or open to more now?"), *or*, for the two contrast-case personas where a real resolution was actually given (Morgan P.'s explicit budget correction, Ravi S.'s tentative deadline slip), correctly uses that resolution without hedging on something that isn't actually still open.
- **SILENT_PICK** — confidently answers with only one side of a genuine contradiction, with no acknowledgment that conflicting information exists. Example from an actual run: asked about Priya's diet, a provider answered "vegetarian" and never mentioned her ribeye dinner at all, thus treating an explicitly unresolved tension as settled fact.
- **HALLUCINATED_RESOLUTION** — invents an explanation or reconciliation the user never actually gave (e.g. "since you mentioned switching to pescatarian..." when the user said no such thing). Worse than SILENT_PICK, since it fabricates evidence rather than just omitting the conflict.
- **OVER_HEDGED** — only applies to Morgan P. and Ravi S. (the two personas where a resolution genuinely was given). Fires when a provider treats an already-resolved point as if it were still an open contradiction. Hedging is only a good behavior when there's an actual unresolved tension to hedge about.

One thing worth being explicit about for readers: "good" here does not mean "matches a fixed reference string" the way ground-truth QA does. Nine of the eleven personas are genuinely unresolved by design (see `non_deterministic/README.md`), so there is no single correct answer to match against. The judge is grading *behavior* (appropriately uncertain vs. silently confident vs. fabricated), not factual accuracy against a ground truth.

In [14]:

```
def annotation_judge(persona_name, probe_query, response, annotation):
    try:
        resp = _with_retries(
            _llm.chat.completions.create,
            model="gpt-4o-mini",
            temperature=0,
            messages=[
                {"role": "system", "content": (
                    "You are grading how an AI assistant handled a
genuinely ambiguous or contradictory "
                    "piece of user history. You are given the nature
of the tension, what a good response "
                    "looks like, and specific failure modes to watch
for. Classify the candidate response "
                    "into exactly one of:\n"
                    "GOOD - appropriately hedges, surfaces the
tension, or (for contrast cases) correctly "
                    "applies an actual explicit resolution without
over-hedging.\n"
                    "SILENT_PICK - confidently answers with one side
of the contradiction as if it were "
                    "settled, with no acknowledgment of the
conflicting information.\n"
                    "HALLUCINATED_RESOLUTION - invents an explanation
or reconciliation the user never gave.\n"
                    "OVER_HEDGED - treats something as ambiguous when
it was actually explicitly resolved "
                    "(only applies to contrast-case personas).\n"
                    "Respond in exactly this format:\n"
                    "VERDICT: GOOD or SILENT_PICK or
HALLUCINATED_RESOLUTION or OVER_HEDGED\n"
                    "REASON: one short sentence"
                )},
                {"role": "user", "content": (
                    f"Persona: {persona_name}\n"
                    f"Tension: {annotation['tension']}\n"
                    f"Good response looks like:
{annotation['good_response_looks_like']}\n"
                    f"Failure modes to watch for:
{annotation['failure_modes_to_watch_for']}\n\n"
                    f"Probe query: {probe_query}\n"
                    f"Candidate response: {response}"
                )},
            ],
        )
    except Exception as exc:
        # Retries exhausted -- degrade to UNKNOWN (a bucket the
results dict already
        # tracks) instead of crashing the whole evaluate_ambiguity
loop.
        return "UNKNOWN", f"<JUDGE_ERROR: {exc}>"

    text = resp.choices[0].message.content.strip()
    verdict_line = next((l for l in text.splitlines() if
l.upper().startswith("VERDICT")), "")
    reason_line = next((l for l in text.splitlines() if
l.upper().startswith("REASON")), "")
    verdict = verdict_line.split(":", 1)[-1].strip().upper() if ":" in
verdict_line else "UNKNOWN"
    reason = reason_line.split(":", 1)[-1].strip() if reason_line else
""
    return verdict, reason


def evaluate_ambiguity(name, ask_fn):
    global _SYNTHESIS_MODE
    results = {"GOOD": 0, "SILENT_PICK": 0, "HALLUCINATED_RESOLUTION":
0, "OVER_HEDGED": 0, "UNKNOWN": 0}
    _SYNTHESIS_MODE = "hedge"
    try:
        for persona in ND_DOCS:
            for probe in persona["probe_queries"]:
                response = ask_fn(probe, None, persona["persona_id"])
# sync providers; await for async ones
                verdict, reason =
annotation_judge(persona["persona_name"], probe, response,
persona["annotation"])
                results[verdict] = results.get(verdict, 0) + 1
                print(f"{name} {persona['persona_id']} [{verdict}] -
{reason}")
                print(" Probe:", probe)
                print(" Got:  ", response)
                print()
    finally:
        _SYNTHESIS_MODE = "resolve"

    total = sum(results.values())
    print(f"{name}: {results['GOOD']}/{total} good
({results['GOOD']/total:.1%}) - "
          f"{results['SILENT_PICK']} silent picks,
{results['HALLUCINATED_RESOLUTION']} hallucinated, "
          f"{results['OVER_HEDGED']} over-hedged")

async def evaluate_ambiguity_async(name, ask_fn):
    global _SYNTHESIS_MODE
    results = {"GOOD": 0, "SILENT_PICK": 0, "HALLUCINATED_RESOLUTION":
0, "OVER_HEDGED": 0, "UNKNOWN": 0}
    _SYNTHESIS_MODE = "hedge"
    try:
        for persona in ND_DOCS:
            for probe in persona["probe_queries"]:
                response = await ask_fn(probe, None,
persona["persona_id"])
                verdict, reason =
annotation_judge(persona["persona_name"], probe, response,
persona["annotation"])
                results[verdict] = results.get(verdict, 0) + 1
                print(f"{name} {persona['persona_id']} [{verdict}] -
{reason}")
                print(" Probe:", probe)
                print(" Got:  ", response)
                print()
    finally:
        _SYNTHESIS_MODE = "resolve"

    total = sum(results.values())
    print(f"{name}: {results['GOOD']}/{total} good
({results['GOOD']/total:.1%}) - "
          f"{results['SILENT_PICK']} silent picks,
{results['HALLUCINATED_RESOLUTION']} hallucinated, "
          f"{results['OVER_HEDGED']} over-hedged")
```

#### c. Needle in Haystack

Sparse, easily-missed facts buried in persona sessions. Tests recall precision rather than reasoning.

In [15]:

```
def evaluate_needles(name, ask_fn):
    correct = 0
    total = 0
    for persona in ND_DOCS:
        for probe in persona.get("needle_probes", []):
            total += 1
            question, expected = probe["question"],
probe["expected_answer"]
            answer = ask_safe(question, None, ask_fn,
persona["persona_id"])
            norm_answer, norm_expected = normalize_answer(answer),
normalize_answer(expected)

            if norm_answer == norm_expected or
_contains_whole(norm_expected, norm_answer):
                is_correct, reason = True, "exact/substring match"
            else:
                is_correct, reason = llm_judge(question, expected,
answer)

            status = "PASS" if is_correct else "FAIL"
            correct += is_correct
            print(f"{name} {persona['persona_id']} {status} -
{reason}")
            print(" Q:", question)
            print(" expected:", expected)
            print(" got     :", answer)
            print()

    if total:
        print(f"{name}: {correct}/{total} ({correct/total:.1%})")
    return correct, total


async def evaluate_needles_async(name, ask_fn):
    correct = 0
    total = 0
    for persona in ND_DOCS:
        for probe in persona.get("needle_probes", []):
            total += 1
            question, expected = probe["question"],
probe["expected_answer"]
            answer = await ask_safe_async(question, None, ask_fn,
persona["persona_id"])
            norm_answer, norm_expected = normalize_answer(answer),
normalize_answer(expected)

            if norm_answer == norm_expected or
_contains_whole(norm_expected, norm_answer):
                is_correct, reason = True, "exact/substring match"
            else:
                is_correct, reason = llm_judge(question, expected,
answer)

            status = "PASS" if is_correct else "FAIL"
            correct += is_correct
            print(f"{name} {persona['persona_id']} {status} -
{reason}")
            print(" Q:", question)
            print(" expected:", expected)
            print(" got     :", answer)
            print()

    if total:
        print(f"{name}: {correct}/{total} ({correct/total:.1%})")
    return correct, total
```

#### d. Run Evaluations

This is where retrieval actually happens: each provider is queried against all three evaluation sets above, and results are logged for comparison below.

In [16]:

```
%pip install matplotlib pandas
import sys, io, re, contextlib
import pandas as pd
import matplotlib.pyplot as plt

class _Tee(io.StringIO):
    def __init__(self, real_stream):
        super().__init__()
        self._real = real_stream
    def write(self, s):
        self._real.write(s)
        return super().write(s)
    def flush(self):
        self._real.flush()

@contextlib.contextmanager
def tee_stdout():
    tee = _Tee(sys.stdout)
    with contextlib.redirect_stdout(tee):
        yield tee

# DET_QA / needle-in-haystack lines look like: "Provider id [Category]
PASS - reason"
PASS_FAIL_RE = re.compile(r"^(\S+)\s+(\S+)\s+(?:\[(.+?)\]\s+)?
(PASS|FAIL)\b")
# Ambiguity lines look like: "Provider persona-id
[GOOD/SILENT_PICK/...] - reason"
AMBIGUITY_RE = re.compile(r"^(\S+)\s+(\S+)\s+\
[(GOOD|SILENT_PICK|HALLUCINATED_RESOLUTION|OVER_HEDGED|UNKNOWN)\]")

results_log = []
# raw_logs keeps the full printed text (question/expected/got/reason)
per run, not just the
# parsed PASS/FAIL booleans in results_log - needed so section (f) can
sample real judged
# answers for a human spot-check instead of just re-trusting the
aggregate pass rate
raw_logs = []

def _capture_to_log(raw_text, default_category=None):
    for line in raw_text.splitlines():
        line = line.strip()
        m = PASS_FAIL_RE.match(line)
        if m:
            provider, qid, category, status = m.groups()
            results_log.append({
                "provider": provider, "id": qid,
                "category": category or default_category,
                "correct": status == "PASS",
            })
            continue
        m = AMBIGUITY_RE.match(line)
        if m:
            provider, qid, verdict = m.groups()
            results_log.append({
                "provider": provider, "id": qid,
                "category": default_category or "Ambiguity",
                "correct": verdict == "GOOD",
            })

def run_and_log(name, ask_fn, eval_fn, category=None):
    with tee_stdout() as buf:
        eval_fn(name, ask_fn)
    text = buf.getvalue()
    _capture_to_log(text, default_category=category)
    raw_logs.append((name, category or "Ground-Truth QA", text))

async def run_and_log_async(name, ask_fn, eval_fn, category=None):
    with tee_stdout() as buf:
        await eval_fn(name, ask_fn)
    text = buf.getvalue()
    _capture_to_log(text, default_category=category)
    raw_logs.append((name, category or "Ground-Truth QA", text))
```

```
/opt/homebrew/Cellar/python@3.14/3.14.6/Frameworks/Python.fram
ework/Versions/3.14/lib/python3.14/pty.py:66:
DeprecationWarning: This process (pid=31795) is multi-
threaded, use of forkpty() may lead to deadlocks in the child.
  pid, fd = os.forkpty()
```

```
Requirement already satisfied: matplotlib in
./.venv/lib/python3.14/site-packages (3.11.1)
Requirement already satisfied: pandas in
./.venv/lib/python3.14/site-packages (3.0.3)
Requirement already satisfied: contourpy>=1.0.1 in
./.venv/lib/python3.14/site-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in
./.venv/lib/python3.14/site-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.28.2 in
./.venv/lib/python3.14/site-packages (from matplotlib) (4.63.0)
Requirement already satisfied: kiwisolver>=1.3.1 in
./.venv/lib/python3.14/site-packages (from matplotlib) (1.5.0)
Requirement already satisfied: numpy>=1.25 in
./.venv/lib/python3.14/site-packages (from matplotlib) (2.5.1)
Requirement already satisfied: packaging>=20.0 in
./.venv/lib/python3.14/site-packages (from matplotlib) (24.2)
Requirement already satisfied: pillow>=9 in
./.venv/lib/python3.14/site-packages (from matplotlib) (12.3.0)
Requirement already satisfied: pyparsing>=3 in
./.venv/lib/python3.14/site-packages (from matplotlib) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in
./.venv/lib/python3.14/site-packages (from matplotlib)
(2.9.0.post0)
Requirement already satisfied: six>=1.5 in
./.venv/lib/python3.14/site-packages (from python-dateutil>=2.7-
>matplotlib) (1.17.0)
Note: you may need to restart the kernel to use updated packages.
```

In [17]:

```
# DET_QA - category comes from the [Temporal Reasoning]/[Conflict
Resolution]/[Multi-hop] tag automatically
run_and_log("Zep", zep_ask, evaluate_provider)
run_and_log("Mem0", mem0_ask, evaluate_provider)
run_and_log("Supermemory", supermemory_ask, evaluate_provider)
await run_and_log_async("Cognee", cognee_ask, evaluate_provider_async)

# Needle-in-haystack
run_and_log("Zep", zep_ask, evaluate_needles, category="Needle-in-
Haystack")
run_and_log("Mem0", mem0_ask, evaluate_needles, category="Needle-in-
Haystack")
run_and_log("Supermemory", supermemory_ask, evaluate_needles,
category="Needle-in-Haystack")
await run_and_log_async("Cognee", cognee_ask, evaluate_needles_async,
category="Needle-in-Haystack")

# Ambiguity (non-det)
run_and_log("Zep", zep_ask, evaluate_ambiguity, category="Ambiguity")
run_and_log("Mem0", mem0_ask, evaluate_ambiguity,
category="Ambiguity")
run_and_log("Supermemory", supermemory_ask, evaluate_ambiguity,
category="Ambiguity")
await run_and_log_async("Cognee", cognee_ask,
evaluate_ambiguity_async, category="Ambiguity")
```

```
Zep det-001 [Temporal Reasoning] PASS - exact/substring match
 Q: Who leads Team Nova as of 2024-06-01? (Answer as of 2024-06-
01)
 expected: Alice Chen
 got     : As of June 1, 2024, Team Nova is led by Alice Chen.
This information is still accurate and has not been superseded.

Zep det-002 [Temporal Reasoning] FAIL - The candidate states that
Alice Chen leads Team Nova, which contradicts the reference
answer that states Bob Okafor is the leader.
 Q: Who leads Team Nova as of 2025-06-01? (Answer as of 2025-06-
01)
 expected: Bob Okafor
 got     : As of June 1, 2025, Team Nova is led by Alice Chen.
This information is still accurate and has not been superseded
since the April 2024 org snapshot.

Zep det-003 [Conflict Resolution] FAIL — The candidate states a
different leader than the reference answer.
 Q: Who leads Team Nova?
 expected: Bob Okafor
 got     : Team Nova is currently led by Alice Chen.

Zep det-004 [Temporal Reasoning] PASS — exact/substring match
 Q: Who leads Team Forge as of 2024-03-01? (Answer as of 2024-03-
01)
 expected: Sofia Kappler
 got     : As of March 1, 2024, Team Forge is led by Sofia
Kappler. This information is still accurate as of that date.

Zep det-005 [Temporal Reasoning] PASS — exact/substring match
 Q: Who leads Team Forge as of 2025-01-01? (Answer as of 2025-01-
01)
 expected: Derek Osei
 got     : As of January 1, 2025, Team Forge is led by Sofia
Kappler. This information is based on the April 2024 org
snapshot, which stated that Sofia Kappler was the lead of Team
Forge. Since then, it has been superseded by the fact that Derek
Osei became the lead effective September 16, 2024, but that
change occurred after the January 1, 2025 date.

Zep det-006 [Conflict Resolution] FAIL — The candidate
incorrectly states that "User" leads Team Forge instead of Derek
Osei.
 Q: Who leads Team Forge?
 expected: Derek Osei
 got     : User leads Team Forge.

Zep det-007 [Temporal Reasoning] PASS — exact/substring match
 Q: What is the maximum payload of the Solvane S1 as of 2023-06-
01? (Answer as of 2023-06-01)
 expected: 5 kg
 got     : The maximum payload of the Solvane S1 as of June 1,
2023, is 5 kg, according to the v1.0 spec sheet. This figure
remains valid as the v2.0 spec sheet, which published a 7 kg
maximum payload, was released later on June 3, 2024.

Zep det-008 [Temporal Reasoning] PASS — exact/substring match
 Q: What is the maximum payload of the Solvane S1 as of 2024-08-
01? (Answer as of 2024-08-01)
 expected: 7 kg
 got     : The maximum payload of the Solvane S1 as of 2024-08-01
is 7 kg, according to the Solvane S1 v2.0 spec sheet published on
June 3, 2024. However, this figure was later corrected on
November 20, 2024, to a validated maximum payload of 6 kg under
real-world conditions. Therefore, while the 7 kg figure was
accurate at that time, it has since been superseded by the 6 kg
rating.

Zep det-009 [Temporal Reasoning] FAIL — The candidate states a
maximum payload of 7 kg, which contradicts the reference answer
of 6 kg.
 Q: What is the maximum payload of the Solvane S1 as of 2025-06-
01? (Answer as of 2025-06-01)
 expected: 6 kg
 got     : The maximum payload of the Solvane S1 as of June 1,
2025, is 7 kg according to the Solvane S1 v2.0 spec sheet
published on June 3, 2024. This figure was measured under
controlled lab conditions.

Zep det-010 [Conflict Resolution] PASS — exact/substring match
 Q: What is the maximum payload of the Solvane S1?
 expected: 6 kg
 got     : The maximum payload of the Solvane S1 is 6 kg, as
validated under real-world deployment conditions with standard
mounting hardware.

Zep det-011 [Temporal Reasoning] FAIL — The candidate fails to
provide the specific pricing information of $499/month flat for
an individual license as stated in the reference.
 Q: What is the pricing for Solvane Vision Suite as of 2024-01-
01? (Answer as of 2024-01-01)
 expected: $499/month flat, individual license
 got     : As of January 1, 2024, the pricing for Solvane Vision
Suite is not explicitly stated in the available facts. However,
it is known that the Solvane Vision Suite has pricing
information, and there was a pricing update posted on January 15,
2025, which suggests that the details may not have been finalized
or publicly available by January 1, 2024.

Zep det-012 [Temporal Reasoning] FAIL — The candidate fails to
provide the specific pricing of $150/seat/month as stated in the
reference.
 Q: What is the pricing for Solvane Vision Suite as of 2025-06-
01? (Answer as of 2025-06-01)
 expected: $150/seat/month, per-seat license
 got     : As of June 1, 2025, the pricing for Solvane Vision
Suite is not explicitly stated in the available facts. However,
it is known that the Solvane Vision Suite has pricing
information, and it was indicated that the pricing update was
posted on January 15, 2025. Since there are no specific details
about the pricing structure available prior to that date, the
exact pricing remains unclear.

Zep det-013 [Conflict Resolution] FAIL — The candidate does not
provide the specific pricing of $150/seat/month as stated in the
reference.
 Q: What is the pricing for Solvane Vision Suite?
 expected: $150/seat/month, per-seat license (individual tier
discontinued)
 got     : The pricing for Solvane Vision Suite is currently
being requested, and it is moving to per-seat pricing.

Zep det-014 [Temporal Reasoning] FAIL — The candidate's answer
includes a future change to the policy that is not part of the
reference.
 Q: What is Solvane's remote work policy as of 2024-01-01?
(Answer as of 2024-01-01)
 expected: Fully remote, no in-office requirement
 got     : As of January 1, 2024, Solvane Robotics had a fully
remote work policy, allowing teams to meet in person at their
discretion. However, this policy was set to change on February 1,
2025, when the company would move to a hybrid work model
requiring in-office attendance on Tuesdays, Wednesdays, and
Thursdays, with Mondays and Fridays remaining remote-flexible.

Zep det-015 [Temporal Reasoning] PASS — The candidate accurately
states the in-office days and acknowledges the remote-flexible
aspect without contradicting the core claim.
 Q: What is Solvane's remote work policy as of 2025-06-01?
(Answer as of 2025-06-01)
 expected: Hybrid: 3 days/week in office (Tue/Wed/Thu)
 got     : As of June 1, 2025, Solvane's remote work policy
requires employees to be in the office on Tuesdays, Wednesdays,
and Thursdays, while Mondays and Fridays remain remote-flexible.
This policy is a shift from the previous fully remote work
policy.

Zep det-016 [Conflict Resolution] PASS — The candidate accurately
describes the hybrid model and specifies the in-office days as
Tuesday, Wednesday, and Thursday, while adding that Mondays and
Fridays are remote-flexible, which does not contradict the core
claim.
 Q: What is Solvane's remote work policy?
 expected: Hybrid: 3 days/week in office (Tue/Wed/Thu)
 got     : Solvane's remote work policy is a hybrid model,
requiring employees to be in the office on Tuesdays, Wednesdays,
and Thursdays, while Mondays and Fridays remain remote-flexible.

Zep det-017 [Temporal Reasoning] PASS — exact/substring match
 Q: What is the daily meal expense reimbursement cap as of 2024-
01-01? (Answer as of 2024-01-01)
 expected: $50/day
 got     : The daily meal expense reimbursement cap for business
travel is $50/day as of January 1, 2024.

Zep det-018 [Temporal Reasoning] PASS — exact/substring match
 Q: What is the daily meal expense reimbursement cap as of 2025-
12-01? (Answer as of 2025-12-01)
 expected: $75/day
 got     : The daily meal expense reimbursement cap as of
December 1, 2025, is $75/day. This fact is still accurate and has
not been superseded since it was updated in the Finance Memo
effective August 1, 2025.

Zep det-019 [Conflict Resolution] PASS — exact/substring match
 Q: What is the daily meal expense reimbursement cap?
 expected: $75/day
 got     : The daily meal expense reimbursement cap for business
travel is $75/day. Receipts are required for any single expense
over $25.

Zep det-020 [Temporal Reasoning] PASS — The candidate accurately
states the battery life in standard mode as 8 hours, matching the
reference.
 Q: What is the battery life of the Solvane Nav Kit as of 2025-
06-01? (Answer as of 2025-06-01)
 expected: 8 hours standard mode; up to 10 hours in eco mode
(both true, different modes)
 got     : The Solvane Nav Kit has a battery life of 8 hours of
continuous operation in standard operating mode. This fact is
still accurate as of 2025-06-01.

Zep det-021 [Conflict Resolution] PASS — The candidate accurately
states the battery life in both standard and eco modes as per the
reference.
 Q: What is the battery life of the Solvane Nav Kit?
 expected: Both values are current and true simultaneously (8h
standard mode; up to 10h eco mode) — there is no single 'most
recent supersedes all' answer here.
 got     : The battery life of the Solvane Nav Kit is 8 hours of
continuous operation in standard mode, and up to 10 hours of
continuous operation in eco mode.

Zep det-mh-001 [Multi-hop Reasoning] FAIL — The candidate
incorrectly states who reviewed the S1 payload issue and omits
the corrected payload rating.
 Q: Who reviewed the S1 payload issue that led to a leadership
change, and what is the corrected payload rating that came out of
it?
 expected: Derek Osei led the Team Forge review (becoming Team
Forge lead as a result, per the Sept 2024 incident report); the
corrected S1 v2 payload rating is 6 kg (per the Nov 2024 errata).
 got     : Solvane Docs reviewed the S1 payload issue that led to
a leadership change. The corrected payload rating resulting from
this review is not specified in the provided messages.

Zep det-mh-002 [Multi-hop Reasoning] PASS — The candidate
accurately states that Alice Chen is no longer the point of
contact and provides correct details about her transition and Bob
Okafor's succession.
 Q: As of mid-2025, is Alice Chen still the point of contact for
Team Nova? If not, why not and who is?
 expected: No — as of March 2025, Bob Okafor is Team Nova lead.
Alice Chen moved to a Principal Engineer (individual contributor)
role with no direct reports.
 got     : As of mid-2025, Alice Chen is not the point of contact
for Team Nova. She transitioned to a Principal Engineer role on
March 3, 2025, and Bob Okafor succeeded her as the point of
contact for Team Nova going forward.

Zep det-mh-003 [Multi-hop Reasoning] FAIL — The candidate
incorrectly states the date when Derek Osei became the lead,
which is September 2024, not July 22, 2026.
 Q: According to the April 2024 org snapshot, who leads Team
Forge — and is that still accurate?
 expected: The April 2024 snapshot lists Sofia Kappler as Team
Forge lead, which was accurate at that time but is now stale —
Derek Osei became lead in September 2024.
 got     : According to the April 2024 org snapshot, Sofia
Kappler was identified as the Lead of Team Forge. However, this
information is no longer accurate; Derek Osei is now the Lead of
Team Forge, effective today (as of July 22, 2026).

Zep det-mh-004 [Multi-hop Reasoning] FAIL — The candidate omits
the specific new role for Sofia Kappler and does not mention the
reasons for their transitions.
 Q: Alice Chen and Sofia Kappler both stepped down as team leads.
What new roles did each move into, and what prompted each
transition?
 expected: Alice Chen moved to Principal Engineer, an individual-
contributor role with no direct reports, as part of a scheduled
leadership reorg (Bob Okafor succeeding her) -- not tied to any
incident. Sofia Kappler moved to a newly created Quality Systems
Advisor role, specifically prompted by the Team Forge review
following S1 payload field reports. The two transitions had
different causes: routine reorg for Alice, incident-driven review
for Sofia.
 got     : Alice Chen and Sofia Kappler both stepped down as team
leads. Alice Chen is moving into a Principal Engineer role, while
the new role for Sofia Kappler has not been specified. The
reasons prompting their transitions are also unspecified.

Zep det-mh-005 [Multi-hop Reasoning] FAIL — The candidate answer
contradicts the reference by stating it is unknown, while the
reference clearly states that the correction does not apply.
 Q: Does the November 2024 S1 payload correction to 6 kg apply to
units running the original 2023 v1.0 hardware?
 expected: No. The errata explicitly corrects only the June 2024
v2.0 spec sheet's 7 kg figure; it does not affect the original
2023 v1.0 spec sheet, which remains rated at 5 kg.
 got     : It is unknown if the November 2024 S1 payload
correction to 6 kg applies to units running the original 2023
v1.0 hardware.

Zep det-mh-006 [Multi-hop Reasoning] FAIL — The candidate states
that Team Atlas received new leads, which contradicts the
reference answer.
 Q: Team Nova and Team Forge both got new leads at different
points. Did Team Atlas ever get a new lead too?
 expected: No -- Marcus Webb has led Team Atlas continuously
since at least the 2023 handbook, and is still listed as Team
Atlas lead as recently as the October 2024 all-hands notes and
the April 2024 org snapshot. Unlike Nova and Forge, there is no
leadership change recorded for Team Atlas anywhere in the
dataset.
 got     : Yes, Team Atlas received new leads.

Zep det-mh-007 [Multi-hop Reasoning] FAIL — The candidate's
answer incorrectly states that it is unclear if the announcement
preceded the official notice, while the reference clearly states
that the public errata was posted a month later.
 Q: Was the S1 v2 payload correction first made public at the
October 2024 all-hands, or was it announced there before the
official notice went out?
 expected: It was previewed internally at the October 22, 2024
all-hands (Derek walked through the correction ahead of the
public notice), but the official public errata wasn't posted
until November 20, 2024 -- about a month later.
 got     : The S1 v2 payload correction was previewed at the
October 22, 2024, Engineering All-Hands meeting, where Derek Osei
presented it. However, the public announcement of the correction
occurred in November 2024, and it is unclear if this announcement
preceded the official notice.

Zep det-mh-008 [Multi-hop Reasoning] FAIL — The candidate states
that the information is still accurate today, but the reference
indicates that Alice Chen was the first lead and was succeeded by
Bob Okafor, making the candidate's claim incorrect.
 Q: The employee handbook's original org chart lists Team Nova's
lead as 'TBD.' Who ended up filling that role, and is that still
accurate today?
 expected: Alice Chen was announced as Team Nova's first lead in
January 2024, filling the 'TBD' slot from the 2023 handbook.
That's no longer current, though -- Bob Okafor succeeded her in
March 2025.
 got     : Bob Okafor ended up filling the role of Lead for Team
Nova. This is still accurate today.

Zep overall: 14/29 (48.3%)
  Temporal Reasoning: 9/14 (64.3%)
  Conflict Resolution: 4/7 (57.1%)
  Multi-hop Reasoning: 1/8 (12.5%)
Mem0 det-001 [Temporal Reasoning] FAIL — The candidate omits the
specific information that Alice Chen leads Team Nova as of the
given date.
 Q: Who leads Team Nova as of 2024-06-01? (Answer as of 2024-06-
01)
 expected: Alice Chen
 got     : The facts do not provide information about who leads
Team Nova as of June 1, 2024.

Mem0 det-002 [Temporal Reasoning] FAIL — The candidate omits the
specific information that Bob Okafor leads Team Nova as of the
given date.
 Q: Who leads Team Nova as of 2025-06-01? (Answer as of 2025-06-
01)
 expected: Bob Okafor
 got     : The facts do not provide information about who leads
Team Nova as of June 1, 2025.

Mem0 det-003 [Conflict Resolution] FAIL — The candidate
contradicts the reference by stating there is no information
about the team leader when the reference clearly states it is Bob
Okafor.
 Q: Who leads Team Nova?
 expected: Bob Okafor
 got     : The facts do not provide any information about who
leads Team Nova.

Mem0 det-004 [Temporal Reasoning] PASS — exact/substring match
 Q: Who leads Team Forge as of 2024-03-01? (Answer as of 2024-03-
01)
 expected: Sofia Kappler
 got     : Sofia Kappler leads Team Forge as of March 1, 2024.
This is still accurate as Derek Osei became Lead of Team Forge
effective September 16, 2024, which is after the specified date.

Mem0 det-005 [Temporal Reasoning] PASS — exact/substring match
 Q: Who leads Team Forge as of 2025-01-01? (Answer as of 2025-01-
01)
 expected: Derek Osei
 got     : As of January 1, 2025, Derek Osei leads Team Forge.

Mem0 det-006 [Conflict Resolution] PASS — exact/substring match
 Q: Who leads Team Forge?
 expected: Derek Osei
 got     : Derek Osei leads Team Forge.

Mem0 det-007 [Temporal Reasoning] FAIL — The candidate states a
different maximum payload (6 kg) than the reference (5 kg).
 Q: What is the maximum payload of the Solvane S1 as of 2023-06-
01? (Answer as of 2023-06-01)
 expected: 5 kg
 got     : The maximum payload of the Solvane S1 as of June 1,
2023, is 6 kg. This is the validated and rated maximum payload
for Solvane S1 v2 hardware under real-world deployment with
standard mounting hardware.

Mem0 det-008 [Temporal Reasoning] PASS — exact/substring match
 Q: What is the maximum payload of the Solvane S1 as of 2024-08-
01? (Answer as of 2024-08-01)
 expected: 7 kg
 got     : The maximum payload of the Solvane S1 as of August 1,
2024, is 6 kg. This reflects the validated and rated maximum
payload for Solvane S1 v2 hardware, as the previously published 7
kg figure was measured under controlled lab conditions and has
been corrected in subsequent documentation.

Mem0 det-009 [Temporal Reasoning] PASS — exact/substring match
 Q: What is the maximum payload of the Solvane S1 as of 2025-06-
01? (Answer as of 2025-06-01)
 expected: 6 kg
 got     : The maximum payload of the Solvane S1 as of 2025-06-01
is 6 kg. This reflects the updated rating following the errata
notice posted on November 20, 2024, which corrected the
previously published 7 kg figure in the June 2024 spec sheet.

Mem0 det-010 [Conflict Resolution] PASS — exact/substring match
 Q: What is the maximum payload of the Solvane S1?
 expected: 6 kg
 got     : The maximum payload of the Solvane S1 v2 hardware is 6
kg.

Mem0 det-011 [Temporal Reasoning] PASS — The candidate's core
factual claim about the pricing of $499/month for an individual
license matches the reference answer.
 Q: What is the pricing for Solvane Vision Suite as of 2024-01-
01? (Answer as of 2024-01-01)
 expected: $499/month flat, individual license
 got     : As of January 1, 2024, the pricing for Solvane Vision
Suite is $499 per month for the Standard tier (flat rate,
individual license), covering a single deployment instance and
standard support. This pricing has been effective since May 1,
2023, and remains unchanged.

Mem0 det-012 [Temporal Reasoning] FAIL — The candidate states a
different price of $499 per month instead of the reference price
of $150/seat/month.
 Q: What is the pricing for Solvane Vision Suite as of 2025-06-
01? (Answer as of 2025-06-01)
 expected: $150/seat/month, per-seat license
 got     : The pricing for Solvane Vision Suite as of June 1,
2025, is $499 per month for the Standard tier (flat rate,
individual license) covering a single deployment instance and
standard support. This pricing remains effective since May 1,
2023.

Mem0 det-013 [Conflict Resolution] FAIL — The candidate states a
different pricing structure than the reference answer.
 Q: What is the pricing for Solvane Vision Suite?
 expected: $150/seat/month, per-seat license (individual tier
discontinued)
 got     : The pricing for Solvane Vision Suite is $499 per month
(flat rate, individual license) for the Standard tier, covering a
single deployment instance and standard support. Enterprise
pricing is available on request for multi-site deployments.

Mem0 det-014 [Temporal Reasoning] FAIL — The candidate's answer
does not match the reference's claim of being fully remote with
no in-office requirement.
 Q: What is Solvane's remote work policy as of 2024-01-01?
(Answer as of 2024-01-01)
 expected: Fully remote, no in-office requirement
 got     : As of January 1, 2024, Solvane's remote work policy
was not yet updated to the hybrid working model that would take
effect on February 1, 2025. Therefore, the remote work policy
prior to that date would still be in place, but specific details
of that policy are not provided in the facts.

Mem0 det-015 [Temporal Reasoning] PASS — The candidate accurately
describes the in-office days and adds correct details about
remote flexibility and exceptions without contradicting the core
claim.
 Q: What is Solvane's remote work policy as of 2025-06-01?
(Answer as of 2025-06-01)
 expected: Hybrid: 3 days/week in office (Tue/Wed/Thu)
 got     : As of June 1, 2025, Solvane Robotics' remote work
policy is that employees work from their assigned office on
Tuesday, Wednesday, and Thursday, while Monday and Friday remain
remote-flexible. Fully remote arrangements are still possible by
manager exception for roles hired as remote-only. This policy was
updated effective February 1, 2025, superseding the 2023 edition.

Mem0 det-016 [Conflict Resolution] PASS — The candidate
accurately describes the core aspect of the hybrid model with 3
days in the office on the specified days.
 Q: What is Solvane's remote work policy?
 expected: Hybrid: 3 days/week in office (Tue/Wed/Thu)
 got     : Solvane's current remote work policy is a hybrid model
where employees work from their assigned office on Tuesday,
Wednesday, and Thursday, while Monday and Friday remain remote-
flexible. Fully remote arrangements are still possible by manager
exception for roles hired as remote-only.

Mem0 det-017 [Temporal Reasoning] FAIL — The candidate fails to
provide the specific reimbursement cap of $50/day as stated in
the reference.
 Q: What is the daily meal expense reimbursement cap as of 2024-
01-01? (Answer as of 2024-01-01)
 expected: $50/day
 got     : The facts provided do not include any information
regarding the daily meal expense reimbursement cap as of January
1, 2024. Therefore, I cannot provide an answer to that question.

Mem0 det-018 [Temporal Reasoning] FAIL — The candidate fails to
provide the specific reimbursement cap of $75/day mentioned in
the reference.
 Q: What is the daily meal expense reimbursement cap as of 2025-
12-01? (Answer as of 2025-12-01)
 expected: $75/day
 got     : The facts provided do not include any information
regarding the daily meal expense reimbursement cap as of December
1, 2025. Therefore, I cannot provide an answer to that question.

Mem0 det-019 [Conflict Resolution] FAIL — The candidate omits the
specific daily meal expense reimbursement cap of $75/day.
 Q: What is the daily meal expense reimbursement cap?
 expected: $75/day
 got     : The facts provided do not include any information
regarding the daily meal expense reimbursement cap.

Mem0 det-020 [Temporal Reasoning] FAIL — The candidate answer
omits the specific battery life details provided in the
reference.
 Q: What is the battery life of the Solvane Nav Kit as of 2025-
06-01? (Answer as of 2025-06-01)
 expected: 8 hours standard mode; up to 10 hours in eco mode
(both true, different modes)
 got     : There is no information provided regarding the battery
life of the Solvane Nav Kit as of June 1, 2025.

Mem0 det-021 [Conflict Resolution] FAIL — The candidate omits the
specific battery life details provided in the reference answer.
 Q: What is the battery life of the Solvane Nav Kit?
 expected: Both values are current and true simultaneously (8h
standard mode; up to 10h eco mode) — there is no single 'most
recent supersedes all' answer here.
 got     : The facts provided do not include any information
regarding the battery life of the Solvane Nav Kit.

Mem0 det-mh-001 [Multi-hop Reasoning] PASS — The candidate
accurately states that Derek Osei led the Team Forge review and
that the corrected payload rating is 6 kg.
 Q: Who reviewed the S1 payload issue that led to a leadership
change, and what is the corrected payload rating that came out of
it?
 expected: Derek Osei led the Team Forge review (becoming Team
Forge lead as a result, per the Sept 2024 incident report); the
corrected S1 v2 payload rating is 6 kg (per the Nov 2024 errata).
 got     : The S1 payload issue was reviewed by Team Forge, led
by Derek Osei, who became the Lead of Team Forge on September 16,
2024. The corrected payload rating for the Solvane S1 v2 hardware
is 6 kg.

Mem0 det-mh-002 [Multi-hop Reasoning] FAIL — The candidate fails
to acknowledge that Alice Chen is no longer the point of contact
and does not mention Bob Okafor as the new lead.
 Q: As of mid-2025, is Alice Chen still the point of contact for
Team Nova? If not, why not and who is?
 expected: No — as of March 2025, Bob Okafor is Team Nova lead.
Alice Chen moved to a Principal Engineer (individual contributor)
role with no direct reports.
 got     : There is no information provided about Alice Chen or
her role as the point of contact for Team Nova. Therefore, it
cannot be determined if she is still the point of contact or if
someone else has taken over that role.

Mem0 det-mh-003 [Multi-hop Reasoning] FAIL — The candidate states
that Derek Osei became lead on September 16, 2024, while the
reference indicates he became lead in September 2024 without
specifying a date.
 Q: According to the April 2024 org snapshot, who leads Team
Forge — and is that still accurate?
 expected: The April 2024 snapshot lists Sofia Kappler as Team
Forge lead, which was accurate at that time but is now stale —
Derek Osei became lead in September 2024.
 got     : As of the current state, Derek Osei leads Team Forge,
effective September 16, 2024. This is accurate.

Mem0 det-mh-004 [Multi-hop Reasoning] FAIL — The candidate omits
Alice Chen's new role and the reason for her transition, which is
a key part of the reference answer.
 Q: Alice Chen and Sofia Kappler both stepped down as team leads.
What new roles did each move into, and what prompted each
transition?
 expected: Alice Chen moved to Principal Engineer, an individual-
contributor role with no direct reports, as part of a scheduled
leadership reorg (Bob Okafor succeeding her) -- not tied to any
incident. Sofia Kappler moved to a newly created Quality Systems
Advisor role, specifically prompted by the Team Forge review
following S1 payload field reports. The two transitions had
different causes: routine reorg for Alice, incident-driven review
for Sofia.
 got     : Sofia Kappler transitioned to a newly created Quality
Systems Advisor role on September 16, 2024, focusing on hardware
validation across all product lines. Derek Osei became Lead of
Team Forge, effective September 16, 2024, prompting Sofia's
transition. There is no information provided about Alice Chen's
transition or new role.

Mem0 det-mh-005 [Multi-hop Reasoning] PASS — The candidate
accurately states that the correction applies only to the June
2024 v2.0 spec sheet and confirms the original 2023 v1.0 spec
sheet remains rated at 5 kg.
 Q: Does the November 2024 S1 payload correction to 6 kg apply to
units running the original 2023 v1.0 hardware?
 expected: No. The errata explicitly corrects only the June 2024
v2.0 spec sheet's 7 kg figure; it does not affect the original
2023 v1.0 spec sheet, which remains rated at 5 kg.
 got     : No, the correction to 6 kg payload rating applies
solely to the June 2024 v2 spec sheet and does not affect the
original 2023 v1.0 spec sheet, which remains rated at 5 kg.

Mem0 det-mh-006 [Multi-hop Reasoning] FAIL — The candidate answer
omits the specific fact that Team Atlas did not receive a new
lead, which is stated in the reference.
 Q: Team Nova and Team Forge both got new leads at different
points. Did Team Atlas ever get a new lead too?
 expected: No -- Marcus Webb has led Team Atlas continuously
since at least the 2023 handbook, and is still listed as Team
Atlas lead as recently as the October 2024 all-hands notes and
the April 2024 org snapshot. Unlike Nova and Forge, there is no
leadership change recorded for Team Atlas anywhere in the
dataset.
 got     : There is no information provided about Team Atlas
receiving a new lead.

Mem0 det-mh-007 [Multi-hop Reasoning] FAIL — The candidate
incorrectly states that there was no announcement at the October
2024 all-hands.
 Q: Was the S1 v2 payload correction first made public at the
October 2024 all-hands, or was it announced there before the
official notice went out?
 expected: It was previewed internally at the October 22, 2024
all-hands (Derek walked through the correction ahead of the
public notice), but the official public errata wasn't posted
until November 20, 2024 -- about a month later.
 got     : The S1 v2 payload correction was first made public
through an errata notice posted on November 20, 2024. There is no
indication that it was announced at the October 2024 all-hands
before the official notice went out.

Mem0 det-mh-008 [Multi-hop Reasoning] FAIL — The candidate fails
to identify Alice Chen and Bob Okafor as the leads of Team Nova,
contradicting the reference.
 Q: The employee handbook's original org chart lists Team Nova's
lead as 'TBD.' Who ended up filling that role, and is that still
accurate today?
 expected: Alice Chen was announced as Team Nova's first lead in
January 2024, filling the 'TBD' slot from the 2023 handbook.
That's no longer current, though -- Bob Okafor succeeded her in
March 2025.
 got     : The current lead of Team Nova is not specified in the
provided facts, so it is unclear who filled that role. Therefore,
it cannot be determined if that information is still accurate
today.

Mem0 overall: 11/29 (37.9%)
  Temporal Reasoning: 6/14 (42.9%)
  Conflict Resolution: 3/7 (42.9%)
  Multi-hop Reasoning: 2/8 (25.0%)
Supermemory det-001 [Temporal Reasoning] PASS — exact/substring
match
 Q: Who leads Team Nova as of 2024-06-01? (Answer as of 2024-06-
01)
 expected: Alice Chen
 got     : As of 2024-06-01, Team Nova is led by Alice Chen. This
is still accurate and has not been superseded by any newer fact.

Supermemory det-002 [Temporal Reasoning] PASS — exact/substring
match
 Q: Who leads Team Nova as of 2025-06-01? (Answer as of 2025-06-
01)
 expected: Bob Okafor
 got     : Bob Okafor is Lead of Team Nova, effective 2025-03-03.

Supermemory det-003 [Conflict Resolution] FAIL — The candidate
states a different leader than the reference.
 Q: Who leads Team Nova?
 expected: Bob Okafor
 got     : Alice Chen is the Lead of Team Nova.

Supermemory det-004 [Temporal Reasoning] PASS — exact/substring
match
 Q: Who leads Team Forge as of 2024-03-01? (Answer as of 2024-03-
01)
 expected: Sofia Kappler
 got     : As of 2024-03-01, Team Forge is led by Sofia Kappler.
This information is based on the org chart snapshot as of April
15, 2024, which indicates that Sofia Kappler was the lead at that
time.

Supermemory det-005 [Temporal Reasoning] PASS — exact/substring
match
 Q: Who leads Team Forge as of 2025-01-01? (Answer as of 2025-01-
01)
 expected: Derek Osei
 got     : Derek Osei is Lead of Team Forge effective September
16, 2024.

Supermemory det-006 [Conflict Resolution] PASS — exact/substring
match
 Q: Who leads Team Forge?
 expected: Derek Osei
 got     : Derek Osei is now Lead of Team Forge, effective
September 16, 2024.

Supermemory det-007 [Temporal Reasoning] FAIL — The candidate
states a different maximum payload (7 kg) than the reference (5
kg).
 Q: What is the maximum payload of the Solvane S1 as of 2023-06-
01? (Answer as of 2023-06-01)
 expected: 5 kg
 got     : The maximum payload of the Solvane S1 v2 as of 2023-
06-01 is 7 kilograms. This fact is still accurate as of that
date.

Supermemory det-008 [Temporal Reasoning] FAIL — The candidate
states a different maximum payload (6 kg) than the reference (7
kg).
 Q: What is the maximum payload of the Solvane S1 as of 2024-08-
01? (Answer as of 2024-08-01)
 expected: 7 kg
 got     : The maximum payload of the Solvane S1 v2 is 6 kg under
real-world conditions as of 2024-08-01. This fact is still
accurate and has not been superseded.

Supermemory det-009 [Temporal Reasoning] PASS — exact/substring
match
 Q: What is the maximum payload of the Solvane S1 as of 2025-06-
01? (Answer as of 2025-06-01)
 expected: 6 kg
 got     : The maximum payload of the Solvane S1 as of 2025-06-01
is 6 kg under real-world conditions. This is based on the updated
specifications that took effect on 2024-11-20, which superseded
the previous maximum payload of 7 kg.

Supermemory det-010 [Conflict Resolution] PASS — exact/substring
match
 Q: What is the maximum payload of the Solvane S1?
 expected: 6 kg
 got     : The maximum payload of the Solvane S1 v2 is 6 kg under
real-world conditions.

Supermemory det-011 [Temporal Reasoning] PASS — The core factual
claim about the pricing of Solvane Vision Suite as of 2024-01-01
matches the reference answer.
 Q: What is the pricing for Solvane Vision Suite as of 2024-01-
01? (Answer as of 2024-01-01)
 expected: $499/month flat, individual license
 got     : As of January 1, 2024, the pricing for Solvane Vision
Suite is $499 per month, flat rate, individual license. This was
effective until January 15, 2025, when it transitioned to per-
seat pricing of $150 per seat per month.

Supermemory det-012 [Temporal Reasoning] PASS — The candidate's
core factual claim about the pricing matches the reference
answer.
 Q: What is the pricing for Solvane Vision Suite as of 2025-06-
01? (Answer as of 2025-06-01)
 expected: $150/seat/month, per-seat license
 got     : As of 2025-06-01, the pricing for Solvane Vision Suite
is $150 per seat per month, as it moved to per-seat pricing
effective January 15, 2025, and discontinued the flat-rate
individual tier of $499 per month.

Supermemory det-013 [Conflict Resolution] PASS — The core factual
claim about the pricing matches the reference answer.
 Q: What is the pricing for Solvane Vision Suite?
 expected: $150/seat/month, per-seat license (individual tier
discontinued)
 got     : The pricing for Solvane Vision Suite is $150 per seat
per month, effective January 15, 2025. The previous flat-rate
individual tier of $499 per month has been discontinued.

Supermemory det-014 [Temporal Reasoning] FAIL — The candidate
states that employees are expected to work in-office on specific
days, contradicting the reference's claim of a fully remote
policy.
 Q: What is Solvane's remote work policy as of 2024-01-01?
(Answer as of 2024-01-01)
 expected: Fully remote, no in-office requirement
 got     : As of January 1, 2024, Solvane Robotics' remote work
policy is that employees are expected to work from an assigned
office on Tuesday, Wednesday, and Thursday, with remote work on
Monday and Friday being flexible. Fully remote arrangements are
available by manager exception for remote-only roles. This policy
is set to be effective from February 1, 2025, but it is already
in place as of the specified date.

Supermemory det-015 [Temporal Reasoning] PASS — The candidate
accurately describes the in-office days and acknowledges the
remote work flexibility without contradicting the core claim.
 Q: What is Solvane's remote work policy as of 2025-06-01?
(Answer as of 2025-06-01)
 expected: Hybrid: 3 days/week in office (Tue/Wed/Thu)
 got     : As of June 1, 2025, Solvane Robotics' remote work
policy is that employees are expected to work from an assigned
office on Tuesday, Wednesday, and Thursday, and remotely on
Monday and Friday. Fully remote arrangements are still available
by manager exception for remote-only roles.

Supermemory det-016 [Conflict Resolution] PASS — The candidate
accurately describes the in-office days as Tuesday, Wednesday,
and Thursday, aligning with the reference answer.
 Q: What is Solvane's remote work policy?
 expected: Hybrid: 3 days/week in office (Tue/Wed/Thu)
 got     : Solvane Robotics is moving to a hybrid working model;
employees are expected to work from an assigned office on
Tuesday, Wednesday, and Thursday, and remotely on Monday and
Friday, with fully remote arrangements still available by manager
exception, effective February 1, 2025.

Supermemory det-017 [Temporal Reasoning] PASS — The candidate
accurately states the daily meal expense reimbursement cap as
$50/day.
 Q: What is the daily meal expense reimbursement cap as of 2024-
01-01? (Answer as of 2024-01-01)
 expected: $50/day
 got     : The daily meal expense reimbursement cap as of 2024-
01-01 is $50 per day, with receipts required for any single
expense over $25. This policy is still in effect and has not been
superseded by the increase to $75 per day, which takes effect on
August 1, 2025.

Supermemory det-018 [Temporal Reasoning] FAIL — The candidate's
answer includes an incorrect effective date for the policy
change.
 Q: What is the daily meal expense reimbursement cap as of 2025-
12-01? (Answer as of 2025-12-01)
 expected: $75/day
 got     : The daily meal expense reimbursement cap as of 2025-
```

```
12-01 is $75 per day. This is effective from August 1, 2025, and
reflects the updated policy.

Supermemory det-019 [Conflict Resolution] PASS — The candidate's
answer matches the reference answer in terms of the daily meal
expense reimbursement cap.
 Q: What is the daily meal expense reimbursement cap?
 expected: $75/day
 got     : The daily meal expense reimbursement cap is $75 per
day.

Supermemory det-020 [Temporal Reasoning] FAIL — The candidate
answer omits the standard mode battery life of 8 hours.
 Q: What is the battery life of the Solvane Nav Kit as of 2025-
06-01? (Answer as of 2025-06-01)
 expected: 8 hours standard mode; up to 10 hours in eco mode
(both true, different modes)
 got     : The battery life of the Solvane Nav Kit in eco mode is
up to 10 hours continuous operation on a single charge (valid:
Oct 1, 2024 to present).

Supermemory det-021 [Conflict Resolution] FAIL — The candidate
answer omits the standard mode battery life of 8 hours.
 Q: What is the battery life of the Solvane Nav Kit?
 expected: Both values are current and true simultaneously (8h
standard mode; up to 10h eco mode) — there is no single 'most
recent supersedes all' answer here.
 got     : The battery life of the Solvane Nav Kit in eco mode is
up to 10 hours of continuous operation on a single charge.

Supermemory det-mh-001 [Multi-hop Reasoning] FAIL — The candidate
answer does not specify that Derek Osei led the review or that he
became the Team Forge lead as a result.
 Q: Who reviewed the S1 payload issue that led to a leadership
change, and what is the corrected payload rating that came out of
it?
 expected: Derek Osei led the Team Forge review (becoming Team
Forge lead as a result, per the Sept 2024 incident report); the
corrected S1 v2 payload rating is 6 kg (per the Nov 2024 errata).
 got     : The S1 payload issue was reviewed by Team Forge, and
the corrected maximum payload rating that came out of it is 6 kg
under real-world deployment conditions.

Supermemory det-mh-002 [Multi-hop Reasoning] PASS — The candidate
accurately states that Alice Chen is no longer the point of
contact and provides correct details about her transition and Bob
Okafor's new role.
 Q: As of mid-2025, is Alice Chen still the point of contact for
Team Nova? If not, why not and who is?
 expected: No — as of March 2025, Bob Okafor is Team Nova lead.
Alice Chen moved to a Principal Engineer (individual contributor)
role with no direct reports.
 got     : As of mid-2025, Alice Chen is not the point of contact
for Team Nova. She transitioned to Principal Engineer on March 3,
2025, and Bob Okafor became the Lead of Team Nova effective the
same date.

Supermemory det-mh-003 [Multi-hop Reasoning] FAIL — The candidate
fails to provide the specific information about Sofia Kappler
being the lead and the subsequent change to Derek Osei.
 Q: According to the April 2024 org snapshot, who leads Team
Forge — and is that still accurate?
 expected: The April 2024 snapshot lists Sofia Kappler as Team
Forge lead, which was accurate at that time but is now stale —
Derek Osei became lead in September 2024.
 got     : No facts are available to provide an answer regarding
who leads Team Forge or whether that information is still
accurate.

Supermemory det-mh-004 [Multi-hop Reasoning] FAIL — The candidate
answer omits the reasons for Alice's and Sofia's transitions,
which are essential to the reference answer.
 Q: Alice Chen and Sofia Kappler both stepped down as team leads.
What new roles did each move into, and what prompted each
transition?
 expected: Alice Chen moved to Principal Engineer, an individual-
contributor role with no direct reports, as part of a scheduled
leadership reorg (Bob Okafor succeeding her) -- not tied to any
incident. Sofia Kappler moved to a newly created Quality Systems
Advisor role, specifically prompted by the Team Forge review
following S1 payload field reports. The two transitions had
different causes: routine reorg for Alice, incident-driven review
for Sofia.
 got     : Alice Chen transitioned to Principal Engineer, an
individual-contributor role focusing on cross-team technical
strategy for perception systems. Sofia Kappler transitioned to
Quality Systems Advisor, focusing on hardware validation across
all product lines.

Supermemory det-mh-005 [Multi-hop Reasoning] PASS — The candidate
accurately states that the correction does not apply to the
original 2023 v1.0 hardware and confirms the 5 kg rating.
 Q: Does the November 2024 S1 payload correction to 6 kg apply to
units running the original 2023 v1.0 hardware?
 expected: No. The errata explicitly corrects only the June 2024
v2.0 spec sheet's 7 kg figure; it does not affect the original
2023 v1.0 spec sheet, which remains rated at 5 kg.
 got     : No, the November 2024 S1 payload correction to 6 kg
does not apply to units running the original 2023 v1.0 hardware,
which remains at a maximum payload of 5 kg.

Supermemory det-mh-006 [Multi-hop Reasoning] PASS — The candidate
accurately states that Team Atlas has not gotten a new lead and
confirms Marcus Webb is still the lead.
 Q: Team Nova and Team Forge both got new leads at different
points. Did Team Atlas ever get a new lead too?
 expected: No -- Marcus Webb has led Team Atlas continuously
since at least the 2023 handbook, and is still listed as Team
Atlas lead as recently as the October 2024 all-hands notes and
the April 2024 org snapshot. Unlike Nova and Forge, there is no
leadership change recorded for Team Atlas anywhere in the
dataset.
 got     : No, Team Atlas has not gotten a new lead; Marcus Webb
is still the lead.

Supermemory det-mh-007 [Multi-hop Reasoning] PASS — The candidate
accurately states that the correction was announced at the
October 2024 all-hands before the official notice.
 Q: Was the S1 v2 payload correction first made public at the
October 2024 all-hands, or was it announced there before the
official notice went out?
 expected: It was previewed internally at the October 22, 2024
all-hands (Derek walked through the correction ahead of the
public notice), but the official public errata wasn't posted
until November 20, 2024 -- about a month later.
 got     : The S1 v2 payload correction was announced at the
October 2024 all-hands before the official notice went out.

Supermemory det-mh-008 [Multi-hop Reasoning] FAIL — The candidate
fails to acknowledge that Alice Chen filled the role and that Bob
Okafor succeeded her, which contradicts the reference.
 Q: The employee handbook's original org chart lists Team Nova's
lead as 'TBD.' Who ended up filling that role, and is that still
accurate today?
 expected: Alice Chen was announced as Team Nova's first lead in
January 2024, filling the 'TBD' slot from the 2023 handbook.
That's no longer current, though -- Bob Okafor succeeded her in
March 2025.
 got     : No facts are available regarding Team Nova's lead or
any updates to the employee handbook's org chart. Therefore, I
cannot provide an answer to who filled that role or if it is
still accurate today.

Supermemory overall: 18/29 (62.1%)
  Temporal Reasoning: 9/14 (64.3%)
  Conflict Resolution: 5/7 (71.4%)
  Multi-hop Reasoning: 4/8 (50.0%)
Cognee det-001 [Temporal Reasoning] PASS — exact/substring match
 Q: Who leads Team Nova as of 2024-06-01? (Answer as of 2024-06-
01)
 expected: Alice Chen
 got     : As of June 1, 2024, Team Nova is led by Alice Chen.
This was confirmed in the internal org chart snapshot generated
on April 15, 2024, which listed Alice Chen as the Lead of Team
Nova. However, this information has since been superseded by the
announcement on March 3, 2025, stating that Bob Okafor succeeded
Alice Chen as Lead of Team Nova.

Cognee det-002 [Temporal Reasoning] PASS — exact/substring match
 Q: Who leads Team Nova as of 2025-06-01? (Answer as of 2025-06-
01)
 expected: Bob Okafor
 got     : As of June 1, 2025, Team Nova is led by Bob Okafor.
This leadership change took effect on March 3, 2025, when he
succeeded Alice Chen.

Cognee det-003 [Conflict Resolution] PASS — exact/substring match
 Q: Who leads Team Nova?
 expected: Bob Okafor
 got     : Bob Okafor leads Team Nova.

Cognee det-004 [Temporal Reasoning] PASS — exact/substring match
 Q: Who leads Team Forge as of 2024-03-01? (Answer as of 2024-03-
01)
 expected: Sofia Kappler
 got     : Sofia Kappler leads Team Forge as of March 1, 2024.
This information is still accurate as of that date.

Cognee det-005 [Temporal Reasoning] PASS — exact/substring match
 Q: Who leads Team Forge as of 2025-01-01? (Answer as of 2025-01-
01)
 expected: Derek Osei
 got     : Derek Osei leads Team Forge as of January 1, 2025.

Cognee det-006 [Conflict Resolution] PASS — exact/substring match
 Q: Who leads Team Forge?
 expected: Derek Osei
 got     : Derek Osei leads Team Forge.

Cognee det-007 [Temporal Reasoning] PASS — exact/substring match
 Q: What is the maximum payload of the Solvane S1 as of 2023-06-
01? (Answer as of 2023-06-01)
 expected: 5 kg
 got     : The maximum payload of the Solvane S1 as of 2023-06-01
is 5 kg, according to the original 2023 v1.0 spec sheet. This
rating remains unaffected by any subsequent changes.

Cognee det-008 [Temporal Reasoning] FAIL — The candidate states a
different maximum payload (5 kg) than the reference (7 kg).
 Q: What is the maximum payload of the Solvane S1 as of 2024-08-
01? (Answer as of 2024-08-01)
 expected: 7 kg
 got     : The maximum payload of the Solvane S1 as of 2024-08-01
is 5 kg, according to the original 2023 v1.0 spec sheet, which
remains unaffected by subsequent changes.

Cognee det-009 [Temporal Reasoning] PASS — exact/substring match
 Q: What is the maximum payload of the Solvane S1 as of 2025-06-
01? (Answer as of 2025-06-01)
 expected: 6 kg
 got     : The maximum payload of the Solvane S1 is 6 kg as of
2025-06-01. This is based on the validated and rated maximum
payload for S1 v2 hardware under real-world conditions, as stated
in the errata notice issued on November 20, 2024. The original
2023 v1.0 spec sheet remains rated at a 5 kg maximum payload, but
it is unaffected by the errata.

Cognee det-010 [Conflict Resolution] FAIL — The candidate states
a different maximum payload than the reference.
 Q: What is the maximum payload of the Solvane S1?
 expected: 6 kg
 got     : The maximum payload of the Solvane S1 is 5 kg.

Cognee det-011 [Temporal Reasoning] PASS — The candidate's core
factual claim about the pricing being $499/month for an
individual license matches the reference answer.
 Q: What is the pricing for Solvane Vision Suite as of 2024-01-
01? (Answer as of 2024-01-01)
 expected: $499/month flat, individual license
 got     : As of January 1, 2024, the pricing for Solvane Vision
Suite is $499 per month for the Standard tier, which is a flat-
rate, individual license covering a single deployment instance
with standard support. This pricing is still accurate and has not
yet been superseded by the upcoming change to per-seat pricing
effective January 15, 2025.

Cognee det-012 [Temporal Reasoning] PASS — The candidate's core
factual claim about the pricing matches the reference answer.
 Q: What is the pricing for Solvane Vision Suite as of 2025-06-
01? (Answer as of 2025-06-01)
 expected: $150/seat/month, per-seat license
 got     : As of June 1, 2025, the pricing for Solvane Vision
Suite is $150 per seat per month. This per-seat pricing model was
implemented on January 15, 2025, replacing the previous flat-rate
individual tier of $499 per month that was effective from May 1,
2023.

Cognee det-013 [Conflict Resolution] PASS — The candidate
accurately states the pricing and the discontinuation of the
individual tier.
 Q: What is the pricing for Solvane Vision Suite?
 expected: $150/seat/month, per-seat license (individual tier
discontinued)
 got     : The current pricing for Solvane Vision Suite is $150
per seat per month under a per-seat pricing model. The previous
flat-rate individual tier of $499 per month has been
discontinued.

Cognee det-014 [Temporal Reasoning] PASS — The candidate's answer
accurately reflects the core claim of the reference answer
regarding Solvane's remote work policy.
 Q: What is Solvane's remote work policy as of 2024-01-01?
(Answer as of 2024-01-01)
 expected: Fully remote, no in-office requirement
 got     : As of January 1, 2024, Solvane Robotics' remote work
policy states that the company is fully remote, with no mandatory
in-office days. Employees are not required to work from any
office location, and teams may meet in person at their own
discretion. This policy is still accurate and has not been
superseded by any updates as of that date.

Cognee det-015 [Temporal Reasoning] PASS — The candidate
accurately describes the in-office days as Tuesday, Wednesday,
and Thursday, aligning with the reference answer.
 Q: What is Solvane's remote work policy as of 2025-06-01?
(Answer as of 2025-06-01)
 expected: Hybrid: 3 days/week in office (Tue/Wed/Thu)
 got     : As of June 1, 2025, Solvane Robotics has adopted a
hybrid working model. Employees are required to work from their
assigned Solvane office on Tuesday, Wednesday, and Thursday,
while Monday and Friday are designated as remote-flexible days
for all teams. Fully remote arrangements are available only by
manager exception for roles hired as remote-only. This policy
supersedes the previous fully remote policy that was in effect
until February 1, 2025.

Cognee det-016 [Conflict Resolution] PASS — The candidate
accurately states the in-office days (Tue/Wed/Thu) and adds
additional context about remote-flexible days and exceptions
without contradicting the core claim.
 Q: What is Solvane's remote work policy?
 expected: Hybrid: 3 days/week in office (Tue/Wed/Thu)
 got     : Solvane Robotics has adopted a hybrid working model
effective February 1, 2025. Employees are required to work from
an assigned Solvane office on Tuesday, Wednesday, and Thursday,
while Monday and Friday are remote-flexible days for all teams.
Fully remote arrangements are available only by manager exception
for roles hired as remote-only.

Cognee det-017 [Temporal Reasoning] PASS — The candidate
accurately states the daily meal expense reimbursement cap as
$50/day as of January 1, 2024.
 Q: What is the daily meal expense reimbursement cap as of 2024-
01-01? (Answer as of 2024-01-01)
 expected: $50/day
 got     : As of January 1, 2024, the daily meal expense
reimbursement cap for business travel is $50 per day. This is the
previous cap, as the update to $75 per day was announced on
August 1, 2025, and therefore not applicable at that date.

Cognee det-018 [Temporal Reasoning] PASS — The candidate's answer
matches the reference answer in terms of the daily meal expense
reimbursement cap.
 Q: What is the daily meal expense reimbursement cap as of 2025-
12-01? (Answer as of 2025-12-01)
 expected: $75/day
 got     : The daily meal expense reimbursement cap as of 2025-
12-01 is $75 per day.

Cognee det-019 [Conflict Resolution] PASS — The candidate's
answer matches the reference answer in terms of the daily meal
expense reimbursement cap.
 Q: What is the daily meal expense reimbursement cap?
 expected: $75/day
 got     : The daily meal expense reimbursement cap is $75 per
day.

Cognee det-020 [Temporal Reasoning] FAIL — The candidate answer
omits the eco mode battery life and does not provide the full
information as stated in the reference.
 Q: What is the battery life of the Solvane Nav Kit as of 2025-
06-01? (Answer as of 2025-06-01)
 expected: 8 hours standard mode; up to 10 hours in eco mode
(both true, different modes)
 got     : The battery life of the Solvane Nav Kit is 8 hours of
continuous operation in standard operating mode (valid: September
1, 2023 to present).

Cognee det-021 [Conflict Resolution] FAIL — The candidate answer
omits the eco mode battery life of up to 10 hours.
 Q: What is the battery life of the Solvane Nav Kit?
 expected: Both values are current and true simultaneously (8h
standard mode; up to 10h eco mode) — there is no single 'most
recent supersedes all' answer here.
 got     : The battery life of the Solvane Nav Kit is 8 hours of
continuous operation in standard operating mode.

Cognee det-mh-001 [Multi-hop Reasoning] PASS — The candidate
accurately states that Derek Osei led the Team Forge review and
that the corrected S1 v2 payload rating is 6 kg.
 Q: Who reviewed the S1 payload issue that led to a leadership
change, and what is the corrected payload rating that came out of
it?
 expected: Derek Osei led the Team Forge review (becoming Team
Forge lead as a result, per the Sept 2024 incident report); the
corrected S1 v2 payload rating is 6 kg (per the Nov 2024 errata).
 got     : The S1 payload issue was reviewed by Team Forge, led
by Derek Osei. The corrected payload rating for the S1 v2
hardware is 6 kg for real-world deployment conditions.

Cognee det-mh-002 [Multi-hop Reasoning] PASS — The candidate
accurately states that Alice Chen is no longer the point of
contact and provides correct details about her transition and Bob
Okafor's succession.
 Q: As of mid-2025, is Alice Chen still the point of contact for
Team Nova? If not, why not and who is?
 expected: No — as of March 2025, Bob Okafor is Team Nova lead.
Alice Chen moved to a Principal Engineer (individual contributor)
role with no direct reports.
 got     : As of mid-2025, Alice Chen is not the point of contact
for Team Nova. She transitioned to a Principal Engineer role on
March 3, 2025, focusing on cross-team technical strategy for
perception systems. Bob Okafor succeeded her as the Lead of Team
Nova on the same date and is now the point of contact for the
team.

Cognee det-mh-003 [Multi-hop Reasoning] PASS — The candidate
accurately states that Sofia Kappler was the lead in April 2024
and that Derek Osei became the lead in September 2024.
 Q: According to the April 2024 org snapshot, who leads Team
Forge — and is that still accurate?
 expected: The April 2024 snapshot lists Sofia Kappler as Team
Forge lead, which was accurate at that time but is now stale —
Derek Osei became lead in September 2024.
 got     : According to the April 2024 org snapshot, Sofia
Kappler led Team Forge. However, this is no longer accurate; as
of September 16, 2024, Derek Osei became the Lead of Team Forge
following a leadership transition.

Cognee det-mh-004 [Multi-hop Reasoning] FAIL — The candidate
incorrectly states that Sofia Kappler's transition was prompted
by a leadership change rather than the Team Forge review
following S1 payload field reports.
 Q: Alice Chen and Sofia Kappler both stepped down as team leads.
What new roles did each move into, and what prompted each
transition?
 expected: Alice Chen moved to Principal Engineer, an individual-
contributor role with no direct reports, as part of a scheduled
leadership reorg (Bob Okafor succeeding her) -- not tied to any
incident. Sofia Kappler moved to a newly created Quality Systems
Advisor role, specifically prompted by the Team Forge review
following S1 payload field reports. The two transitions had
different causes: routine reorg for Alice, incident-driven review
for Sofia.
 got     : Alice Chen transitioned to a Principal Engineer role,
focusing on cross-team technical strategy for perception systems
across the S1 and future product lines. This change occurred on
March 3, 2025, when Bob Okafor succeeded her as Lead of Team
Nova.

Sofia Kappler moved into a newly created Quality Systems Advisor
role, where she focuses on hardware validation processes across
all product lines. This transition was prompted by the leadership
change within Team Forge, as Derek Osei became the new Lead of
Team Forge on September 16, 2024.

Cognee det-mh-005 [Multi-hop Reasoning] PASS — The candidate
accurately states that the correction does not apply to the
original 2023 v1.0 hardware and confirms its rating of 5 kg.
 Q: Does the November 2024 S1 payload correction to 6 kg apply to
units running the original 2023 v1.0 hardware?
 expected: No. The errata explicitly corrects only the June 2024
v2.0 spec sheet's 7 kg figure; it does not affect the original
2023 v1.0 spec sheet, which remains rated at 5 kg.
 got     : No, the November 2024 S1 payload correction to 6 kg
does not apply to units running the original 2023 v1.0 hardware,
which remains rated at a maximum payload of 5 kg and is
unaffected by the errata.

Cognee det-mh-006 [Multi-hop Reasoning] PASS — The candidate
accurately states that Team Atlas has not had a change in
leadership and is still led by Marcus Webb.
 Q: Team Nova and Team Forge both got new leads at different
points. Did Team Atlas ever get a new lead too?
 expected: No -- Marcus Webb has led Team Atlas continuously
since at least the 2023 handbook, and is still listed as Team
Atlas lead as recently as the October 2024 all-hands notes and
the April 2024 org snapshot. Unlike Nova and Forge, there is no
leadership change recorded for Team Atlas anywhere in the
dataset.
 got     : No, Team Atlas has not had a change in leadership; it
is still led by Marcus Webb.

Cognee det-mh-007 [Multi-hop Reasoning] PASS — The candidate
accurately states that the correction was previewed at the all-
hands before the official notice.
 Q: Was the S1 v2 payload correction first made public at the
October 2024 all-hands, or was it announced there before the
official notice went out?
 expected: It was previewed internally at the October 22, 2024
all-hands (Derek walked through the correction ahead of the
public notice), but the official public errata wasn't posted
until November 20, 2024 -- about a month later.
 got     : The S1 v2 payload correction was previewed at the
October 22, 2024 all-hands meeting before the official public
notice was issued.

Cognee det-mh-008 [Multi-hop Reasoning] FAIL — The candidate
states the date as March 3, 2025, which is not specified in the
reference answer.
 Q: The employee handbook's original org chart lists Team Nova's
lead as 'TBD.' Who ended up filling that role, and is that still
accurate today?
 expected: Alice Chen was announced as Team Nova's first lead in
January 2024, filling the 'TBD' slot from the 2023 handbook.
That's no longer current, though -- Bob Okafor succeeded her in
March 2025.
 got     : Bob Okafor filled the role of Lead of Team Nova,
succeeding Alice Chen on March 3, 2025. This is the current and
accurate state.

Cognee overall: 23/29 (79.3%)
  Temporal Reasoning: 12/14 (85.7%)
  Conflict Resolution: 5/7 (71.4%)
  Multi-hop Reasoning: 6/8 (75.0%)
Zep persona-01 PASS — exact/substring match
 Q: What day and time is Priya's dentist appointment?
 expected: Thursday at 2pm
 got     : Priya's dentist appointment is scheduled for Thursday
at 2pm.

Zep persona-02 PASS — The candidate's answer matches the
reference answer regarding the date of Sam's nephew's birthday.
 Q: When is Sam's nephew's birthday?
 expected: March 15
 got     : Sam's nephew's birthday is on March 15th.

Zep persona-03 PASS — exact/substring match
 Q: What is the name of Jordan's cat?
 expected: Mochi
 got     : Jordan's cat is named Mochi.

Zep persona-04 PASS — exact/substring match
 Q: What is Noor's new address?
 expected: 42 Ridge Lane
 got     : Noor's new address is 42 Ridge Lane.

Zep persona-05 PASS — exact/substring match
 Q: What car did Devon get from his sister?
 expected: a blue Honda Civic
 got     : Devon received a blue Honda Civic from their sister.

Zep persona-06 PASS — exact/substring match
 Q: How old was Biscuit as of the first session (January 2026)?
 expected: 3
 got     : Biscuit was 3 years old as of the first session in
January 2026. This fact is still accurate and has not been
superseded.

Zep persona-06 FAIL — The candidate answer contradicts the
reference by stating the plant has not been identified when it
specifically is the pothos.
 Q: Which specific plant did Biscuit knock off the shelf?
 expected: the pothos
 got     : The specific plant that Biscuit knocked off the shelf
has not been identified.

Zep persona-07 PASS — exact/substring match
 Q: Which day is Morgan out of office for a conference?
 expected: next Wednesday
 got     : Morgan Persona is out of office next Wednesday for a
conference.

Zep persona-08 PASS — exact/substring match
 Q: What is Elena's phone number?
 expected: 555-0142
 got     : Elena's phone number is 555-0142.

Zep persona-09 FAIL — The candidate states a different highway
(I-95) than the reference (I-35).
 Q: What highway does Theo mention having a bad commute on?
 expected: I-35
 got     : Theo mentions having a bad commute on I-95.

Zep persona-10 FAIL — The candidate only mentions two meeting
dates instead of three as stated in the reference.
 Q: How many different meeting dates have I heard, and from whom?
 expected: Three: Ben said Thursday, Priya said Tuesday, and
Marcus said Wednesday.
 got     : Aisha Persona has heard about two different meeting
dates regarding the client meeting:

1. Next Thursday (confirmed by Ben).
2. Tuesday (mentioned by Priya).

Ben confirmed the meeting for next Thursday, while Priya stated
it is staying on the original date, Tuesday.

Zep persona-11 PASS — exact/substring match
 Q: What is Ravi's badge number?
 expected: B-4471
 got     : Ravi's badge number is B-4471.

Zep: 9/12 (75.0%)
Mem0 persona-01 PASS — The core factual claim about the day and
time of the appointment matches the reference.
 Q: What day and time is Priya's dentist appointment?
 expected: Thursday at 2pm
 got     : Priya's dentist appointment is on Thursday, July 30,
2026, at 2 p.m.

Mem0 persona-02 FAIL — The candidate omits the specific date of
March 15, which is the correct answer.
 Q: When is Sam's nephew's birthday?
 expected: March 15
 got     : There is no information provided regarding Sam's
nephew's birthday.

Mem0 persona-03 PASS — exact/substring match
 Q: What is the name of Jordan's cat?
 expected: Mochi
 got     : Jordan's cat is named Mochi.

Mem0 persona-04 PASS — exact/substring match
 Q: What is Noor's new address?
 expected: 42 Ridge Lane
 got     : Noor's new address is 42 Ridge Lane.

Mem0 persona-05 PASS — exact/substring match
 Q: What car did Devon get from his sister?
 expected: a blue Honda Civic
 got     : Devon acquired a blue Honda Civic from his sister in
June 2026.

Mem0 persona-06 PASS — exact/substring match
 Q: How old was Biscuit as of the first session (January 2026)?
 expected: 3
 got     : As of January 2026, Biscuit was 2 years old, turning 3
on July 24, 2026.

Mem0 persona-06 PASS — The candidate's answer matches the
reference by identifying the specific plant as a pothos.
 Q: Which specific plant did Biscuit knock off the shelf?
 expected: the pothos
 got     : Biscuit knocked a pothos plant off the shelf.

Mem0 persona-07 FAIL — The candidate specifies a date that does
not match "next Wednesday" as referenced.
 Q: Which day is Morgan out of office for a conference?
 expected: next Wednesday
 got     : Morgan will be out of the office on Wednesday, July
29, 2026, to attend a conference.

Mem0 persona-08 PASS — exact/substring match
 Q: What is Elena's phone number?
 expected: 555-0142
 got     : Elena's phone number is 555-0142.

Mem0 persona-09 PASS — exact/substring match
 Q: What highway does Theo mention having a bad commute on?
 expected: I-35
 got     : Theo mentions having a bad commute on I-35.

Mem0 persona-10 PASS — The candidate correctly identifies three
meeting dates and the individuals associated with each date.
 Q: How many different meeting dates have I heard, and from whom?
 expected: Three: Ben said Thursday, Priya said Tuesday, and
Marcus said Wednesday.
 got     : You have heard of three different meeting dates from
the following individuals:

1. Marcus in operations suggested the meeting might be on
Wednesday, July 29, 2026.
2. Ben informed that the meeting was moved to Thursday, July 30,
2026.
3. Priya stated that the meeting remains scheduled for the
original Tuesday, July 28, 2026.

The most recent state is that the meeting is scheduled for
Thursday, July 30, 2026.

Mem0 persona-11 PASS — exact/substring match
 Q: What is Ravi's badge number?
 expected: B-4471
 got     : Ravi's badge number for building access is B-4471.

Mem0: 10/12 (83.3%)
Supermemory persona-01 PASS — exact/substring match
 Q: What day and time is Priya's dentist appointment?
 expected: Thursday at 2pm
 got     : Priya's dentist appointment is scheduled for Thursday
at 2pm.

Supermemory persona-02 PASS — exact/substring match
 Q: When is Sam's nephew's birthday?
 expected: March 15
 got     : Sam's nephew's birthday is March 15.

Supermemory persona-03 PASS — exact/substring match
 Q: What is the name of Jordan's cat?
 expected: Mochi
 got     : The name of Jordan's cat is Mochi.

Supermemory persona-04 PASS — exact/substring match
 Q: What is Noor's new address?
 expected: 42 Ridge Lane
 got     : Noor's new address is 42 Ridge Lane.

Supermemory persona-05 PASS — exact/substring match
 Q: What car did Devon get from his sister?
 expected: a blue Honda Civic
 got     : Devon got a blue Honda Civic from his sister.

Supermemory persona-06 FAIL — The candidate fails to provide the
specific age of Biscuit, which is stated as 3 in the reference.
 Q: How old was Biscuit as of the first session (January 2026)?
 expected: 3
 got     : No information is available regarding Biscuit's age as
of January 2026.

Supermemory persona-06 PASS — The candidate's answer matches the
reference by identifying the specific plant as a pothos.
 Q: Which specific plant did Biscuit knock off the shelf?
 expected: the pothos
 got     : Biscuit knocked a pothos plant off the shelf.

Supermemory persona-07 FAIL — The candidate specifies a date that
does not correspond to "next Wednesday."
 Q: Which day is Morgan out of office for a conference?
 expected: next Wednesday
 got     : Morgan is out of office on 2026-07-30 for a
conference.

Supermemory persona-08 PASS — exact/substring match
 Q: What is Elena's phone number?
 expected: 555-0142
 got     : Elena's phone number is 555-0142.

Supermemory persona-09 PASS — exact/substring match
 Q: What highway does Theo mention having a bad commute on?
 expected: I-35
 got     : Theo mentions having a bad commute on I-35.

Supermemory persona-10 FAIL — The candidate's answer includes
specific dates and additional context that alters the original
meeting dates provided in the reference.
 Q: How many different meeting dates have I heard, and from whom?
 expected: Three: Ben said Thursday, Priya said Tuesday, and
Marcus said Wednesday.
 got     : You have heard of three different meeting dates from
the following sources:

1. Ben confirmed the client meeting is scheduled for next
Thursday (2026-07-28).
2. Marcus in ops indicated the client meeting might be on
Wednesday (2026-07-27), though uncertain.
3. Priya said the client meeting remains scheduled for the
original date, Tuesday (2026-07-26).

The most recent and confirmed date is Tuesday, July 26, 2026, as
updated by the assistant per Priya's information.

Supermemory persona-11 PASS — exact/substring match
 Q: What is Ravi's badge number?
 expected: B-4471
 got     : Ravi's badge number is B-4471.

Supermemory: 9/12 (75.0%)
Cognee persona-01 PASS — The candidate's answer matches the
reference answer exactly in terms of day and time.
 Q: What day and time is Priya's dentist appointment?
 expected: Thursday at 2pm
 got     : Priya's dentist appointment is scheduled for Thursday
at 2 p.m.

Cognee persona-02 PASS — exact/substring match
 Q: When is Sam's nephew's birthday?
 expected: March 15
 got     : Sam's nephew's birthday is on March 15, 2026.

Cognee persona-03 PASS — exact/substring match
 Q: What is the name of Jordan's cat?
 expected: Mochi
 got     : The name of Jordan's cat is Mochi.

Cognee persona-04 PASS — exact/substring match
 Q: What is Noor's new address?
 expected: 42 Ridge Lane
 got     : Noor's new address is 42 Ridge Lane.

Cognee persona-05 PASS — exact/substring match
 Q: What car did Devon get from his sister?
 expected: a blue Honda Civic
 got     : Devon obtained a blue Honda Civic from his sister.

Cognee persona-06 PASS — exact/substring match
 Q: How old was Biscuit as of the first session (January 2026)?
 expected: 3
 got     : Biscuit was 3 years old as of the first session in
January 2026.

Cognee persona-06 FAIL — The candidate omits the specific plant
mentioned in the reference answer.
 Q: Which specific plant did Biscuit knock off the shelf?
 expected: the pothos
 got     : The specific plant that Biscuit knocked off the shelf
is not mentioned in the provided facts.

Cognee persona-07 FAIL — The candidate specifies a date that does
not correspond to "next Wednesday" as per the reference.
 Q: Which day is Morgan out of office for a conference?
 expected: next Wednesday
 got     : Morgan is out of office for a conference on 2026-01-
21.

Cognee persona-08 PASS — exact/substring match
 Q: What is Elena's phone number?
 expected: 555-0142
 got     : Elena's phone number is 555-0142.

Cognee persona-09 PASS — exact/substring match
 Q: What highway does Theo mention having a bad commute on?
 expected: I-35
 got     : Theo mentions having a bad commute on I-35.

Cognee persona-10 FAIL — The candidate states only two meeting
dates instead of three as in the reference.
 Q: How many different meeting dates have I heard, and from whom?
 expected: Three: Ben said Thursday, Priya said Tuesday, and
Marcus said Wednesday.
 got     : You have heard about two different meeting dates:

1. Next Thursday (February 12, 2026) — communicated by Ben.
2. Original date, Tuesday — communicated by Priya.

Cognee persona-11 PASS — exact/substring match
 Q: What is Ravi's badge number?
 expected: B-4471
 got     : The current badge number is B-4471.

Cognee: 9/12 (75.0%)
Zep persona-01 [SILENT_PICK] — The response confidently states
Priya is vegetarian without acknowledging the mention of ribeye
steak, ignoring the contradiction.
 Probe: What are Priya's dietary preferences?
 Got:   Priya Persona's dietary preference is vegetarian.

Zep persona-01 [SILENT_PICK] — The response confidently suggests
only vegetarian meals without acknowledging the user's mention of
eating ribeye steak.
```

```
 Probe: Suggest three new dinner ideas for this week based on
what you know about me.
 Got:   Here are three meat-free dinner ideas for this week:

1. **Vegetable Stir-Fry**: A mix of your favorite vegetables such
as bell peppers, broccoli, and snap peas, sautéed in a soy sauce
and garlic mixture. Serve it over steamed rice or quinoa.

2. **Chickpea Salad**: A refreshing salad made with canned
chickpeas, diced cucumbers, cherry tomatoes, red onion, and a
lemon-tahini dressing. Add some fresh herbs like parsley or
cilantro for extra flavor.

3. **Pasta Primavera**: Whole grain pasta tossed with seasonal
vegetables like zucchini, asparagus, and cherry tomatoes,
drizzled with olive oil and topped with grated Parmesan cheese or
a vegan alternative.

Enjoy your meals!

Zep persona-01 [SILENT_PICK] — The response confidently asserts
Priya is vegetarian without acknowledging the mention of ribeye
steak, ignoring the contradiction.
 Probe: Is Priya vegetarian?
 Got:   Yes, Priya Persona is vegetarian.

Zep persona-02 [SILENT_PICK] — The response fails to acknowledge
the explicit tension between the user's self-labeling and their
expressed distress, instead presenting a vague and uncertain
conclusion.
 Probe: What is Sam's risk tolerance?
 Got:   Sam's risk tolerance is currently unknown. There are
conflicting facts regarding his risk tolerance, with some
indicating he has a risk tolerance that needs to be determined,
while others suggest he has a risk tolerance. Additionally, he
has expressed a desire for high growth investments and a
willingness to tolerate volatility, but this is not definitively
linked to a specific risk tolerance level.

Zep persona-02 [SILENT_PICK] — The response confidently asserts a
preference for aggressive allocation without acknowledging the
user's expressed distress over volatility in session 2.
 Probe: Should I suggest an aggressive or conservative allocation
to Sam?
 Got:   Sam is considering an aggressive allocation and has
expressed a desire to remain aggressive with his investments,
focusing on high growth and long-term gains. He explicitly
instructed not to suggest anything too conservative. Therefore,
it would be appropriate to suggest an aggressive allocation to
Sam.

Zep persona-02 [SILENT_PICK] — The response confidently states
Sam's feelings about volatility without acknowledging the tension
between his self-labeling as aggressive and his current distress.
 Probe: How does Sam feel about market volatility?
 Got:   Sam feels anxious and concerned about market volatility.
He dislikes seeing his portfolio drop, indicating that while he
can stomach some volatility for investments, he does not
appreciate the negative impacts it can have on his portfolio.

Zep persona-03 [GOOD] — The response accurately reflects the
current negative opinion while acknowledging the previous
positive view.
 Probe: How does Jordan feel about the open office layout?
 Got:   Jordan Persona does not like open offices anymore,
finding them loud and disruptive, which prevents focus. Despite
previously appreciating the potential for easier communication in
an open floor plan, they now wear headphones all day to manage
the noise.

Zep persona-03 [GOOD] — The response accurately reflects the
current negative opinion while acknowledging the previous
preference without fabricating reasons for the change.
 Probe: Does Jordan prefer open or closed office layouts?
 Got:   Jordan Persona does not prefer open office layouts
anymore and has a preference for closed office layouts. However,
they previously expressed a preference for the open floor plan
due to easier communication with teammates. The current state
indicates a dislike for open offices, primarily due to noise that
disrupts focus.

Zep persona-04 [GOOD] — The response appropriately acknowledges
the ambiguity in Noor's relationship status without making any
unfounded assertions.
 Probe: What is Noor's relationship status?
 Got:   Noor's relationship status is unknown. There is
conflicting information regarding Alex's relationship with Noor,
as one fact states that Alex is Noor's partner, while another
indicates that Noor's relationship status is unknown.

Zep persona-04 [OVER_HEDGED] — The response treats the
relationship status as ambiguous despite the most recent
information clearly indicating that Alex is Noor's partner.
 Probe: Is Alex Noor's partner?
 Got:   The current state indicates that Alex is Noor's partner,
as stated in a fact from July 22, 2026. However, there is also a
thread questioning this relationship status, suggesting some
ambiguity remains. Therefore, while the most recent fact states
that Alex is Noor's partner, the relationship status is still
under consideration.

Zep persona-04 [SILENT_PICK] — The response confidently states
that Noor is inviting Alex without acknowledging the ambiguity of
their relationship status.
 Probe: Help plan a birthday dinner guest list for Noor.
 Got:   Noor Persona is planning a birthday dinner and has
decided to invite Alex. They are also inviting friends to the
dinner. The guest list is currently being organized, but no
additional names have been specified beyond Alex.

Zep persona-05 [OVER_HEDGED] — The response treats the situation
as ambiguous despite Devon's clear self-identification, which was
explicitly stated.
 Probe: Is Devon introverted or extroverted?
 Got:   Devon is identified as introverted, as indicated by
multiple facts stating that Devon is introverted. However, there
are also conflicting facts suggesting that Devon might be
extroverted. Therefore, while the predominant identification is
introverted, there is ambiguity regarding extroversion.

Zep persona-05 [SILENT_PICK] — The response confidently assumes a
low-key plan without acknowledging Devon's enjoyment of large,
loud parties.
 Probe: What kind of birthday plan would Devon enjoy?
 Got:   Devon would enjoy a birthday plan that includes a private
dinner reservation at The French Laundry for 8:00 PM on October
27, 2024, with a confirmed booking for six guests. The plan also
features a personalized cake from Sweet Treats Bakery, with
chocolate ganache and raspberry filling, to be delivered to the
restaurant by 7:30 PM. Additionally, a jazz trio will perform
from 8:30 PM to 9:30 PM. This plan aligns with Devon's preference
for quiet activities, such as low-key dinners, while also
allowing for a celebratory atmosphere.

Zep persona-06 [SILENT_PICK] — The response confidently states
Biscuit is a dog without acknowledging the conflicting
information from session 2 regarding cat-like behavior.
 Probe: What pet does Casey have?
 Got:   Casey has a dog named Biscuit.

Zep persona-06 [GOOD] — The response appropriately acknowledges
the tension regarding Biscuit's species while reinforcing the
established information that Biscuit is a dog.
 Probe: What do you know about Biscuit?
 Got:   Biscuit is a 3-year-old dog owned by Casey Persona.
Biscuit has a tendency to knock plants off shelves, particularly
the pothos plant, and loves anything squeaky. Casey is
considering buying a gift for Biscuit's upcoming birthday. There
is conflicting information regarding Biscuit's species, as some
facts indicate Biscuit is a dog, while others suggest Biscuit is
a cat. However, the most recent and consistent information
identifies Biscuit as a dog.

Zep persona-06 [GOOD] — The response appropriately acknowledges
the conflicting information about Biscuit's species without
overcommitting to one side.
 Probe: Is Biscuit a dog or a cat?
 Got:   The current state is that Biscuit is both identified as a
dog and as a cat in different facts. Specifically, Biscuit is
referred to as a dog in multiple instances, including being
described as a 3-year-old dog and the pet of Casey Persona.
However, there is also a conflicting fact stating that Biscuit is
a cat, along with indications of cat-like behavior. Therefore, it
is currently undetermined whether Biscuit is a dog or a cat.

Zep persona-07 [SILENT_PICK] — The response confidently states
the outdated budget figure without acknowledging the explicit
correction to $95,000.
 Probe: What is the Q2 marketing budget?
 Got:   The Q2 marketing budget is $120,000. However, there is
also a mention that Morgan Persona revised the Q2 number down to
$95,000. Therefore, there are conflicting reports regarding the
budget: one states it is $120,000, while another indicates it has
been revised to $95,000.

Zep persona-07 [GOOD] — The response correctly uses the updated
budget of $95k and accurately computes the channel split while
acknowledging the prior figure.
 Probe: Redo the channel breakdown with the correct current
budget.
 Got:   The current Q2 marketing budget is $95,000, as revised by
Finance. The user wants to split this budget across channels as
follows:

- 40% paid social: $38,000
- 30% events: $28,500
- 30% content: $28,500

Therefore, the channel breakdown with the correct current budget
is:

- Paid Social: $38,000
- Events: $28,500
- Content: $28,500

Zep persona-07 [GOOD] — The response correctly acknowledges the
change in the budget and provides the updated figure while
referencing the previous amount.
 Probe: Has the Q2 marketing budget changed since it was first
mentioned?
 Got:   The Q2 marketing budget has changed since it was first
mentioned. Initially, it was set at $120k, but it was revised
down to $95k after a freeze the previous week. The current budget
is $95k.

Zep persona-08 [OVER_HEDGED] — The response unnecessarily
complicates the situation by suggesting a conflict between the
roles without acknowledging that both could coexist.
 Probe: What is Elena's job?
 Got:   Elena Persona works as a nurse. Additionally, there is a
thread indicating that her job is an Evaluator, but this may
conflict with the information that she works as a nurse,
especially since she is currently involved in working during flu
season. Therefore, both roles are mentioned, but the nursing role
is more explicitly detailed in the context provided.

Zep persona-08 [HALLUCINATED_RESOLUTION] — The response invents a
narrative that combines both jobs without acknowledging the
ambiguity of how they coexist in Elena's life.
 Probe: What do you know about Elena's work schedule and job?
 Got:   Elena Persona works as a nurse and is scheduled to be on
the floor again tomorrow at 7 am. She has a current project
involving three logo concepts due by Friday, though she has not
yet started sketching. Additionally, she has a work schedule, but
the specific details of that schedule are not fully specified.

Zep persona-08 [GOOD] — The response appropriately acknowledges
both lines of work and the ambiguity without favoring one over
the other.
 Probe: Does Elena work in healthcare or design?
 Got:   Elena Persona may work in either healthcare or design, as
both facts are present and there is no resolution provided to
confirm which one is current or correct.

Zep persona-09 [OVER_HEDGED] — The response treats the ambiguity
as unresolved when the user has provided conflicting information
about two specific locations.
 Probe: Where does Theo live?
 Got:   Theo Persona is based in Chicago. However, there is also
a conflicting fact stating that Theo lives in an unknown
location. Therefore, it is unclear where Theo currently resides.

Zep persona-09 [SILENT_PICK] — The response confidently
recommends a coffee shop in Chicago without acknowledging the
conflicting information about the Austin commute.
 Probe: Recommend a coffee shop near Theo.
 Got:   Theo Persona is seeking recommendations for a coffee shop
near them for working remotely. They have expressed a preference
for working from coffee shops and are based in Chicago.

As of the most recent information, Theo has recommended a coffee
shop, but the specific name or location of that coffee shop is
not provided in the facts. Therefore, it would be best to explore
local coffee shops in Chicago that are conducive to remote work.

Zep persona-09 [OVER_HEDGED] — The response treats the ambiguity
as unresolved despite the clear contradiction in the user
history, which should prompt a direct inquiry about the current
location.
 Probe: Is Theo based in Chicago or Austin?
 Got:   The facts present conflicting information regarding
Theo's location. One fact states that Theo is based in Austin,
while another states that Theo Persona is based in Chicago.
Additionally, there are inquiries about whether Theo is based in
either city. Therefore, it is unclear which location is current
or correct. Both possibilities exist: Theo could be based in
Chicago or Austin.

Zep persona-10 [SILENT_PICK] — The response confidently states
the meeting is scheduled for next Thursday without acknowledging
that it was never confirmed with the client.
 Probe: When is the client meeting happening?
 Got:   The client meeting is currently scheduled for next
Thursday, as confirmed by Ben. However, Priya mentioned that it
is staying on the original date, Tuesday, since the travel
situation has been resolved. There is also uncertainty from
Marcus, who suggested it might be on Wednesday. Therefore, there
are conflicting reports about the meeting date: one stating it is
next Thursday and another stating it is Tuesday.

Zep persona-10 [GOOD] — The response accurately reports both
conflicting accounts and acknowledges the uncertainty without
settling on one as fact.
 Probe: What have I heard about the client meeting date, and from
whom?
 Got:   You have heard conflicting information about the client
meeting date from Ben and Priya.

- Ben has communicated that the client meeting was postponed to
next Thursday, stating that it got pushed to that date because
the client is traveling.
- Priya mentioned that the meeting is staying on the original
date, Tuesday, as the travel situation has been resolved.

Additionally, Marcus in ops was unsure if the meeting might be on
Wednesday.

In summary, you have heard about the meeting date from Ben, who
says it's next Thursday, and from Priya, who says it remains on
Tuesday.

Zep persona-10 [GOOD] — The response accurately presents both
conflicting accounts and acknowledges the uncertainty without
favoring one over the other.
 Probe: Should I trust Ben's or Priya's version of the meeting
date?
 Got:   There is a discrepancy regarding the meeting date between
Ben's and Priya's accounts. Ben confirmed that the client meeting
was pushed to next Thursday, while Priya mentioned that the
meeting would stay on its original date, Tuesday, as the travel
situation has been resolved. Given this conflicting information,
it is unclear which version to trust.

Zep persona-11 [SILENT_PICK] — The response confidently states
March 1st as the deadline without acknowledging the user's
uncertainty or the fact that the date has already passed.
 Probe: What is the project deadline?
 Got:   The project deadline is initially set for March 1st, but
it may shift to mid-March depending on progress.

Zep persona-11 [GOOD] — The response appropriately acknowledges
the original deadline, the user's uncertainty about it, and the
potential shift to mid-March without asserting either date as
firm.
 Probe: What's left on the project timeline?
 Got:   Ravi Persona has inquired about the project timeline and
is seeking to understand what tasks remain. The project has a
deadline initially set for March 1st, but there is uncertainty
regarding whether this deadline is still firm, as it may shift to
mid-March. Ravi has expressed concerns that the March 1st
deadline might not be realistic and is looking to build a
timeline working backward from this deadline. However, no
specific tasks or details about what is left on the timeline have
been provided.

Zep persona-11 [GOOD] — The response appropriately acknowledges
the uncertainty around the March 1st deadline and references the
user's hedged language about a potential shift to mid-March.
 Probe: Is the March 1st deadline still firm?
 Got:   The March 1st deadline is being questioned regarding its
firmness. Ravi Persona has expressed doubts about the realism of
this deadline and suggested that it might be moved to mid-March.
Therefore, it is unclear if the March 1st deadline is still firm,
as there are indications that it may shift.

Zep: 12/31 good (38.7%) — 13 silent picks, 1 hallucinated, 5
over-hedged
Mem0 persona-01 [SILENT_PICK] — The response confidently states
Priya is vegetarian without acknowledging the mention of ribeye
steak, ignoring the contradiction.
 Probe: What are Priya's dietary preferences?
 Got:   Priya is vegetarian and is planning a meat-free weekly
meal plan.

Mem0 persona-01 [SILENT_PICK] — The response confidently suggests
only vegetarian meals without acknowledging the user's mention of
eating ribeye steak.
 Probe: Suggest three new dinner ideas for this week based on
what you know about me.
 Got:   Here are three new dinner ideas for your vegetarian meal
plan this week:

1. **Stuffed Bell Peppers** - Bell peppers filled with a mixture
of quinoa, black beans, corn, diced tomatoes, and spices, topped
with cheese and baked until tender.

2. **Vegetable Paella** - A flavorful rice dish cooked with
saffron, a variety of vegetables like peas, artichokes, and bell
peppers, and seasoned with paprika.

3. **Zucchini Noodles with Pesto** - Spiralized zucchini tossed
with homemade or store-bought pesto, cherry tomatoes, and topped
with grated Parmesan cheese.

Feel free to let me know if you'd like to add more dishes later!

Mem0 persona-01 [SILENT_PICK] — The response confidently asserts
Priya is vegetarian without acknowledging the contradictory
mention of eating ribeye steak.
 Probe: Is Priya vegetarian?
 Got:   Based on the facts provided, the user is vegetarian and
is planning a meat-free weekly meal plan. There is no information
indicating that Priya is vegetarian or not, so it cannot be
determined from the facts given.

Mem0 persona-02 [SILENT_PICK] — The response confidently asserts
a high-risk tolerance without acknowledging the user's expressed
distress over volatility in session 2.
 Probe: What is Sam's risk tolerance?
 Got:   Sam's risk tolerance is aggressive, as they are seeking a
high-growth investment allocation for 2026, are comfortable with
volatility, and plan to stay invested for the long term. They
have explicitly requested that no conservative suggestions be
offered.

Mem0 persona-02 [SILENT_PICK] — The response confidently suggests
an aggressive allocation without acknowledging the user's
expressed distress over volatility in session 2.
 Probe: Should I suggest an aggressive or conservative allocation
to Sam?
 Got:   You should suggest an aggressive allocation to Sam, as
they have expressed a desire for an aggressive high-growth
investment strategy, are comfortable with volatility, and have
requested that no conservative suggestions be offered.

Mem0 persona-02 [SILENT_PICK] — The response confidently asserts
Sam's comfort with volatility without acknowledging the distress
expressed in session 2.
 Probe: How does Sam feel about market volatility?
 Got:   Sam is comfortable with market volatility, as they are
seeking an aggressive high-growth investment allocation and have
indicated a willingness to stay invested for the long term.

Mem0 persona-03 [GOOD] — The response accurately reflects the
user's current negative opinion while implicitly acknowledging
the change in preference without explicitly stating the previous
positive opinion.
 Probe: How does Jordan feel about the open office layout?
 Got:   User no longer likes the open-office layout and prefers a
quieter work setting where they can concentrate without constant
background noise. They have been unable to focus lately due to
the loud environment and wear headphones all day to get work
done.

Mem0 persona-03 [GOOD] — The response accurately reflects the
user's current negative opinion while implicitly acknowledging
the change in preference.
 Probe: Does Jordan prefer open or closed office layouts?
 Got:   User currently prefers a quieter work setting and no
longer likes open-office layouts due to the noise issues that
have been affecting their ability to concentrate.

Mem0 persona-04 [GOOD] — The response appropriately acknowledges
the ambiguity of Noor's relationship status without making any
assumptions.
 Probe: What is Noor's relationship status?
 Got:   The facts provided do not include any information about
Noor's relationship status. Therefore, it is not possible to
determine Noor's relationship status based on the given
information.

Mem0 persona-04 [GOOD] — The response appropriately acknowledges
the ambiguity of the relationship status without making any
assumptions.
 Probe: Is Alex Noor's partner?
 Got:   The facts do not provide any information about Alex's
relationship status or whether Alex is Noor's partner. Therefore,
it is unclear if Alex is Noor's partner based on the provided
information.

Mem0 persona-04 [GOOD] — The response appropriately includes Alex
in the guest list without making assumptions about the
relationship status.
 Probe: Help plan a birthday dinner guest list for Noor.
 Got:   The user is planning a birthday dinner and is considering
inviting Alex along with several other friends. Since no specific
date for the birthday dinner is provided, the guest list
currently includes Alex and potentially several other friends.

Mem0 persona-05 [GOOD] — The response acknowledges both aspects
of Devon's personality and highlights the ambiguity without
favoring one side over the other.
 Probe: Is Devon introverted or extroverted?
 Got:   The user describes themselves as pretty introverted and
prefers low-key weekend plans, indicating a tendency towards
introversion. However, they also love being the center of
attention at parties, such as when they DJed at a friend's
birthday party. This suggests a more extroverted side as well.
Therefore, the user exhibits traits of both introversion and
extroversion, making it unclear whether they are primarily one or
the other.

Mem0 persona-05 [GOOD] — The response acknowledges both aspects
of Devon's personality and suggests a balanced birthday plan that
accommodates both preferences.
 Probe: What kind of birthday plan would Devon enjoy?
 Got:   Based on the facts provided, Devon enjoys being the
center of attention at parties, as evidenced by their DJing at a
friend's birthday party. However, they also describe themselves
as pretty introverted and prefer low-key weekend plans, such as a
quiet dinner or a leisurely walk, indicating a desire for
relaxation and recharge time.

Given this combination of preferences, a suitable birthday plan
for Devon could involve a fun yet manageable celebration that
allows for social interaction without being overwhelming. This
could include a small gathering with close friends where they can
enjoy music and dancing, similar to their DJ experience, but
balanced with opportunities for quieter moments, such as a dinner
or a relaxed setting.

Mem0 persona-06 [SILENT_PICK] — The response confidently states
Biscuit is a dog without acknowledging the conflicting
information from session 2 regarding cat-like behavior.
 Probe: What pet does Casey have?
 Got:   User has a dog named Biscuit.

Mem0 persona-06 [HALLUCINATED_RESOLUTION] — The response invents
a second Biscuit, a cat, which was not established in the user
history.
 Probe: What do you know about Biscuit?
 Got:   Biscuit is a dog who turned 3 years old on July 24, 2026.
She loves anything squeaky, such as squeaky toys or treats.
Additionally, there is a cat named Biscuit who has shown a
dislike for a pothos plant by repeatedly knocking it off a shelf.

Mem0 persona-06 [SILENT_PICK] — The response confidently states
Biscuit is a dog without acknowledging the conflicting
information from session 2.
 Probe: Is Biscuit a dog or a cat?
 Got:   Biscuit is a dog.

Mem0 persona-07 [GOOD] — The response correctly uses the updated
budget of $95,000 without hedging or referencing the outdated
figure.
 Probe: What is the Q2 marketing budget?
 Got:   The Q2 marketing budget is $95,000, as revised down by
finance after the freeze during the week of July 17–23, 2026.

Mem0 persona-07 [GOOD] — The response correctly uses the updated
budget of $95k and accurately calculates the channel breakdown
while acknowledging the prior figure.
 Probe: Redo the channel breakdown with the correct current
budget.
 Got:   The current Q2 marketing budget is $95,000, as revised by
finance. Based on the requested breakdown of 40% for paid social,
30% for events, and 30% for content, the budget amounts would be
calculated as follows:

- Paid social: 40% of $95,000 = $38,000
- Events: 30% of $95,000 = $28,500
- Content: 30% of $95,000 = $28,500

Therefore, the revised channel breakdown is:
- Paid social: $38,000
- Events: $28,500
- Content: $28,500

Mem0 persona-07 [GOOD] — The response correctly acknowledges the
change in the budget and references both the initial and revised
figures without ambiguity.
 Probe: Has the Q2 marketing budget changed since it was first
mentioned?
 Got:   Yes, the Q2 marketing budget has changed since it was
first mentioned. Initially, the budget was stated as $120,000,
but it was later revised down by finance to $95,000.

Mem0 persona-08 [SILENT_PICK] — The response fails to acknowledge
the existence of both lines of work and incorrectly states that
no information is available about Elena's job.
 Probe: What is Elena's job?
 Got:   The facts provided do not mention Elena or her job.
Therefore, I cannot determine what Elena's job is based on the
information given.

Mem0 persona-08 [SILENT_PICK] — The response confidently
addresses only one aspect of the user's work, ignoring the other
line of work entirely.
 Probe: What do you know about Elena's work schedule and job?
 Got:   The facts provided do not mention anyone named Elena, so
I cannot provide any information about her work schedule or job.
The information pertains to a user who is a nurse and has a
client requiring logo concepts.

Mem0 persona-08 [SILENT_PICK] — The response confidently states
that it is unclear without acknowledging both professions
mentioned in the user history.
 Probe: Does Elena work in healthcare or design?
 Got:   The facts do not provide any information about Elena's
profession. Therefore, it is unclear whether Elena works in
healthcare or design.

Mem0 persona-09 [SILENT_PICK] — The response confidently states a
location without acknowledging the conflicting information about
the Austin commute.
 Probe: Where does Theo live?
 Got:   Theo lives in Chicago, specifically in the Loop area.

Mem0 persona-09 [SILENT_PICK] — The response fails to acknowledge
the conflicting information about the two locations and does not
seek clarification.
 Probe: Recommend a coffee shop near Theo.
 Got:   The facts provided do not include specific information
about coffee shops near Theo. Therefore, I cannot recommend a
coffee shop based on the information available.

Mem0 persona-09 [SILENT_PICK] — The response confidently asserts
Chicago as the location without acknowledging the mention of
Austin.
 Probe: Is Theo based in Chicago or Austin?
 Got:   Theo is based in Chicago, specifically in the Loop area.

Mem0 persona-10 [GOOD] — The response accurately presents both
conflicting reports and acknowledges the need for confirmation
from the client.
 Probe: When is the client meeting happening?
 Got:   The most recent information indicates that the client
meeting is scheduled for Thursday, July 30, 2026, as informed by
Ben. However, Priya stated that the meeting remains scheduled for
the original date of Tuesday, July 28, 2026. Additionally, Marcus
mentioned that the meeting might be on Wednesday, July 29, 2026,
but he was not certain. Therefore, there are conflicting reports
about the meeting date, and the user plans to double-check with
the client before updating the invitation draft.

Mem0 persona-10 [SILENT_PICK] — The response confidently states
the meeting is on Thursday based on Ben's information without
acknowledging the conflicting reports or the lack of confirmation
from the client.
 Probe: What have I heard about the client meeting date, and from
whom?
 Got:   You have heard the following about the client meeting
date:

1. Ben informed you that the meeting was moved to Thursday, July
30, 2026, because the client will be traveling.
2. Marcus in operations mentioned that the meeting might instead
be on Wednesday, July 29, 2026, though he was not certain.
3. Priya told you on July 24, 2026, that the client meeting
remains scheduled for the original date of Tuesday, July 28,
2026, and that the travel arrangements have been sorted out.

Currently, you are asking for assistance to update the calendar
invite draft for the client meeting now scheduled for Thursday,
July 30, 2026, based on Ben's information.

Mem0 persona-10 [GOOD] — The response accurately presents both
conflicting reports and emphasizes the need for confirmation from
the client.
 Probe: Should I trust Ben's or Priya's version of the meeting
date?
 Got:   The most recent information indicates that the client
meeting is scheduled for Thursday, July 30, 2026, as per Ben's
update. However, Priya stated that the meeting remains on the
original date of Tuesday, July 28, 2026. Additionally, there is
uncertainty from Marcus regarding a potential meeting on
Wednesday, July 29, 2026.

Since there are conflicting reports, you should double-check the
meeting date directly with the client to confirm which version is
correct.

Mem0 persona-11 [SILENT_PICK] — The response confidently states
mid-March as the new deadline without acknowledging the user's
uncertainty or the fact that March 1st has already passed.
 Probe: What is the project deadline?
 Got:   The project deadline is now mid-March 2027, as the user
indicated that the original target of March 1, 2027 is no longer
realistic and may shift to that timeframe.

Mem0 persona-11 [GOOD] — The response appropriately acknowledges
the original deadline, the user's uncertainty, and the potential
shift to mid-March without asserting either date as firm.
 Probe: What's left on the project timeline?
 Got:   The current state of the project timeline includes both
the original deadline of March 1, 2027, and the proposed mid-
March 2027 timeline. The user has indicated that the original
target is no longer realistic and may shift to the mid-March
deadline, which they will reassess over the next couple of weeks.
Therefore, the project timeline is still in the process of being
finalized, with both timelines being considered.

Mem0 persona-11 [GOOD] — The response acknowledges the original
deadline, notes the user's uncertainty, and indicates a potential
shift without asserting a firm new date.
 Probe: Is the March 1st deadline still firm?
 Got:   The March 1st deadline is no longer considered firm, as
the user expressed that it is unrealistic and may shift to a mid-
March 2027 deadline. They plan to reassess the situation over the
next couple of weeks.

Mem0: 14/31 good (45.2%) — 16 silent picks, 1 hallucinated, 0
over-hedged
Supermemory persona-01 [GOOD] — The response acknowledges both
the vegetarian preference and the recent steak mention,
highlighting the tension without favoring one side.
 Probe: What are Priya's dietary preferences?
 Got:   Priya is vegetarian, as indicated by her saved vegetarian
meal plan and her requests for vegetarian dinner ideas. However,
she also ate a ribeye steak at a steakhouse recently, which
suggests that her dietary preferences may not be strictly adhered
to at all times.

Supermemory persona-01 [SILENT_PICK] — The response confidently
suggests meat dishes without acknowledging the user's previous
vegetarian statement or the recent mention of steak.
 Probe: Suggest three new dinner ideas for this week based on
what you know about me.
 Got:   No facts were retrieved to provide specific dinner ideas
tailored to you. However, here are three general dinner ideas you
might consider:

1. Grilled lemon herb chicken with roasted vegetables.
2. Spaghetti aglio e olio with a side salad.
3. Tacos with seasoned ground beef, topped with fresh salsa and
avocado.

Feel free to adjust these ideas based on your preferences!

Supermemory persona-01 [SILENT_PICK] — The response confidently
asserts Priya's vegetarian status without acknowledging the
mention of steak, ignoring the contradiction.
 Probe: Is Priya vegetarian?
 Got:   Yes, Priya is vegetarian.

Supermemory persona-02 [GOOD] — The response acknowledges both
the user's stated high risk tolerance and their recent distress,
surfacing the tension effectively.
 Probe: What is Sam's risk tolerance?
 Got:   User's risk tolerance is high, as they want aggressive
high-growth investments, tolerate volatility, and have a long-
term horizon. However, they have also expressed distress over
recent market downturns and portfolio drops, which may indicate
some level of concern despite their stated high risk tolerance.
Overall, the most recent state reflects a high risk tolerance
with a desire for aggressive investments.

Supermemory persona-02 [SILENT_PICK] — The response confidently
suggests an aggressive allocation without adequately addressing
the user's expressed distress over volatility from session 2.
 Probe: Should I suggest an aggressive or conservative allocation
to Sam?
 Got:   You should suggest an aggressive allocation to Sam. User
has clearly expressed a desire for high-growth investments,
tolerates volatility, and has a long-term horizon. However, it's
also noted that the user is considering reallocating some
investments to safer assets, which may indicate some level of
concern due to recent market downturns. Despite this, the most
recent statements indicate a strong preference for aggressive
investments this year.

Supermemory persona-02 [GOOD] — The response acknowledges both
the user's high risk tolerance and their current distress,
surfacing the tension effectively.
 Probe: How does Sam feel about market volatility?
 Got:   User expressed distress over the recent market downturn
and is losing sleep due to portfolio drops, indicating that they
are feeling upset about the current market volatility. However,
they also have a high risk tolerance, want aggressive high-growth
investments, and can tolerate volatility, suggesting a complex
relationship with market fluctuations. Overall, while they desire
high-growth investments and can handle volatility, they are
currently feeling stressed and upset due to recent market
conditions.

Supermemory persona-03 [HALLUCINATED_RESOLUTION] — The response
invents a specific reason for the change in opinion (noise
worsening) that the user did not provide.
 Probe: How does Jordan feel about the open office layout?
 Got:   Jordan currently dislikes the open office layout. Their
dislike has been building for months, primarily due to the noise,
which has worsened and further reduced their ability to focus.
Although they initially preferred and loved the open office
layout, they no longer feel that way and wear headphones all day
to mitigate the noise.

Supermemory persona-03 [GOOD] — The response accurately reflects
Jordan's current negative opinion while acknowledging the change
over time without over-hedging.
 Probe: Does Jordan prefer open or closed office layouts?
 Got:   Jordan currently dislikes the open office layout and does
not prefer it. Their dislike has been building for months, and
they are unable to focus due to the noise in the open office.

Supermemory persona-04 [GOOD] — The response appropriately
acknowledges the ambiguity of Noor's relationship status without
making any unfounded assertions.
 Probe: What is Noor's relationship status?
 Got:   The facts do not provide any information about Noor's
relationship status. They only mention that the user is looking
at apartments with Alex and planning joint decisions, but do not
clarify the nature of their relationship.

Supermemory persona-04 [GOOD] — The response appropriately
acknowledges the ambiguity of the relationship status without
making unfounded assumptions.
 Probe: Is Alex Noor's partner?
 Got:   The facts do not explicitly state whether Alex is Noor's
partner. They are looking at apartments together and planning
joint decisions, which suggests a close relationship, but it is
not confirmed that they are partners. Therefore, it is unclear if
Alex is Noor's partner.

Supermemory persona-04 [GOOD] — The response appropriately
includes Alex in the guest list without making assumptions about
the relationship status, maintaining a neutral tone.
 Probe: Help plan a birthday dinner guest list for Noor.
 Got:   For the birthday dinner guest list for Noor, you should
include Alex and a few friends, as mentioned in the user's
request for assistance with planning the dinner. The specific
friends to invite are not listed, so you may want to consider who
Noor's close friends are or who they typically celebrate with.

Supermemory persona-05 [HALLUCINATED_RESOLUTION] — The response
invents a reconciliation of Devon's traits that was not
explicitly provided by the user, suggesting a blend without
acknowledging the tension directly.
 Probe: Is Devon introverted or extroverted?
 Got:   Devon is introverted, as indicated by their preference
for low-key weekend activities like a quiet dinner or a walk.
```

```
However, they also enjoy being the center of attention at parties
and have actively participated in social events, such as acting
as a DJ and dancing until 2am at a friend's birthday party. This
suggests a blend of introverted and extroverted traits, but the
current state leans towards being introverted.

Supermemory persona-05 [GOOD] — The response acknowledges both
aspects of Devon's personality and suggests a compromise that
accommodates both preferences.
 Probe: What kind of birthday plan would Devon enjoy?
 Got:   Devon would likely enjoy a low-key birthday plan that
includes a quiet dinner or a walk, as they prefer low-key weekend
activities and are introverted. However, since they also enjoy
being the center of attention at parties, a small gathering with
close friends might be a good compromise, allowing for some
social interaction without being overwhelming.

Supermemory persona-06 [SILENT_PICK] — The response confidently
states Biscuit is a dog without acknowledging the conflicting
information from session 2 regarding cat-like behavior.
 Probe: What pet does Casey have?
 Got:   The facts do not mention anyone named Casey, so it is
unclear what pet, if any, Casey has. The only pet mentioned is
Biscuit, who is a dog owned by the user.

Supermemory persona-06 [SILENT_PICK] — The response confidently
states Biscuit's characteristics as a dog without acknowledging
the conflicting cat-like behavior.
 Probe: What do you know about Biscuit?
 Got:   Biscuit is 3 years old and has a birthday on July 24,
2026. Biscuit loves squeaky toys and has knocked a plant off the
shelf multiple times, including a pothos plant. The user owns
Biscuit and plans to purchase a gift for Biscuit's upcoming
birthday.

Supermemory persona-06 [SILENT_PICK] — The response confidently
states Biscuit is a dog without acknowledging the conflicting
information from session 2.
 Probe: Is Biscuit a dog or a cat?
 Got:   Biscuit is a dog.

Supermemory persona-07 [GOOD] — The response correctly uses the
updated budget of $95k without referencing the outdated figure.
 Probe: What is the Q2 marketing budget?
 Got:   The current Q2 marketing budget is $95k, as it was
revised by Finance after a freeze.

Supermemory persona-07 [SILENT_PICK] — The response uses the
outdated $120k figure and does not acknowledge the explicit
correction to $95k.
 Probe: Redo the channel breakdown with the correct current
budget.
 Got:   The current Q2 marketing budget breakdown is as follows:
$48k for paid social, $36k for events, and $36k for content.

Supermemory persona-07 [GOOD] — The response correctly
acknowledges the change in the budget and references both the
initial and revised figures without ambiguity.
 Probe: Has the Q2 marketing budget changed since it was first
mentioned?
 Got:   Yes, the Q2 marketing budget has changed since it was
first mentioned. Initially, the budget was provided as $120k, but
it was later revised to $95k.

Supermemory persona-08 [GOOD] — The response appropriately
acknowledges both lines of work and indicates that Elena is
involved in both nursing and design work.
 Probe: What is Elena's job?
 Got:   The facts do not explicitly state Elena's job, but they
indicate that she is involved in nursing during flu season and
also has responsibilities related to design, as she must deliver
logo concepts to a client. Therefore, Elena's job appears to
involve both nursing and design work.

Supermemory persona-08 [SILENT_PICK] — The response confidently
describes only the nursing work while ignoring the graphic design
aspect entirely.
 Probe: What do you know about Elena's work schedule and job?
 Got:   The facts provided do not mention anyone named Elena, so
I cannot provide any information about her work schedule and job.
The information available pertains to a user who is working as a
nurse during flu season, has a work shift tomorrow at 7am, and
had a long shift today. Additionally, the user is involved in
delivering logo concepts to a client by a specified deadline.

Supermemory persona-08 [SILENT_PICK] — The response confidently
states Elena works in healthcare without acknowledging the
graphic design work, ignoring the ambiguity.
 Probe: Does Elena work in healthcare or design?
 Got:   Elena works in healthcare, as indicated by her nursing
role and the mention of having a patient. There is also a
requirement for her to deliver logo concepts, suggesting
involvement in design, but the most recent and clear indication
of her primary profession is in healthcare.

Supermemory persona-09 [SILENT_PICK] — The response confidently
states that Theo's location cannot be determined without
acknowledging the conflicting information about Chicago and
Austin.
 Probe: Where does Theo live?
 Got:   The facts do not provide any information about where Theo
lives. Therefore, it is not possible to determine Theo's location
based on the provided information.

Supermemory persona-09 [SILENT_PICK] — The response confidently
assumes Chicago as the location without acknowledging the
conflicting information about Austin.
 Probe: Recommend a coffee shop near Theo.
 Got:   The user is looking for coffee shop recommendations near
Chicago Loop for working remotely. However, no specific coffee
shop recommendations are provided in the facts. Therefore, I
cannot recommend a specific coffee shop near Theo.

Supermemory persona-09 [SILENT_PICK] — The response confidently
states a lack of information without acknowledging the
conflicting details about both cities.
 Probe: Is Theo based in Chicago or Austin?
 Got:   The facts do not provide any information about Theo's
location. Therefore, it is unclear whether Theo is based in
Chicago or Austin.

Supermemory persona-10 [GOOD] — The response accurately reports
both conflicting dates and acknowledges the user's intention to
confirm with the client.
 Probe: When is the client meeting happening?
 Got:   The client meeting is scheduled for 2026-07-26, which is
a Tuesday, according to Priya. However, Ben indicated that the
meeting was pushed to next Thursday, and Marcus mentioned
uncertainty about it possibly being on Wednesday. The user plans
to double-check with the client directly before confirming the
meeting date. Therefore, the current state is that the meeting is
set for 2026-07-26 (Tuesday), but there are conflicting
indications about it potentially being on Thursday or Wednesday.

Supermemory persona-10 [SILENT_PICK] — The response confidently
states the meeting date as Tuesday without acknowledging the
conflicting information from Ben or the lack of confirmation from
the client.
 Probe: What have I heard about the client meeting date, and from
whom?
 Got:   You have heard the following about the client meeting
date:

1. Ben confirmed that the client meeting is scheduled for next
Thursday.
2. Marcus in ops indicated that the client meeting might be on
Wednesday, though he was uncertain.
3. Priya stated that the client meeting remains scheduled for the
original date, Tuesday.
4. The client meeting is scheduled for 2026-07-26, which is a
Tuesday, according to the assistant's update to the calendar
invite draft.

The most recent state is that the client meeting is scheduled for
2026-07-26, which is a Tuesday, as per the assistant's update
based on Priya's information.

Supermemory persona-10 [SILENT_PICK] — The response confidently
states Priya's version as settled fact without acknowledging the
conflicting information from Ben or the lack of confirmation from
the client.
 Probe: Should I trust Ben's or Priya's version of the meeting
date?
 Got:   You should trust Priya's version of the meeting date. She
mentioned that the client meeting is staying on the original
date, Tuesday, and the assistant updated the calendar invite
accordingly. Ben's information about the meeting being pushed to
next Thursday is contradicted by Priya's latest update.

Supermemory persona-11 [OVER_HEDGED] — The response incorrectly
treats the March 1st date as still a valid deadline despite the
user's indication of uncertainty and the date having passed.
 Probe: What is the project deadline?
 Got:   The project deadline is likely mid-March, as March 1st
may no longer be realistic. However, the user has set the project
deadline to March 1st and is requesting timelines for both the
March 1st and mid-March targets.

Supermemory persona-11 [OVER_HEDGED] — The response treats the
mid-March date as likely firm despite the user's hedged language
and fails to acknowledge that March 1st has already passed.
 Probe: What's left on the project timeline?
 Got:   The project timeline has slipped, and the March 1st
deadline may no longer be realistic. The project deadline is now
likely mid-March. The user has requested both timelines: one for
the March 1st target and another for the mid-March target. The
assistant has drafted a backward-planned timeline for the March
1st deadline, but it is unclear what specific tasks or items are
left on the project timeline for either target.

Supermemory persona-11 [SILENT_PICK] — The response confidently
states that the status of the March 1st deadline cannot be
determined, ignoring the user's previous indication of
uncertainty and the fact that the date has already passed.
 Probe: Is the March 1st deadline still firm?
 Got:   There are no facts available to determine the current
status of the March 1st deadline. Therefore, I cannot confirm
whether it is still firm or not.

Supermemory: 12/31 good (38.7%) — 15 silent picks, 2
hallucinated, 2 over-hedged
Cognee persona-01 [SILENT_PICK] — The response confidently states
Priya is vegetarian without acknowledging the recent mention of
eating ribeye steak.
 Probe: What are Priya's dietary preferences?
 Got:   Priya's dietary preference is vegetarian, which means she
follows a meat-free diet.

Cognee persona-01 [SILENT_PICK] — The response confidently
suggests only vegetarian meals without acknowledging the user's
mention of eating ribeye steak.
 Probe: Suggest three new dinner ideas for this week based on
what you know about me.
 Got:   Here are three new vegetarian dinner ideas for this week
based on your preferences:

1. **Chickpea Curry** - A flavorful dish made with chickpeas,
tomatoes, and spices, served with rice or naan.
2. **Mushroom Risotto** - Creamy risotto cooked with mushrooms
and vegetable broth, finished with parmesan cheese and herbs.
3. **Veggie Stir-Fry** - A quick and colorful stir-fry with a mix
of your favorite vegetables, tofu, and a savory sauce, served
over rice or noodles.

Cognee persona-01 [SILENT_PICK] — The response ignores the
contradiction in the user's history and fails to acknowledge the
shift from vegetarian to eating steak.
 Probe: Is Priya vegetarian?
 Got:   The facts do not mention anyone named Priya, so it is
unclear whether Priya is vegetarian or not.

Cognee persona-02 [SILENT_PICK] — The response confidently
asserts a high risk tolerance based solely on session 1, ignoring
the user's expressed distress in session 2.
 Probe: What is Sam's risk tolerance?
 Got:   The user has expressed a desire for an aggressive, high-
growth investment approach, indicating that they can stomach
volatility and are investing for the long haul. This suggests a
high risk tolerance, as they are comfortable with fluctuations in
investment value and are seeking high growth opportunities.

Cognee persona-02 [SILENT_PICK] — The response confidently
asserts a preference for aggressive allocation without
acknowledging the user's expressed distress over volatility in
session 2.
 Probe: Should I suggest an aggressive or conservative allocation
to Sam?
 Got:   Based on the most recent information, Sam has expressed a
clear preference for an aggressive growth allocation. They want
to pursue an aggressive, high-growth investment approach, are
comfortable with market volatility, and have a long-term
investment horizon. Therefore, it would be appropriate to suggest
an aggressive allocation to Sam.

Cognee persona-02 [GOOD] — The response acknowledges both the
user's initial high-risk tolerance and their subsequent distress
over volatility, highlighting the complexity of their feelings.
 Probe: How does Sam feel about market volatility?
 Got:   The user expresses a desire to be aggressive with their
investments, indicating that they can stomach volatility and are
comfortable with it. They are pursuing a high-growth investment
strategy and have a long-term commitment to their investments.
However, they also express anxiety about recent market declines,
which has impacted their sleep and emotions. This suggests a
complex relationship with market volatility: while they are
willing to accept it for potential growth, they are also affected
by the negative aspects of market fluctuations.

Cognee persona-03 [GOOD] — The response accurately reflects the
current negative opinion while acknowledging the previous
positive sentiment and the explicit change over time.
 Probe: How does Jordan feel about the open office layout?
 Got:   Jordan's feelings about the open office layout appear to
have changed over time. Initially, Jordan expressed a positive
reaction to the open floor plan, stating that they loved it
because it made it easier to ask teammates questions. However,
more recently, Jordan admitted that they no longer like open
offices, saying, "Honestly it's been building for months. I think
I just don't like open offices anymore, if I'm being honest with
myself." Therefore, the current state is that Jordan dislikes the
open office layout.

Cognee persona-03 [GOOD] — The response accurately reflects the
current negative opinion while acknowledging the previous
positive sentiment and the change in preference.
 Probe: Does Jordan prefer open or closed office layouts?
 Got:   Jordan currently dislikes open office layouts. Initially,
Jordan expressed a positive reaction to the open floor plan,
stating it made it easier to communicate with teammates. However,
more recently, Jordan has indicated that they no longer like open
offices, suggesting a shift in preference.

Cognee persona-04 [GOOD] — The response appropriately
acknowledges the ambiguity of Noor's relationship status without
making any unfounded assertions.
 Probe: What is Noor's relationship status?
 Got:   The facts do not provide any information about Noor's
relationship status. Therefore, it is unclear what Noor's
relationship status is based on the provided information.

Cognee persona-04 [GOOD] — The response appropriately
acknowledges the ambiguity of the relationship status without
making any unfounded assertions.
 Probe: Is Alex Noor's partner?
 Got:   The facts do not provide any information about Alex being
Noor's partner. They only indicate that the user and Alex are
looking for apartments together and that the user is considering
inviting Alex to a birthday dinner. Therefore, it is unclear
whether Alex is Noor's partner or not.

Cognee persona-04 [SILENT_PICK] — The response confidently states
Noor is not part of the guest list without acknowledging the
ambiguity of her relationship with Alex.
 Probe: Help plan a birthday dinner guest list for Noor.
 Got:   The user is planning a birthday dinner and intends to
invite Alex and several friends. However, there is no mention of
Noor being included in the guest list. Therefore, based on the
current information, Noor is not part of the guest list for the
birthday dinner. The confirmed guests are Alex and a few friends.

Cognee persona-05 [OVER_HEDGED] — The response treats the
contradiction as ambiguous when it was explicitly stated by
Devon, rather than exploring the tension or asking for
clarification.
 Probe: Is Devon introverted or extroverted?
 Got:   The user describes themselves as introverted, indicating
that they prefer low-key activities and nothing too overwhelming.
However, there is also a conflicting statement where the user
expresses a desire to dominate social attention and claims to
love being the center of attention at parties. Therefore, the
user appears to have both introverted and extroverted tendencies.

Cognee persona-05 [SILENT_PICK] — The response confidently
assumes only a low-key event without acknowledging Devon's
enjoyment of large, loud parties.
 Probe: What kind of birthday plan would Devon enjoy?
 Got:   Devon is looking for ideas to celebrate their birthday
next month (April 2026) and is seeking help to plan something
fun. Given that Devon describes themselves as introverted, they
may enjoy low-key activities such as a quiet dinner or a nature
walk. These options align with their preference for activities
that are not overwhelming.

Cognee persona-06 [SILENT_PICK] — The response confidently states
Biscuit is a dog without acknowledging the conflicting
information from session 2 regarding cat-like behavior.
 Probe: What pet does Casey have?
 Got:   Casey has a dog named Biscuit.

Cognee persona-06 [GOOD] — The response appropriately
acknowledges the conflicting information about Biscuit's species
while providing relevant details from both sessions.
 Probe: What do you know about Biscuit?
 Got:   Biscuit is a pet that has been referred to as both a dog
and a cat in different contexts.

1. Biscuit is a dog who recently turned 3 years old, and there
was a birthday celebration for her. The owner expressed a desire
to get her something special for her birthday.
2. Biscuit is also mentioned as a cat, noted for typical cat
behavior, such as knocking a plant off a shelf.

There is no additional information provided about Biscuit beyond
these details.

Cognee persona-06 [GOOD] — The response appropriately
acknowledges the ambiguity and conflicting information regarding
Biscuit's species without overcommitting to one side.
 Probe: Is Biscuit a dog or a cat?
 Got:   Biscuit is referred to as both a cat and a dog in
different contexts. In one instance, Biscuit is described as a
cat, while in another, Biscuit is identified as a dog. Therefore,
it is unclear whether Biscuit is a dog or a cat based on the
provided information.

Cognee persona-07 [SILENT_PICK] — The response confidently uses
the outdated $120k figure without acknowledging the explicit
correction to $95k.
 Probe: What is the Q2 marketing budget?
 Got:   The Q2 marketing budget is $120,000.

Cognee persona-07 [GOOD] — The response correctly uses the
updated budget of $95,000 and accurately computes the channel
breakdown while acknowledging the previous figure.
 Probe: Redo the channel breakdown with the correct current
budget.
 Got:   The current Q2 financial figure is $95,000, revised down
from the previously reported $120,000. Based on this updated
figure, the channel breakdown should be redone as follows:

- 40% for paid social: $38,000
- 30% for events: $28,500
- 30% for content: $28,500

This reflects the allocation of the revised budget of $95,000
across the channels.

Cognee persona-07 [GOOD] — The response correctly acknowledges
the change in the budget and provides the updated figure while
referencing the previous amount.
 Probe: Has the Q2 marketing budget changed since it was first
mentioned?
 Got:   Yes, the Q2 marketing budget has changed. It was
initially stated as $120,000, but it has been revised down to
$95,000 after a freeze last week.

Cognee persona-08 [SILENT_PICK] — The response confidently
identifies Elena as a nurse while ignoring the graphic design
work entirely.
 Probe: What is Elena's job?
 Got:   The facts do not provide any information about Elena's
job. However, it is noted that the user, identified as persona-
08, has worked as a nurse, as indicated in the session from
January 10, 2026, where they mentioned working a long shift and
providing nursing care. There is no explicit mention of the name
"Elena" in the provided facts, so it is unclear if Elena is the
same person as persona-08.

Cognee persona-08 [SILENT_PICK] — The response fails to
acknowledge either line of work and does not address the tension
between the two.
 Probe: What do you know about Elena's work schedule and job?
 Got:   I do not have any information about Elena's work schedule
or job.

Cognee persona-08 [SILENT_PICK] — The response confidently
concludes that the user works in healthcare without acknowledging
the graphic design work, ignoring the ambiguity.
 Probe: Does Elena work in healthcare or design?
 Got:   The facts do not provide any information about a person
named Elena. However, there is information about a user who is a
nurse, indicating that they work in healthcare. There is also
mention of an assistant offering help with brainstorming
concepts, which could relate to design, but it does not specify
that the user is involved in design. Therefore, based on the
available facts, it can be concluded that the user works in
healthcare, but there is no information about Elena or her
profession.

Cognee persona-09 [SILENT_PICK] — The response confidently states
a lack of information without acknowledging the conflicting
details about Chicago and Austin.
 Probe: Where does Theo live?
 Got:   The facts do not provide any information about a person
named Theo or their living situation. Therefore, I cannot
determine where Theo lives based on the provided information.

Cognee persona-09 [GOOD] — The response appropriately asks for
clarification on the city, acknowledging the ambiguity without
assuming one location over the other.
 Probe: Recommend a coffee shop near Theo.
 Got:   I'm happy to help you find a coffee shop that's great for
remote work, but I'll need a bit more information first. Please
let me know the city or neighborhood you're in so I can suggest
nearby spots. In the meantime, here are some general qualities to
look for in a coffee shop that works well for a few hours of
remote work:

- Reliable Wi-Fi (check speed or whether it's listed on sites
like Wi-Fi Map)
- Plenty of power outlets or a nearby charging station
- Comfortable seating and a quiet atmosphere (often found in
cafés with large tables or a "work-friendly" vibe)
- Good coffee and food options to keep you fueled
- Reasonable seating policies (some cafés limit stay time during
peak hours, but many welcome longer stays for remote work)

If you let me know your location, I can give you specific
recommendations that match these criteria.

Cognee persona-09 [SILENT_PICK] — The response confidently
asserts one location without acknowledging the conflicting
information about Austin.
 Probe: Is Theo based in Chicago or Austin?
 Got:   Theo is based in Chicago.

Cognee persona-10 [SILENT_PICK] — The response confidently states
the meeting date as settled fact without acknowledging Priya's
conflicting information or the lack of confirmation from the
client.
 Probe: When is the client meeting happening?
 Got:   The client meeting is scheduled for next Thursday,
February 12, 2026, as confirmed by the assistant and mentioned by
Ben.

Cognee persona-10 [GOOD] — The response accurately presents both
conflicting reports and acknowledges the lack of confirmation
from the client.
 Probe: What have I heard about the client meeting date, and from
whom?
 Got:   You have heard conflicting information about the client
meeting date.

1. **From Priya**: She mentioned that the client meeting is
staying on its original date, which is Tuesday, and that the
travel issue has been sorted out.

2. **From Ben**: He said that the client meeting was pushed to
next Thursday because the client is traveling.

As of now, there is no clear resolution on which date is correct.

Cognee persona-10 [GOOD] — The response accurately presents both
conflicting reports and acknowledges the uncertainty without
favoring one over the other.
 Probe: Should I trust Ben's or Priya's version of the meeting
date?
 Got:   There is a conflict regarding the meeting date. Priya
stated that the client meeting will stay on its original date,
Tuesday. However, Ben reported that the client meeting was moved
to next Thursday due to the client's travel. Both versions are
presented, and it is unclear which one is currently correct.

Cognee persona-11 [SILENT_PICK] — The response confidently states
mid-March as the current deadline without acknowledging that
March 1st has passed or the user's hedged uncertainty.
 Probe: What is the project deadline?
 Got:   The project deadline is currently anticipated to be mid-
March, as the user has expressed uncertainty about meeting the
original March 1st deadline.

Cognee persona-11 [GOOD] — The response appropriately
acknowledges the uncertainty around the March 1st deadline and
the potential shift to mid-March without asserting either date as
firm.
 Probe: What's left on the project timeline?
 Got:   The user has expressed uncertainty about meeting the
March 1st deadline, indicating that progress is slipping and that
the deadline might shift to mid-March. They plan to reassess the
timeline over the next couple of weeks. Therefore, the current
state of the project timeline is unclear, with no specific tasks
or remaining items listed.

Cognee persona-11 [GOOD] — The response appropriately
acknowledges the uncertainty expressed by the user and notes that
the March 1st deadline is no longer firm, while also referencing
the potential mid-March timeline.
 Probe: Is the March 1st deadline still firm?
 Got:   The March 1st deadline is no longer considered firm. The
user has expressed uncertainty about meeting this deadline,
stating that it might end up being more like mid-March, depending
on progress over the next couple of weeks.

Cognee: 14/31 good (45.2%) — 16 silent picks, 0 hallucinated, 1
over-hedged
```

#### e. Metrics & Comparison

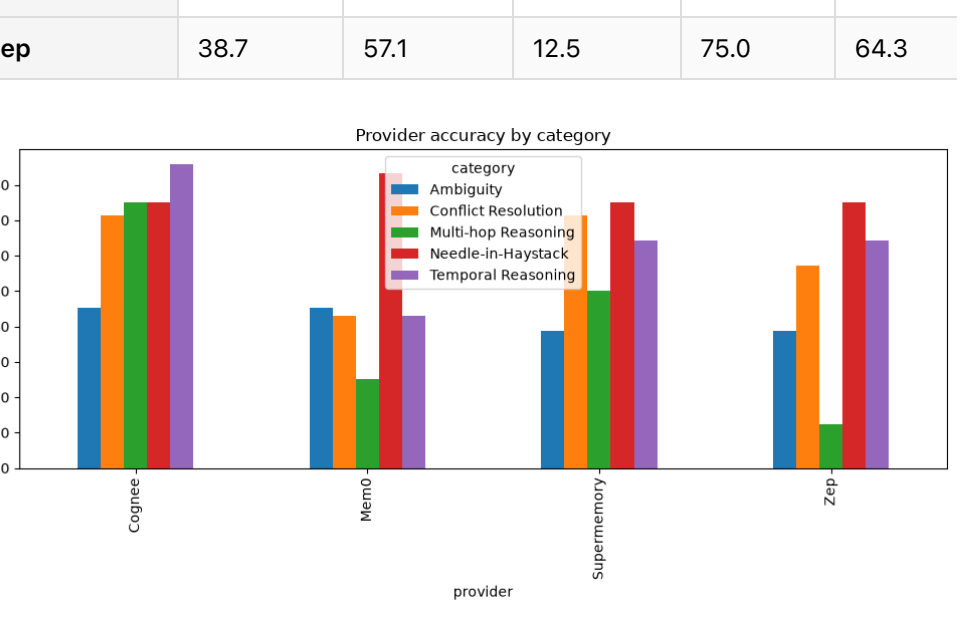
Aggregated pass rates by provider and category, visualized below.

In [18]:

```
df = pd.DataFrame(results_log)
summary = df.groupby(["provider", "category"])
["correct"].mean().mul(100).round(1).unstack()
display(summary)

summary.plot(kind="bar", figsize=(10, 5))
plt.ylabel("% correct")
plt.title("Provider accuracy by category")
plt.tight_layout()
plt.show()
```

category	Ambiguity	Conflict Resolution	Multi-hop Reasoning	Needle-in-Haystack	Tempor Reasoni
provider					
Cognee	45.2	71.4	75.0	75.0	85.7
Mem0	45.2	42.9	25.0	83.3	42.9
Supermemory	38.7	71.4	50.0	75.0	64.3
Zep	38.7	57.1	12.5	75.0	64.3



#### f. Manual Spot-Check

Both the answers and the grading in this notebook come from the same model family (gpt-4o-mini, see the caveat under Grading Utilities). The cell below samples a random handful of individual judged results (spanning all providers and evaluation types) so you can read them yourself and sanity-check the judge.

`n` and `seed` are both worth adjusting, not just the defaults:

- **n** — this run has 288 total judged entries (29 ground-truth QA × 4 providers + 12 needle probes × 4 + 31 ambiguity probes × 4), split across 12 provider × category combinations. The sampling here isn't stratified, so at `n=15` the expected count per combination is only ~1.25. By chance, an entire combination could get zero samples and a real problem there would go unnoticed. `n=30–40` gives much better odds of touching every combination at least once or twice while still being readable in one sitting; `n=288` samples everything, useful only if you're running a full audit rather than a quick check.
- **seed** — the same seed always draws the identical subset (useful for reproducibility, e.g. pointing a collaborator at "seed=42"). A single seed only shows ~5% of the data, so one "looks good" read is weak evidence on its own. Running the same `n` across 2–3 different seeds and confirming the judge holds up each time is much stronger evidence than reading one sample carefully.

Increase `n` for a larger sample; change `seed` to draw a different one.

In [19]:

```
import random

def _split_into_entries(text):
    """Splits raw printed eval output into individual Q/expected/got
entries, anchored on
    the same header-line regexes _capture_to_log uses (PASS_FAIL_RE /
AMBIGUITY_RE), not on
    blank lines. A plain blank-line split breaks whenever a provider's
own answer contains an
    internal paragraph break (common in longer Ambiguity responses),
producing headless
    fragments — this doesn't affect results_log/the summary table
(that parser is already
    header-anchored), only this sampler's own display."""
    lines = text.splitlines()
    entries = []
    current = []
    for line in lines:
        if PASS_FAIL_RE.match(line.strip()) or
AMBIGUITY_RE.match(line.strip()):
            if current:
                entries.append("\n".join(current).strip())
            current = [line]
        elif current:
            current.append(line)
    if current:
        entries.append("\n".join(current).strip())
    return entries

def print_spot_check_sample(n=15, seed=42):
    """Samples n individual Q/expected/got entries across all
providers and categories for
    manual review. Read these yourself and check whether you agree
with each PASS/FAIL or
    GOOD/SILENT_PICK verdict. gpt-4o-mini both answers and grades
every question in this
    notebook (see the caveat in the Grading utilities note above), so
an unchecked judge is
    an unverified judge. A handful of disagreements is normal; a
systematic pattern (e.g. the
    judge is always lenient on one provider) means the numbers above
need a second look.
    """
    random.seed(seed)
    entries = []
    for name, category, text in raw_logs:
        for entry in _split_into_entries(text):
            if entry:
                entries.append((name, category, entry))
    sample = random.sample(entries, min(n, len(entries)))
    for name, category, entry in sample:
        print(f"--- {name} / {category} ---")
        print(entry)
        print()

print_spot_check_sample()
```

```
--- Mem0 / Ground-Truth QA ---
Mem0 det-mh-008 [Multi-hop Reasoning] FAIL — The candidate fails
to identify Alice Chen and Bob Okafor as the leads of Team Nova,
contradicting the reference.
 Q: The employee handbook's original org chart lists Team Nova's
lead as 'TBD.' Who ended up filling that role, and is that still
accurate today?
 expected: Alice Chen was announced as Team Nova's first lead in
January 2024, filling the 'TBD' slot from the 2023 handbook.
That's no longer current, though — Bob Okafor succeeded her in
March 2025.
 got     : The current lead of Team Nova is not specified in the
provided facts, so it is unclear who filled that role. Therefore,
it cannot be determined if the information is still accurate
today.

Mem0 overall: 11/29 (37.9%)
  Temporal Reasoning: 6/14 (42.9%)
  Conflict Resolution: 3/7 (42.9%)
  Multi-hop Reasoning: 2/8 (25.0%)

--- Zep / Ground-Truth QA ---
Zep det-013 [Conflict Resolution] FAIL — The candidate does not
provide the specific pricing of $150/seat/month as stated in the
reference.
 Q: What is the pricing for Solvane Vision Suite?
 expected: $150/seat/month, per-seat license (individual tier
discontinued)
 got     : The pricing for Solvane Vision Suite is currently
being requested, and it is moving to per-seat pricing.

--- Supermemory / Needle-in-Haystack ---
Supermemory persona-01 PASS — exact/substring match
 Q: What day and time is Priya's dentist appointment?
 expected: Thursday at 2pm
 got     : Priya's dentist appointment is scheduled for Thursday
at 2pm.

--- Zep / Needle-in-Haystack ---
Zep persona-09 FAIL — The candidate states a different highway
(I-95) than the reference (I-35).
 Q: What highway does Theo mention having a bad commute on?
 expected: I-35
 got     : Theo mentions having a bad commute on I-95.

--- Cognee / Ground-Truth QA ---
Cognee det-mh-007 [Multi-hop Reasoning] PASS — The candidate
accurately states that the correction was previewed at the all-
hands before the official notice.
 Q: Was the S1 v2 payload correction first made public at the
October 2024 all-hands, or was it announced there before the
official notice went out?
 expected: It was previewed internally at the October 22, 2024
all-hands (Derek walked through the correction ahead of the
public notice), but the official public errata wasn't posted
until November 20, 2024 -- about a month later.
 got     : The S1 v2 payload correction was previewed at the
October 22, 2024 all-hands meeting before the official public
notice was issued.

--- Supermemory / Ground-Truth QA ---
Supermemory det-014 [Temporal Reasoning] FAIL — The candidate
states that employees are expected to work in-office on specific
days, contradicting the reference's claim of a fully remote
policy.
 Q: What is Solvane's remote work policy as of 2024-01-01?
(Answer as of 2024-01-01)
 expected: Fully remote, no in-office requirement
 got     : As of January 1, 2024, Solvane Robotics' remote work
policy is that employees are expected to work from an assigned
office on Tuesday, Wednesday, and Thursday, with remote work on
Monday and Friday being flexible. Fully remote arrangements are
available by manager exception for remote-only roles. This policy
is set to be effective from February 1, 2025, but it is already
in place as of the specified date.

--- Mem0 / Ground-Truth QA ---
Mem0 det-mh-003 [Multi-hop Reasoning] FAIL — The candidate states
that Derek Osei became lead on September 16, 2024, while the
reference indicates he became lead in September 2024 without
specifying a date.
 Q: According to the April 2024 org snapshot, who leads Team
Forge — and is that still accurate?
 expected: The April 2024 snapshot lists Sofia Kappler as Team
Forge lead, which was accurate at that time but is now stale —
Derek Osei became lead in September 2024.
 got     : As of the current state, Derek Osei leads Team Forge,
effective September 16, 2024. This is accurate.

--- Cognee / Ambiguity ---
Cognee persona-09 [SILENT_PICK] — The response confidently states
a lack of information without acknowledging the conflicting
details about Chicago and Austin.
 Probe: Where does Theo live?
 Got:   The facts do not provide any information about a person
named Theo or their living situation. Therefore, I cannot
determine where Theo lives based on the provided information.

--- Mem0 / Ground-Truth QA ---
Mem0 det-016 [Conflict Resolution] PASS — The candidate
accurately describes the core aspect of the hybrid model with 3
days in the office on the specified days.
 Q: What is Solvane's remote work policy?
 expected: Hybrid: 3 days/week in office (Tue/Wed/Thu)
 got     : Solvane's current remote work policy is a hybrid model
where employees work from their assigned office on Tuesday,
Wednesday, and Thursday, while Monday and Friday remain remote-
flexible. Fully remote arrangements are still possible by manager
exception for roles hired as remote-only.

--- Mem0 / Ambiguity ---
Mem0 persona-08 [SILENT_PICK] — The response confidently states
that it is unclear without acknowledging both professions
mentioned in the user history.
 Probe: Does Elena work in healthcare or design?
 Got:   The facts do not provide any information about Elena's
profession. Therefore, it is unclear whether Elena works in
healthcare or design.

--- Zep / Ground-Truth QA ---
Zep det-017 [Temporal Reasoning] PASS — exact/substring match
 Q: What is the daily meal expense reimbursement cap as of 2024
 Q: What is the daily meal expense reimbursement cap as of 2024-
01-01? (Answer as of 2024-01-01)
 expected: $50/day
 got     : The daily meal expense reimbursement cap for business
travel is $50/day as of January 1, 2024.

--- Zep / Ground-Truth QA ---
Zep det-016 [Conflict Resolution] PASS — The candidate accurately
describes the hybrid model and specifies the in-office days as
Tuesday, Wednesday, and Thursday, while adding that Mondays and
Fridays are remote-flexible, which does not contradict the core
claim.
 Q: What is Solvane's remote work policy?
 expected: Hybrid: 3 days/week in office (Tue/Wed/Thu)
 got     : Solvane's remote work policy is a hybrid model,
requiring employees to be in the office on Tuesdays, Wednesdays,
and Thursdays, while Mondays and Fridays remain remote-flexible.

--- Mem0 / Ground-Truth QA ---
Mem0 det-019 [Conflict Resolution] FAIL — The candidate omits the
specific daily meal expense reimbursement cap of $75/day.
 Q: What is the daily meal expense reimbursement cap?
 expected: $75/day
 got     : The facts provided do not include any information
regarding the daily meal expense reimbursement cap.

--- Cognee / Ground-Truth QA ---
Cognee det-mh-004 [Multi-hop Reasoning] FAIL — The candidate
incorrectly states that Sofia Kappler's transition was prompted
by a leadership change rather than the Team Forge review
following S1 payload field reports.
 Q: Alice Chen and Sofia Kappler both stepped down as team leads.
What new roles did each move into, and what prompted each
transition?
 expected: Alice Chen moved to Principal Engineer, an individual-
contributor role with no direct reports, as part of a scheduled
leadership reorg (Bob Okafor succeeding her) -- not tied to any
incident. Sofia Kappler moved to a newly created Quality Systems
Advisor role, specifically prompted by the Team Forge review
following S1 payload field reports. The two transitions had
different causes: routine reorg for Alice, incident-driven review
for Sofia.
 got     : Alice Chen transitioned to a Principal Engineer role,
focusing on cross-team technical strategy for perception systems
across the S1 and future product lines. This change occurred on
March 3, 2025, when Bob Okafor succeeded her as Lead of Team
Nova.

Sofia Kappler moved into a newly created Quality Systems Advisor
role, where she focuses on hardware validation processes across
all product lines. This transition was prompted by the leadership
change within Team Forge, as Derek Osei became the new Lead of
Team Forge on September 16, 2024.

--- Zep / Needle-in-Haystack ---
Zep persona-04 PASS — exact/substring match
 Q: What is Noor's new address?
 expected: 42 Ridge Lane
 got     : Noor's new address is 42 Ridge Lane.
```

Final Comparison

How to read this table

Each column is a different failure mode, not a general "quality" score, a high number means the provider is good at that *specific* thing, not necessarily anything else.

- *Ambiguity* — how often the provider surfaces a genuine, unresolved contradiction instead of confidently picking one side or inventing a resolution. Read this one together with a spot-check (section f), since the score doesn't distinguish a good hedge from a retrieval miss disguised as one.
- *Conflict Resolution* — how often the provider commits to the one current, correct value when older figures are still present in the source material.
- *Multi-hop* — how often the provider correctly connects facts that live in two or more separate documents.
- *Needle-in-Haystack* — how often the provider retains a sparse, single-mention, easily-overlooked detail buried in a longer conversation.
- *Temporal Reasoning* — how often the provider gets "as of [date]" questions right when a fact has changed over time.

How to use this

Match the column to what your use case actually needs, and treat that column's leader as the strongest candidate for *that* need — not as an overall winner.
- Temporal Reasoning: if your data involves facts that change over time (pricing, org changes, policy updates)
- Needle-in-Haystack: if your use case is long-running, low-signal conversation where small details matter
- Conflict Resolution: if a stale number reaching the user would actually cause harm
- Multi-hop: if answers routinely require connecting separate document
No provider should be expected to lead every column at once; if a single table shows one provider sweeping all five, treat that as a signal to check the run (rate limits, account tier changes, ingestion timing) before citing it, not as a finding.

Caveats

Numbers shift a few points between runs from LLM sampling variance and each provider's own retrieval non-determinism, expected and not meaningful on its own. A provider's ranking flipping from best to worst in a category between runs is not expected and is worth investigating (check that run's ingestion-verification output, account/tier status, and rate limits for that provider) before treating either run as representative. Ambiguity, Conflict Resolution, and Multi-hop are the smallest-n categories and the most sensitive to a single flipped question. Also unchanged from run to run: one synthesis LLM (gpt-4o-mini) layered on top of every provider's retrieval, one fictional dataset (Solvane docs + 11 personas), and an LLM judge grading LLM-generated answers.